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FILOZOFICKÁ FAKULTA
ÚAM

UNIVERSITÉ DE
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NEW APPROACHES FOR THE ACQUISITION, SYSTEMATISATION AND INTERPRETATION OF ARCHAEOLOGICAL ARTEFACTS

Doctoral thesis

Josef Wilczek



Supervisors: PhDr. Natalie Venclová, DrSc.

Prof. Fabrice Monna

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I declare that this doctoral thesis is original work and was carried out by myself except where specific contributions by other persons are acknowledged.

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Chapter 1

Introduction

The universe around us is composed of innumerable elements. At the largest scale, it is composed of galaxies, solar systems, and planets. At a more human scale, it is composed of natural structures, biological organisms, and anthropogenic artefacts. At the smallest scale, it is composed of molecules, atoms, protons, electrons, and quarks. All these systems and structures are interdependent, constantly undergoing profound changes in their characteristics. Even structures considered as stable are affected by change over time (Reeves et al., 2016).

Identifying, characterising, and naming structures can be seen as activities specific to humans, which are then shared, copied, and transmitted, within a given civilization or culture, through collective perception, in a similar way to the transmission of genetic material, in the form of genes. This phenomenon was introduced by Dawkins (1976) in the 1970s, with the notion of “meme”, which corresponds to “an element of a culture or system of behaviour passed from one person to another by imitation or other non-genetic means” (English Oxford Living Dictionaries, 2017; see also Blackmore, 1999, p. 46). At any given moment, all the memes existing within the population form a set of thoughts named “the meme pool” (Dawkins, 1976). Thus, in Dawkins’ theory, humans copy ideas, skills and behaviours through imitation and learning. This process generates variation, either because of error, or by voluntary changes, or through creativity.

In the context of the history of science, the notion of meme can be compared with that of paradigm, defined as “...what the members of a scientific community share” (Kuhn, 1977), or “... the universally recognized scientific discoveries that, for a time, provide typical problems and solutions to a community of researchers” (Kuhn 2008, p.11). Within the archaeological community, several memes or paradigms (romantic, evolutionary, cultural-historical, processual and post-processual) can coexist in the meme pool.

There is one aspect that links all paradigms in archaeology – shape. The importance of shape for humans comes from their cognitive perceptions – the shape of an object in the real world can be deduced mainly by sight and touch. Identifying and classifying natural objects and structures by their colour and shape has constantly been essential for human survival.

In archaeology, artefact shape provides valuable information about chronology, socio-economic patterns, and socio-cultural identity. This information can be obtained by identifying structural and relational patterns between artefacts, based on their spatial and temporal position and their intrinsic formal aspects (e.g. shape, decoration, method of fabrication, and material composition).

The fundamental tool to assess the formal organisation of artefacts is taxonomy, through classification, and then typology. The essential element in typology is the type, “a class of artefacts defined by the clustering of attributes” (Renfrew and Bahn, 2012, p.584), traditionally supposed to represent an ideal shape, in relation to which all artefacts must be compared (Lestrel, 2000, p. 86).

Although coherent typologies have been adopted and are successfully used to organise many different classes of artefacts (e.g. ceramics, lithics, and jewellery), both their conception and application are still sometimes the subject of debate (e.g. Rouse, 1960; Hayden, 1984; Adams and Adams, 1991).

Typologies are initially intuitive, based on the researcher’s experience, acquired over time, through exposure to many artefacts, and thus difficult to reproduce by other researchers, who do not share exactly the same experience and exposure, even within the same meme pool. Typologies, even for the same class of artefacts, may also vary from region to region, as many different archaeological memes or paradigms may coexist. The set of descriptors used to establish a typology is generally based on a given corpus of artefacts, and has been built to take account of the specificities and variation existing within that corpus, to provide an exhaustive classification that includes every artefact in the corpus. As archaeology advances, and more artefacts are unearthed, it often becomes necessary to extend the existing typology by introducing new

descriptors or variants. The creation of typologies is therefore to some extent a never-ending process.

Illustration techniques may vary from one expert to another, thus rendering difficult the effective sharing of interpreted data between communities. Manual data acquisition allowing artefacts to be classified in a typology is a tedious and time-consuming process. The time required to record exhaustively the numerous artefacts resulting from archaeological excavation often means that not all artefacts are recorded, thus limiting even further the information available to the research community. These issues are particularly critical for the study of ceramics, as the information used for analysis is based on profiles drawn by experts.

Although thousands of artefacts have been recorded in large databases, there is still no analytical tool that can be used to classify them in existing typological classes (types), or even to identify the best analogy for a given fragment. Furthermore, for many artefacts, notably ceramics, the level of preservation makes it difficult to assign a type to a fragment, thus reducing the amount of information available for interpretation.

The aims of this doctoral thesis are therefore to investigate the acquisition and systematic shape-based organisation of artefacts, for their archaeological interpretation. Several innovative methodological and technical solutions are proposed for the many issues mentioned in the preceding paragraphs.

This volume is composed of an introduction (this chapter), followed by three chapters presenting the methods proposed here to simplify several key tasks necessary for artefact illustration, identification, reconstruction, and interpretation. The remainder of this volume is composed of three chapters presenting case-studies dealing with the shape-based classification of different types of artefacts, followed by a general conclusion discussing the importance of geometric morphometrics for archaeology.

The second chapter provides an overview of geometric morphometrics, the overarching theme and focus of this shape-based study. This chapter also includes a brief description of many of the mathematic and statistical methods used in archaeological analysis to study artefact shape.

The third chapter describes the DACORD software system, developed to semi-automatically or automatically orient and draw archaeological pottery, based on 3D model geometry. Given the Declaration of Inventorship, this chapter will appear blank in the freely available version of this dissertation.

The fourth chapter presents a series of workflows to classify pottery fragments, two of which are probabilistic.

The fifth chapter is a case study comparing geometric morphometrics with two traditional approaches for the morpho-typological classification of ceramics.

The sixth chapter is a case study of Middle Bronze Age axes, confirming the validity of the existing generic typology, and presenting a finer-grained, shape-based automatic classification for the same corpus.

The seventh chapter is a case study of La Tène brooches, reconstructing their shape evolution over time.

The eighth and final chapter summarises the most important aspects of this exploration of shape in archaeological artefacts, in the context of geometric morphometrics.

Each of the five central chapters (chapters 3-7) is presented in the form and format of a scientific article, two of which have already been published (chapters 5 and 6, presented as published). Illustrations and references are therefore integrated chapter by chapter. A complete list of references is provided at the end, including references to artefacts in the corpuses, together with the list of figures, tables and supplementary materials. All supplementary materials (including those already published) are presented on a CD, chapter by chapter. To harmonise the graphical

layout of the dissertation with the previously published articles, all links to figures, references and Supplementary materials in the dissertation are highlighted in blue.

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Chapter 2

Geometric morphometrics in archaeology

1. Introduction

The goal of this chapter is to provide a brief introduction to morphometrics and its applications for the study of artefact shape in archaeology, from the acquisition to the statistical analysis of shape data. The focus here is not the evolution of morphometrics as a scientific methodology, although a brief history of its development is presented. This chapter can thus serve as an overview of the literature on the use of geometric morphometrics in archaeology. More information about traditional and modern morphometrics, especially in the context of life sciences, can be found elsewhere (Rohlf 1990; Bookstein, 1991; Lestrel, 1997, 2000; Adams et al., 2004; Zelditch et al., 2004; Slice, 2005; Reyment, 2010).

1.1. Traditional morphometrics

Morphometrics is often considered as the study of shape variation and its covariation with other variables (Bookstein, 1991; Dryden and Mardia, 1998; Adams et al., 2004).

Traditional morphometrics, sometimes called multivariate morphometrics (Blackith and Reyment, 1971), is based on the study of shape, generally expressed by linear distances (length, width, height, etc.), angles, and/or ratios; or by a set of pre-defined categories (triangular, rounded, oval, etc.; Fig. 1). Morphological variables are usually treated by multivariate statistical analysis in order to identify any structuration pattern(s), and differences between samples, or to investigate relationships between shape and size (Marcus, 1990; Slice, 2005).

Interest in traditional morphometric analysis for archaeology greatly increased as advances in biometrics and numerical taxonomy were proposed and developed by biologists and statisticians in the 1960s and 1970s (e.g. Sokal and Sneath, 1963). Shape contained measurable information that could be used for computer-aided classifications and seriations of archaeological artefacts. Methods of “typometrics” (Djindjian, 1991, p. 74) were developed by researchers from the numerical taxonomy movement – many of them present at the “Anglo-Romanian Conference

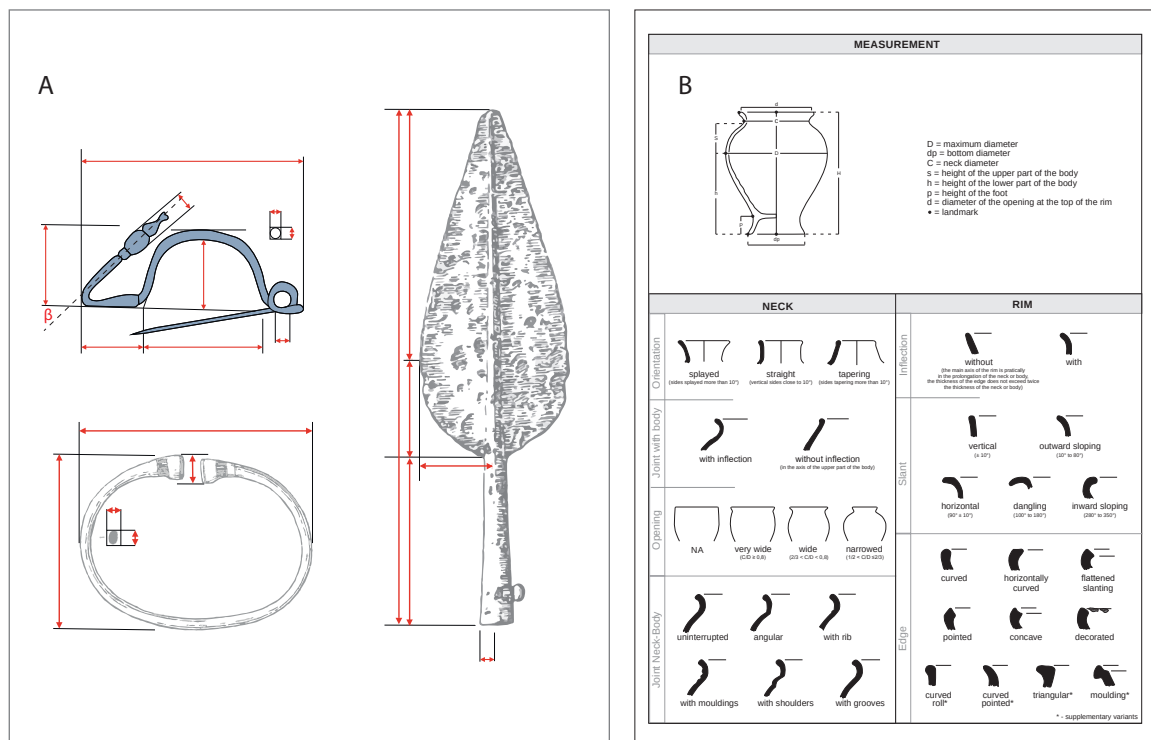


Fig. 1. Traditional morphometrics in archaeology. A) Measurements. B) Descriptive system (set of pre-defined categories).

on Mathematics in the Archaeological and Historical Sciences in Mamaia (Romania)”, held in 1968 (Hodson et al., 1971). Their methods were rapidly adapted by the broader archaeological community and increasingly used to classify a wide range of artefacts, notably brooches (Hodson et al., 1966, 1971), lithic artefacts (Boutin et al., 1977; Monnier and Etienne, 1978; Vigneron, 1979), bronze axes (Vuaillat and Massonnie, 1974), and ceramics (e.g. Hardy-Smith, 1974; Orton, 1974; Shennan and Wilcock, 1975; for a general overview see Orton, 1980 and Orton et al., 1993).

1.2. Modern geometric morphometrics

The need to rethink traditional morphometrics arose in the 1980s when a series of problems appeared that could not be solved by approaches based on linear measurements, ratios and angles. For instance, it became evident that (i) it was not possible to assess the homology of measurements on objects without precisely defined homologous points, (ii) a set of measurements did not automatically express the spatial relationship existing between them (an object with a maximum height of 1 cm and maximum length of 0.5 cm can be an oval, or a teardrop, or another shape entirely), (iii) measurements taken from a single object were inter-correlated (if starting from the same point), and (iv) no definitive solution to remove the size effect and thus observe only the shape had yet been found. Other limitations were that (v) generating datasets required a considerable amount of time, (vi) the operator effect on data acquisition and/or treatment was rarely discussed or quantified, and that (vii) some scarce and/or incomplete objects were often removed from the study (e.g. Adams et al., 2004; Zelditch et al., 2004; Urbanová and Králík, 2009).

New-wave morphometrics stated that most of these limitations could be overcome by studying the shape of objects, expressed not as sets of measurements, but by means of geometric relationships. The key aspect of modern morphometrics lies in the definition of shape as “all the geometric information that remains when location, scale and rotational effects are filtered out from an object” (Zelditch et al., 2004, p. 11; Kendall, 1977). The basic idea behind this is that, if any two or more objects are positioned in space, observable differences between them are caused by four factors – differences in position, size, orientation, and shape. When the first three factors are removed from the equation, objects differ only by their shape.

Although the idea of shape is simple, a considerable amount of work was required to transpose theoretical considerations about shape into shape theory, expressed in terms of mathematics (notably geometry), physics, statistics, and computer graphics. These aspects were developed more specifically by F. J. Rohlf, F. L. Bookstein, N. MacLeod, D. E. Slice, L. Marcus, P. E. Lestrel, F. P. Kuhl, and G. R. Giardina, through a variety of methods known together as geometric morphometrics.

The common element for all these methods is that they represent the shape of an object in a quantitative continuous way, in the form of shape variables. These shape variables characterise the shape of the original object, just as its chemical composition characterises the elements that compose it. They can also express the amount and direction(s) of shape transformation needed to metamorphose the shape of a given object into that of another.

Modern geometric morphometric methods are nowadays widely applied in life and other natural sciences. Shape variables are generally used to investigate both shape variation in the samples studied, and its covariation with any other discrete (e.g. gender, type, region) or continuous (e.g. size, weight, time) variables.

Although modern geometric morphometrics has not yet become a mainstream technique for the analysis of archaeological artefacts, archaeology was nevertheless in the forefront of key events related to the birth and development of modern morphometrics. Kendall, who defined

the modern concept of shape (Kendall, 1977, 1981, 1984, 1986), largely contributed to research in archaeology, notably through the application of computational Seriation (Kendall, 1963, 1971) and Multidimensional Scaling (Kendall, 1971). Kendall claimed that his interest in shape theory “was prompted by a statistical topic on the fringes of archaeology” (Kendall, 1989, p. 87). One of the earliest demonstrations of modern morphometric analysis was the study of 52 megalithic stones located near Land’s End (England), in order to find out whether they were aligned linearly or randomly (Kendall and Kendall, 1980; Kendall, 1989).

Although the term, concept, and terminology of geometric morphometrics as defined and used in the life sciences were not initially applied to the study of archaeological artefacts, some morphometric methods were gradually adopted, as computational power increased. Many computer applications and artefact databases were developed during the 1980s. Outline-based morphometric methods could thus be used more easily (e.g. B-spline curves, Bezier curves, cubic splines, tangent representation, Fourier Analysis; Jones and Wilcock, 1974; Main, 1978, 1986, 1988; Leese and Main, 1983; Gero and Mazzullo, 1984; Hall and Laflin, 1984; Laflin, 1986), and several tools were developed to compare artefact outlines (e.g. Jičín and Vašíček, 1971; Orton, 1974, 1987; Wilcock and Shennan, 1975; Gero and Mazzullo, 1984). However, these studies did not overtly set out to inspect shape variation, but rather sought to identify the closest analogy to the artefact in question in a referential database. This enthusiasm for computer-based archaeology shifted during the 1990s to applications based on imaging and Geographic Information Systems (GIS), thus focusing less on artefact shape and more on spatial organisation (Conolly and Lake, 2006).

The use of geometric morphometric methods in archaeology remained chiefly the domain of the life sciences, and biological remains were the main object of investigation. Pollen was used to study and model plant domestication and diffusion (e.g. Bouby et al., 2005; Terral et al., 2010), while human and animal skeletons (mainly skulls, pelvises, and long bones) were used to investigate sexual dimorphism, age and differences between samples of different geographic and temporal origin (e.g. Bookstein et al., 2003; Bruner, 2004; Gunz et al., 2009; Ottoni et al., 2013; Cucchi et al., 2017; Weissbrod et al., 2017).

Archaeology in the 21st century has now adopted geometric morphometric methods for the study of archaeological artefacts. The most successful applications to date have been with lithic artefacts (e.g. Lycett et al., 2006; Costa, 2010; Eren and Lycett, 2012), ceramics (e.g. Saragusti et al., 2005; Karasik and Smilansky, 2011), brooches (Small, 1996; Dryden, 2000), statues (Buxeda i Garrigós and Gordaliza, 2011; Urbanová et al., 2011), and Bronze Age axes (Forel et al., 2009; Monna et al., 2013).

2. Role of morphometrics in the study of archaeological artefacts

Archaeology is largely based on the study of artefacts, i.e. the intentional products of our ancestors. By definition, artefacts are “dead” objects, inert, motionless, mute, found either buried or lying on the surface of the Earth. Archaeological knowledge is based on the analysis of these dead items, by observing their attributes and spatial relationships, identifying any meaningful structure, and then interpreting these findings in terms of current paradigms or memes in archaeology (Spaulding, 1960; Neustupný, 2007, 2010).

The key question here is how geometric morphometrics can contribute to the archaeological investigation of artefacts. Generally, geometric morphometrics focuses on the quantitative analysis of shape information. In archaeology, geometric morphometrics combined with other statistical techniques can be used for:

Objective	Research question	Statistics	Literature/Chapters
(Semi-)automatic data acquisition)	What is the best method?	normal vectors circle fitting profile fitting rim fitting	Halif, 1997, 1999; Halif and Flusser, 1997; Cao and Mumford, 2002; Kampel et al., 2005; Mara and Sablatnig, 2006; Mara et al., 2007; Karasik and Smilansky, 2008; Mara, 2009; Karasik, 2010; Han and Hahn, 2014
		-	Chapter 3
Creating morpho-typology	Are artefacts organised by shape?	PCA CA	Gilboa et al., 2004; Smilansky et al., 2004; Karasik et al., 2005, 2014; Saragusti et al., 2005; Grosman et al., 2008; Karasik and Smilansky, 2011
	How many groups exist? Can morpho-typology be automatically generated?	PCA/MBCA	Chapter 4 Chapter 5 (Wilczek et al., 2014) Chapter 6 (Wilczek et al., 2015)
Shape comparison between pre-classified artefacts	Is there any shape difference between groups? Can shape difference between groups be visualised?	PCA LDA MANOVA	Lycett et al., 2006; Adan-Bayewitz et al., 2009; Forel et al., 2009; Costa, 2010; Karasik, 2010; Grosman et al., 2011; Karasik and Smilansky, 2011; Charlin and Rolando, 2012; Eren and Lycett, 2012; Lycett and Von Cramon-Taubadel, 2013; Okumura and Araujo, 2014; Eren et al., 2015
		PCA MBDA LDA MANOVA SOM	Chapter 5 (Wilczek et al., 2014) Chapter 6 (Wilczek et al., 2015) Chapter 7
Identification of best analogy for complete artefacts	What is the closest analogy to a given artefact?	calculation of similarities between artefacts	Martínez-Carillo et al., 2009, 2010; Lucena et al., 2014
	To which group does this artefact belong?	MBDA	Chapter 6 (Wilczek, Monna et al. 2015)
Identification of best analogy for incomplete artefacts (fragments)	What is the closest analogy to a given fragment?	calculation of similarities between a fragment and artefacts or fragment	Sablatnig and Menard, 1998; Kampel et al., 2001; Adler et al., 2002; Kampel and Sablatnig, 2002; Maiza and Gaildrat, 2005; Karasik and Smilansky, 2011; Picolli et al., 2015
	To which group does this fragment belong?	calculation of similarities between a fragment and existing artefacts or “potentially existing” artefacts projection of a fragment into a morphospace of all possible fragments	Chapter 4
Assessing variability between classes of artefacts	Which class of artefacts presents the greatest variability?	coefficients of variation Mahalanobis distance	Eren and Lycett, 2012; Monna et al., 2013
		convex hull calculation	Chapter 5 (Wilczek et al., 2014)
Assessing symmetry, regularity, and deformation in artefacts	How symmetrical or regular is this artefact?	calculating residual distances between expected and observed shape	Karasik et al., 2010, 2014; Mara et al., 2004; Saragusti et al., 2005; Lycett et al., 2006; Mara and Sablatnig, 2006, 2007; Lycett, 2008
		idem	Chapter 3
Testing and visualising shape changes over (continuous) time	Are artefact shape changes related to time?	-	-
		MANCOVA Multivariate Regression	Chapter 7

Table 1. General overview of research questions and statistical methods in the literature (chapters dealing with the same topics are also indicated). PCA – Principal Component Analysis; CA – Cluster Analysis; MBCA – Model-Based Cluster Analysis; MBDA – Model-Based Discriminant Analysis; LDA – Linear Discriminant Analysis; MANOVA – Multivariate Analysis of Variance; MANCOVA – Multivariate Analysis of Covariance; SOM – Self-Organised Maps.

- (Semi-)automatic data acquisition;
- Creating morpho-typologies, i.e. typologies based solely on shape; if required, classification can also introduce other descriptors (e.g. decoration, composition, or mode of fabrication);
- Shape comparison between pre-classified artefacts;
- Identification of best analogy for complete artefacts and prediction of class (“automatic classification”) for complete artefacts;
- Identification of best analogy for incomplete artefacts (fragments) and prediction of class (“automatic classification”) for incomplete artefacts;
- Assessing variability between classes of artefacts, notably for comparison of level of standardisation between classes of artefacts;
- Assessing symmetry, regularity, and deformation in artefacts;
- Testing and visualising shape changes over (continuous) time.

The choice of method depends on the research question, data selection, and on the logical and methodological research design developed in response to this question (e.g. [Legendre and Legendre, 1998](#)). In the absence of a focused research question, applying morphometrics to data is unlikely to provide any meaningful answers.

The following table ([Table 1](#)) presents a general overview of several previously studied research questions from the literature, and the statistical methods used to answer them. In addition, chapters in this thesis that deal with the same topics are also indicated.

Results obtained with geometric morphometrics and statistics are based on mathematical concepts, and are therefore reproducible. The crucial question is to what extent the results obtained can be considered as objective. The degree of objectivity depends on the selection criteria, the choice of model and method, and the quality and quantity of the artefacts studied. The availability of artefacts is not a matter of choice, but rather the objective result of archaeological surveys and excavations. Choosing models, methods, and selection criteria is the responsibility of the research team, and such choices will pre-condition the objectivity of the results. Systematically recording the choices made and the reasons for those choices will enhance reproducibility and thus contribute to the objectivity of the study.

3. Applying geometric morphometrics to archaeological artefacts

The workflow to analyse artefacts using geometric morphometrics can be summarised as follows. Once the research question and hypotheses have been defined, the appropriate shape data are acquired, and processed by the best adapted method to obtain shape variables. These shape variables are then analysed by statistical techniques to provide coherent scientifically based answers to the research question.

This section provides a general description of the methods used in the studies presented here.

3.1. Data acquisition

The initial step in archaeological geometric morphometrics consists in acquiring shape data, (i) directly on the artefact (positioned in a fixed coordinate system), (ii) from any 2D image (e.g. drawings, orthogonal photographs, etc.), or (iii) from 3D models (see [Fig. 2](#) for a general overview of 3D acquisition techniques). Artefact shape can be expressed as landmarks and/or outlines.

3.1.1. Landmarks

Most modern morphometric studies are based on the treatment of landmark data. Landmarks, or homologous points, are “discrete anatomical loci that are arguably homologous among all the individuals under analysis” (Benítez et al., 2014, p. 1) and are expressed by two- or three-dimensional Cartesian coordinates (e.g. Rohlf and Marcus, 1993; Slice, 1996). A set of landmarks on a given artefact is called a configuration. Three types of landmarks are often distinguished, based on the degree of precision with which each point can be located on the artefact (Bookstein, 1991). Landmarks of Type 1 correspond to the point where three structures meet. Landmarks of Type 2 correspond to the maxima of curvature. Landmarks of Type 3 correspond to the most external points on the artefact (Fig. 3).

The landmarks identified on the artefact: (i) should be homologous loci, (ii) should not alter typological positions relative to other landmarks, (iii) should provide adequate coverage of the morphology of the object, (iv) should be found repeatedly and reliably, and (v) should lie within the same plane (Zelditch et al., 2004).

A case study using landmarks on La Tène brooches is presented in Chapter 7.

3.1.2. Outlines

Identifying landmarks on archaeological artefacts is often problematic. Problems with artefacts arise (i) from their state of preservation, (ii) from their great variability in shape, due to many different factors (e.g. functional, chronological, stylistic, regional, and social differences in production), and (iii) from the absence of internal structure, as most artefacts are recorded in the literature in the form of external outlines. However, morphometric analysis can be performed on outline data. Outlines can be open or closed, i.e. with a beginning and end, or forming a continuous contour. Outlines are usually represented by an ordered set of discrete 2D/3D Cartesian, or polar coordinates or by what is known as chaincode (Slice, 2005).

Case studies using outlines can be found in Chapters 4 and 5 (ceramics), and 6 (Bronze Age axes).

3.2. Processing

If two artefacts are positioned in space, differences between them are caused by size, position, orientation, and shape. In order to observe differences in shape alone, the first three factors must be removed to allow the artefacts to be superimposed. The distance between two or more objects can be minimised either by direct calculation or by optimisation. These minimised distances are usually called shape variables. Shape variables in outline-based methods are coefficients obtained by mathematical transformation.

3.2.1. Size

To observe differences between artefact shapes, size correction must precede the calculation of shape variables. The commonest way to remove the size effect in landmark-based approaches is by scaling configurations by their corresponding centroid size, i.e. the sum of Euclidean distances between the centroid and every landmark in the configuration (Gower, 1971). In outline-based approaches, size can be removed by scaling: (i) relative to the perimeter (Schmittbuhl et al., 2003), or (ii) relative to the surface bounded by the outline (Sundberg, 1996; Hurth et al., 2003; Navarro et al., 2004). In the first case, the centred coordinates are divided by the perimeter length, in the second by the square root of the area bounded by the outline. In the case of very convoluted outlines, which have a longer perimeter than smooth outlines, it is not recommended to use the perimeter for size removal (Lestrel, 2000).

Several different methods of size removal for archaeological pottery are discussed in Chapter 5.

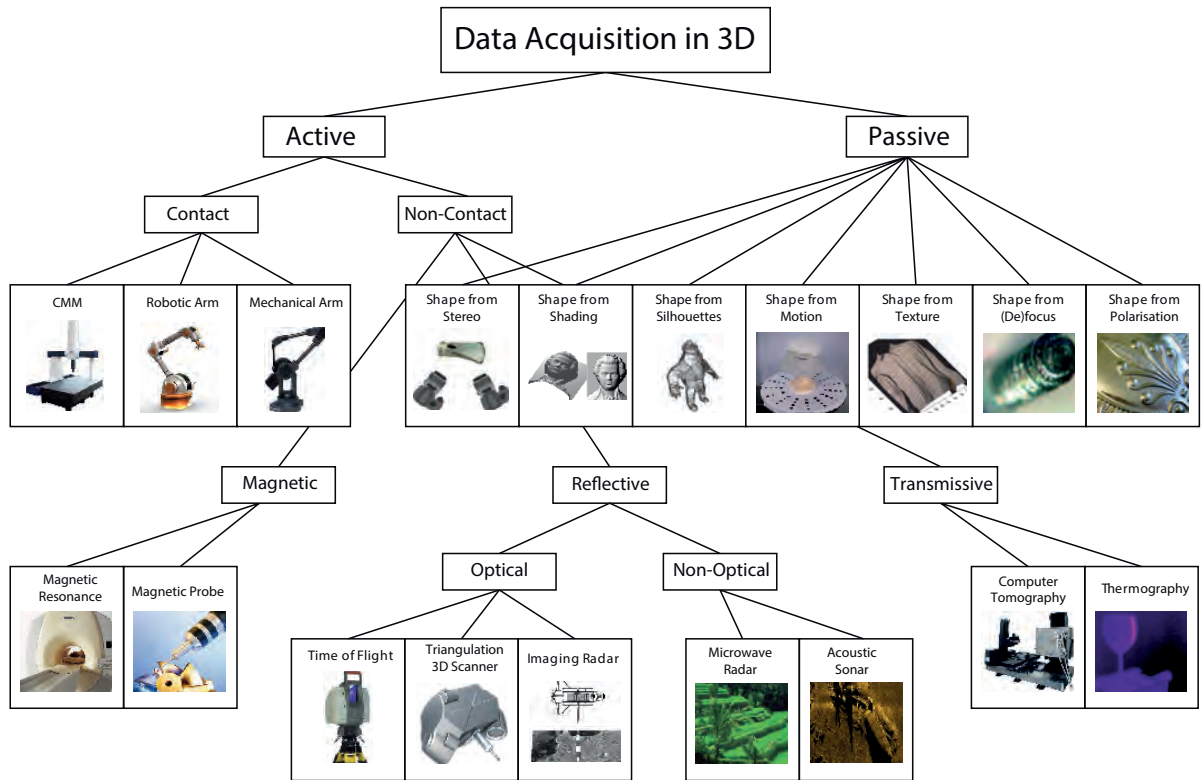


Fig. 2. General overview of 3D acquisition techniques (from Wilczek, 2013, Fig. 1; based on Curless, 1997, Fig. 1.1, 2000, Fig. 1; Lorient, 2009, Fig. 2.1, 2.2; Várady et al., 1997, Fig. 2; Vukašinović et al., 2007, Fig. 2).

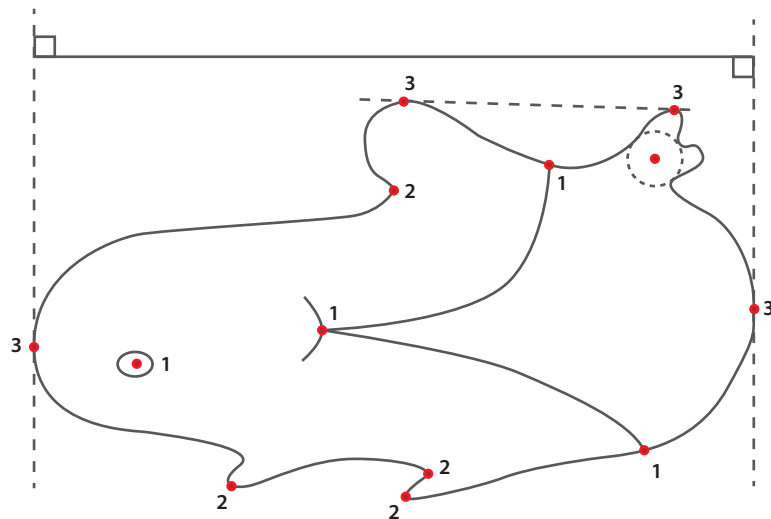


Fig. 3. The three types of landmarks defined by Bookstein, 1991. 1. Point where three structures meet (Type 1), 2. Maxima of curvature (Type 2), 3. External points (Type 3). From Bookstein, 1991, Fig. 3.3.1.

3.2.2. Baseline registration

Landmark configurations or outlines with two homologous points can be superimposed directly by Baseline registration (Bookstein, 1984, 1986, 1991). The two homologous points are fixed at the ends of a baseline, defined arbitrarily, often as (0,0) and (1,0), resulting in Bookstein coordinates or Bookstein shape coordinates (Bookstein, 1996). If the points used for baseline registration are too close, results may be affected, especially regarding group differentiation (Slice, 2005).

A case study using baseline registration on archaeological pottery is presented in Chapter 5.

3.2.3. Procrustes superimposition

The most common landmark-based method for shape superimposition is Procrustes superimposition. The adjustment process in Procrustes superimposition, using the least-square approach, is based on landmark matching. Analysis is often performed in three consecutive steps: the first eliminates the effect of position, the second removes the effect of size, and the third removes the effect of orientation (Rohlf, 1990; Rohlf and Slice, 1990; Zelditch et al., 2004; Slice, 2005). In more detail:

- To remove the position effect, the centroids of configurations are translated to the origin of the coordinate system, ($x=0$; $y=0$; and in 3D, $z=0$). Distances between configurations will then reflect differences in size, orientation, and shape.
- To remove the size effect, configurations need to be scaled to the same unit size (Gower, 1971), by dividing the coordinates of each landmark by the centroid size of the corresponding configuration. Distances between configurations will then reflect differences in orientation and shape.
- To remove the effect of orientation, the summed squared distances between corresponding landmarks are minimised. One configuration is fixed and labelled as reference (R). The second configuration, labelled target (T), is then rotated to minimise the distances between corresponding landmarks. The coordinates of the configurations aligned in this way are Procrustes coordinates. The sum of distances between corresponding landmarks (Procrustes distances) will then reflect differences in shape.

By removing position, size and orientation, the final dimensionality (degrees of freedom) of the shape space is $2p-4$ for 2D data, and $3p-7$ for 3D data, where p is the number of landmarks.

Procrustes superimposition uses the least-squares criterion, and is therefore sensitive to “outliers”, or imprecisely placed landmarks. The best-known problem is the Pinocchio effect: if one point is positioned at a great distance from the others, it will affect the alignment of all the other landmarks. For this reason, resistant fitting approaches were developed, using the median instead of the mean in alignment (Siegel and Benson, 1982; Rohlf and Slice, 1990; Slice, 2005).

Procrustes coordinates can be used to calculate the mean shape (Procrustes mean), which reflects the average shape of all the configurations. The vector of displacement between mean shape landmarks and each configuration (Procrustes residuals) can also be calculated. Mean shape and Procrustes residuals can be used to visualise differences between artefacts.

The most common method used to superimpose a set of objects is the General Procrustes Analysis (GPA), or General Procrustes Superimposition (GPS). The GPA consists in an iterative alignment process, in which one configuration is arbitrarily chosen as reference (mean shape). All configurations are then aligned with this reference. From these superimposed configurations, the new mean shape, i.e. new reference, and Procrustes distance are calculated. The process of fitting configurations to the reference continues until the difference in subsequently calculated Procrustes distances does not fall below an arbitrarily chosen threshold (Gower, 1975; Slice, 2005).

When the calculation of superimposition consists only in translation and rotation (without

size adjustment), the method is called Partial Procrustes Analysis. If the points can be mirrored in the analysis, the method is called Reflection Procrustes (Urbanová, 2009). If only two forms are superimposed, the analysis is called Ordinary Procrustes Analysis (OPA).

A case study using GPA on La Tène brooches is presented in Chapter 7.

3.2.4. *The Iterative Closest Point algorithm*

Superimposition is also widely used in computer vision for 3D model alignment. The Iterative Closest Point (ICP) algorithm is used for this purpose (Besl and McKay, 1992; Turk and Levoy, 1994; Zhang, 1994; Fitzgibbon, 2003; Szeliski, 2011; Pomerleau et al., 2013). The difference between Procrustes analysis and ICP is that the ICP algorithm does not necessarily require landmarks for alignment. Any other features, such as non-corresponding point sets (point clouds), triangle sets (meshes), line segments (polylines), curves, or surfaces can be used (Besl and McKay, 1992).

The basic tenets of ICP are similar to Procrustes analysis, i.e. minimising distances between objects by modifying their position, size and orientation. An artefact (target) is given a set position in the coordinate system and another artefact (source) is iteratively scaled, translated and rotated in order to minimise the distance between them (Besl and McKay, 1992).

The use of ICP to identify the best analogy for pottery fragments is presented in Chapter 4.

3.2.5. *Fourier analysis (harmonic or spectral analysis)*

Many outline-based geometric morphometric methods use Fourier descriptors calculated through Fourier analysis (Lestrel, 1997). Fourier analysis consists in transforming the original outline, expressed by (i) Cartesian coordinates, (ii) polar coordinates, (iii) tangent angles, or (iv) by chaincode (Claude, 2008), into an infinite series, called harmonics. Three types of Fourier Analysis are often presented in the literature: conventional Fourier analysis (FA), Elliptic Fourier Analysis (EFA) and Discrete Cosine Transform (DCT).

Conventional Fourier Analysis

Conventional Fourier Analysis transforms an outline by a trigonometric function into a set of harmonics, which can be represented as circles. Each harmonic is represented by two Fourier coefficients, which can serve as shape variables. Harmonics are ordered hierarchically, so that lower harmonics (i.e. larger circles) express overall shape, while higher harmonics (smaller circles) express local outline features. By summing harmonics (i.e. sets of coefficients), the shape of the artefact can be reconstructed by harmonic synthesis (inverse Fourier transformation). The greater the number of harmonics in the transformation, the more precise the reconstruction of the outline.

The Nyquist frequency criterion limits the number of harmonics to no more than half the number of points along the outline (Lestrel, 1989, 1997; Lestrel et al., 2005).

Elliptic Fourier Analysis

Elliptic Fourier Analysis (EFA), as proposed by Kuhl and Giardina (1982) decomposes an outline into a set of harmonic ellipses by separating the x and y coordinates, which are then expressed as a function of a third variable (t). Elliptic analysis can be used on both 2D and 3D outlines.

Case studies using EFA can be found in Chapters 4 and 5 (ceramics), and 6 (Bronze Age axes).

Discrete Cosine Transform

Another Fourier-based method is Discrete Cosine Transform (DCT; Rao and Yip, 1990), originally developed for audio, video and image data compression. It is used to decompose an open outline into a set of Fourier harmonics, expressed by pairs of coefficients, which can be treated as shape variables (Dommergues et al., 2007; Forel et al., 2009).

Case studies using DCT on ceramics can be found in Chapters 4 and 5.

3.2.6. Other outline-based methods

Other outline-based methods used to transform open outlines into sets of shape variables are Tangent transform (e.g. [Main, 1978, 1986](#); [Karasik and Smilansky, 2011](#)), Polynomials (e.g. [Monna et al., 2013](#)), Bezier curves (e.g. [Engel, 1986](#); [Rohlf, 1990a, 1990b](#)), and Spline methods (e.g. [Evans et al., 1985](#); [Lafin, 1986](#); [Hlaváčková-Schindler et al., 2001](#); [Kampel and Sablatnig, 2002](#)), such as Basis Splines (B-splines), Cubic Splines, Parametric Cubic Splines, and Bell-shaped Splines (for a general overview see [Rohlf, 1990](#)).

3.3. Post-processing

3.3.1. Investigation of shape variability – Principal Component Analysis

Principal Component Analysis (PCA) is the most frequently used multivariate method in geometric morphometrics. The main purpose of PCA is to reduce the dimensionality of the initial shape data, by replacing the original shape variables with Principal Components or Axes. These Principal Components (PC) are linear combinations of the original shape variables, and are mutually independent ([Jolliffe, 2002](#)). Principal Components are ordered in relation to the variance of the shape data they account for. A relatively low number of PC (from 3 to 10) is often sufficient to represent most (e.g. 90%) of the shape information in the dataset, and can thus be adequate for data interpretation.

In geometric morphometrics, the space thus defined by PCA is sometimes termed morphospace. The mean shape occupies the centre of the morphospace, around which all artefacts are distributed, based on their shape similarities.

In morphometrics, PCA is generally used to (i) transform original shape variables into linear combinations, (ii) inspect shape variation between artefacts, and (iii) identify shape-related organisation of artefacts. When the distribution of artefacts in a PCA-based morphospace represents spatially distinct classes, PCA can also serve to (i) create a classification, (ii) differentiate between classes, and (iii) integrate newly found artefacts into an existing classification.

3.3.2. Classification

Cluster Analysis

Cluster Analysis (CA) is a generic family of methods designed to identify clusters (i.e. discrete loci) in a corpus of artefacts, based on discontinuities in the data ([Legendre and Legendre, 1998](#)). Cluster Analysis is the statistical technique most frequently used in archaeology for the creation of typologies (see e.g. [Orton, 1980](#); [Djindjian, 1991](#); [Baxter, 1994, 2003](#)). Clusters are formed based on similarities between artefacts. Results of hierarchical cluster analysis are usually represented as a dendrogram showing the distances between clustering artefacts.

Model-based cluster analysis

Model-Based Cluster Analysis (MBCA) is a modern classification method using probability models. The main difference between hierarchical and model-based clustering is that clusters are not defined by the similarities (i.e. distances) between artefacts, but rather by their shape. The advantage of model-based clustering over other classification techniques is that it allows (i) easier determination of the optimal number of clusters, (ii) a wide range of classification designs, (iii) more robust treatment of outliers, and (iv) probability values to be used to classify artefacts ([Fraley and Raftery, 1998, 2002](#); [McLachlan and Peel, 2000](#)).

In model-based clustering, the data are supposed to form groups, defined by the mean (corresponding to location) and the covariance (corresponding to shape, size and orientation). These parameters are usually estimated by a two-step iterative Expectation-Maximisation (EM) algorithm ([Dempster et al., 1977](#)).

The optimal classification model in MBCA and the number of typological groups are usually

determined by the Akaike or Bayesian Information Criterion (AIC, BIC; [Akaike, 1973](#); [Schwarz, 1978](#); [Wehrens, 2011](#)).

3.3.3. Investigation of differences between classes

Differences between classes can be investigated when membership is known in advance. Several statistical methods can be used to investigate differences between classes: (i) (Multivariate) Analysis of Variance, (ii) Linear Discriminant Analysis, or (iii) Canonical Variate Analysis.

(Multivariate) Analysis of Variance

Shape expressed by original or PCA-transformed shape variables is continuous in nature. The most appropriate statistical tests used to compare shape differences between groups are One-Way Analysis of Variance (ANOVA), Hotelling's T^2 test, and One-Way Multivariate Analysis of Variance (MANOVA).

One-Way ANOVA tests if any difference exists between the means of two or more classes on one continuous dependent variable. The F statistic used for ANOVA calculation is the ratio between the variance explained by the categorical variable, and the variance that is unexplained. One-Way ANOVA in geometric morphometrics is generally used to compare differences between classes, based on size, or the selected Principal Component (e.g. [Gharaibeh, 2005](#)).

Hotelling's T^2 tests if any difference exists between the vectors of the means of two classes on several continuous variables (e.g. shape expressed by a set of shape variables, or by a set of Principal Components; [Hotelling, 1931](#); [McLachlan, 1992](#); [Reyment, 2010](#)).

One-Way MANOVA has the same purpose as Hotelling's T^2 test, but can be used to compare more than two groups.

The results of One-Way ANOVA and MANOVA indicate whether three or more classes differ statistically. Further information about which class differs significantly from the others can be obtained by additional post hoc multiple comparison procedures, such as LSD, Bonferroni, or Tukey.

The assumptions when using these methods are (i) the independence of observations, (ii) normally distributed data, (iii) the equality of variance, and (iv) the absence of outliers. A MANOVA also requires an adequate sample size (i.e. the number of artefacts within each class is higher than the number of dependent variables). In the case of assumption violation, differences between classes can be tested by using non-parametric alternatives: for example, Kruskal-Wallis ANOVA for One-Way ANOVA ([McKnight and Najab, 2010](#)), Two-group permutation test for Hotelling's T^2 ([Urbanová, 2009](#)), or Non-parametric One-Way MANOVA for One-Way MANOVA ([Anderson, 2001](#)).

Linear Discriminant Analysis

The purpose of Linear Discriminant Analysis (LDA), is to optimally separate two or more classes of artefacts. Discrimination is achieved by identifying a linear transformation that maximises between-group variation while minimising within-group variation. The linear discriminant function calculated from an initial dataset can be used to classify newly found artefacts ([McLachlan, 1992](#)).

The LDA can be calculated only when the class for all artefacts is known in advance (*a priori*), and each object can be a member of only one class. Although LDA also assumes multivariate normal distribution, this technique remains reasonably robust even when the assumption of normality is not satisfied ([Klecka, 1980](#); [Lestrel, 2000](#)).

Canonical Variate Analysis

Canonical Variate Analysis (CVA) in geometric morphometrics serves as a powerful exploratory method to visually represent differentiation between classes of artefacts in a multidimensional space ([Lestrel, 2000](#)). Although based on the same principles as (M)ANOVA (i.e. maximising

between-group variation while minimising within-group variation), CVA calculation involves procedures that alter the multidimensional space, and thus do not allow statistical hypothesis testing (Zelditch et al., 2004).

4. Conclusion

This chapter has introduced all the morphometric and statistical methods that will be applied in the case studies presented in Chapters 4 to 7. It can thus serve as an overview of the literature on the use of geometric morphometrics in archaeology and provide a sound basis for the development of future morphometric applications.

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Chapter 3

Computer-Assisted Drawing of Archaeological Pottery

The quality of data used in the morphometric analyses presented in the previous chapter is of crucial importance. Drawing artefacts, especially pottery fragments, is a tedious and time-consuming process, thus leaving many fragments unexploited. The goal of this chapter is therefore to present and test a (semi-)automatic method of pottery shape data acquisition, to improve both the quality and quantity of data available. These methods are included in a software system, DACORD, recently developed in collaboration between researchers at the University of Burgundy Franche-Comté in Dijon (France) and Masaryk University in Brno (Czech Republic).

Given the Declaration of Inventorship, this chapter is only available to those persons having signed a confidentiality agreement, and will thus appear blank in the freely available version of this dissertation.

Chapter 4

Reconstructing the past from fragments

The main motivation for the DACORD system, presented in the previous chapter, was to acquire precisely oriented profile outlines of pottery fragments. These outlines can then be used for fragment attribution and to create classifications.

The goal of this chapter is to present three different workflows by which profile outlines can be morpho-typologically classified. These workflows are tested on a corpus of 47 complete vessels from Moravia, dated to the middle La Tène period (LT B–C1).

Abstract

The automatic attribution of incomplete objects to pre-defined or potential morphological classifications, with probabilistic and visual control, is sought by many archaeologists, notably those working with ceramics. Innumerable fragments have been collected through archaeological excavation, but sufficient time is rarely available for full analysis and interpretation. Sometimes, it is possible to treat, classify and catalogue only those fragments possessing a set of “typical/significant” characteristics, while the smaller fragments remain unanalysed, unclassified, and may eventually be discarded.

The goal of this paper is to propose a solution for these problems, by presenting three workflows, based on a combination of morphometrics, statistics and modern optimisation techniques. Together, these methods identify the best analogy for the fragment, provide probabilistic classification for the fragment, and also suggest a complete shape reconstruction, based on the overall shape information contained in the corpus. Results show that the multiplication of partial information contained within small and non-rim fragments can considerably increase the overall information available for archaeological interpretations. The entire process is tested on a corpus of complete vessels from Moravia (Czech Republic), dated from the middle La Tène period (LT B–C1).

Keywords: Archaeology; (Semi-)automatic classification; (Semi-)automatic attribution; Ceramics; Morphometrics; Closed contour; Open contour; Iterative Closest Point; Model-Based Cluster Analysis; Model-Based Discriminant Analysis; Simulated Annealing

1. Introduction

Pottery is the most abundant material found in archaeological excavations. Besides information about chronology, and the evolution of technique and style, it contains evidence of social organisation. Unlike more precious artefacts dedicated to the elite, ceramics have been used by all social strata. The socio-economic dynamics of ancient populations can be reconstructed, based on the observation of intrinsic pottery features (clay, fabrication technique, shape, and decoration) and spatial distribution (e.g. [Orton, 1980](#); [Rice, 1987](#); [Buko, 1990](#); [Orton et al., 1993](#)). However, ceramics suffer from strong post-depositional degradation caused by climate, soil acidity, and other factors. It is estimated that almost 95% of artefacts have vanished over time and, of the remaining 5%, only 10 to 20% of fragments are used in analysis and interpretation (i.e. mainly those containing the rim or base).

Traditional typological techniques suffer from a recurring problem of subjectivity, and acquiring sufficient expertise to overcome this problem takes many years (e.g. [Hodson et al., 1966](#); [Hermon and Niccolucci, 2002](#); [Hermon et al., 2004](#)). Adapting typological schemes to another geographic region or chronological period may be challenging. The pottery specialist often tries to find the best analogy (i.e. candidate or match) for the fragment shape, from within a well-established typological system, or through similarity with another previously discovered artefact. This attribution process is time-consuming, particularly when thousands of images have to be examined, and compared to the fragment to be classified.

Since the 1980s, numerous methods have been published for the automatic attribution and classification of ceramics based on profile geometry. These approaches often seek to identify landmarks along the profile outline, dividing it into segments. Such segments (e.g. rim, body, or base) are then expressed mathematically, either as tangents (e.g. [Leese and Main, 1983](#); [Main, 1986, 1988](#); [Gilboa et al., 2004](#); [Saragusti et al., 2005](#); [Karasik and Smilansky, 2011](#); [Smith et al., 2014](#)), or cubic and B-spline curves (e.g. [Lafin, 1986](#); [Hlaváčková-Schindler et al., 2001](#); [Kampel and Sablatnig, 2003, 2007](#)). Similarities between a given fragment and the segments in the database can then be calculated to identify the best analogy for that fragment (the process of attribution or matching). Similarity can also be calculated by minimising distances between points on profiles or their skeletons ([Liu et al., 2005](#); [Maiza and Gaildrat, 2005](#); [Picolli et al., 2015](#)).

These methods can be used to classify incomplete pottery fragments. However, (i) they often require profile segmentation, (ii) they can be sensitive to the quality of the initial (outline) data, (iii) they do not provide the degree of certainty by which fragments are classified, and (iv) they have rarely been evaluated on archaeological data. These methods are often applied only to rim fragments, while the identification potential of body and base fragments remains unexplored. Questions that still remain unanswered relate to (i) the size of a useful fragment, and (ii) the usefulness of non-rim fragments.

Here, we propose three different methods of typological attribution and classification for pottery fragments, to reduce the effects of these limitations, and to attempt to provide answers for these two remaining questions.

The first workflow follows the traditional method of attribution, i.e. identifying the most similar analogy for a given fragment in the database. This method ranks all vessels in the database, from most to least similar to the fragment (Fig. 1:A; Martínez-Carrillo al., 2010). The pottery specialist can then identify the best candidates for the classification process, with visual and quantitative control of best match.

The second workflow seeks to overcome the quantitative and qualitative limitations of the existing corpus of artefacts. The workflow is based on two assumptions. In a PCA-based morphospace, calculated from the shape of all the vessels in the database (Fig. 1:B): (i) all shapes can exist, (ii) the mean shape of the database occupies the centre of the morphospace. All existing vessels surround the mean shape, thus limiting the spatial distribution of potential vessel shapes that are sufficiently similar to the corpus. These potential vessel shapes can serve as a referential database within which the most similar analogy for a fragment (best match) can be identified based on similarity calculation (as in workflow 1). The location of this best match can be highlighted in the morphospace. The real vessel closest to this best match determines the typological class of the fragment (Fig. 1:B). When a probabilistic method (i.e. Model-Based Cluster Analysis or Model-Based Discriminant Analysis) is used for classification (Fig. 1:C), the degree of probability of this best match can be calculated. The typological class of the fragment can thus be defined with a quantifiable degree of certainty.

The third workflow consists in projecting the fragment into a PCA-based morphospace of all possible fragments, obtained by mathematically fragmenting all previously classified complete vessel shapes in the database (Fig. 1:D). As in workflow 2, the application of a probabilistic method serves to quantify the degree of certainty.

All three workflows are demonstrated and assessed on a corpus of complete vessels from Central Europe, dated to the middle La Tène period (LT B–C1).

2. Materials and methods

2.1. Corpus

The test corpus consists of 47 visually similar complete ceramic vessels from Second Iron Age funerary areas in Moravia (Czech Republic), all dated to LT B–C1 (420–175 BC). Their morphology has been described in catalogues as “vases”, “biconical vases” and/or “situlae” (e.g. Filip, 1956; Čížmářová, 2005, 2009, 2011, 2013). Some of these vessels have been used to define La Tène chronology (Meduna, 1980). Unfortunately, no uniform typology has yet been established. Model-based Cluster Analysis (MBCA) is used here for unsupervised standardised classification of vessels.

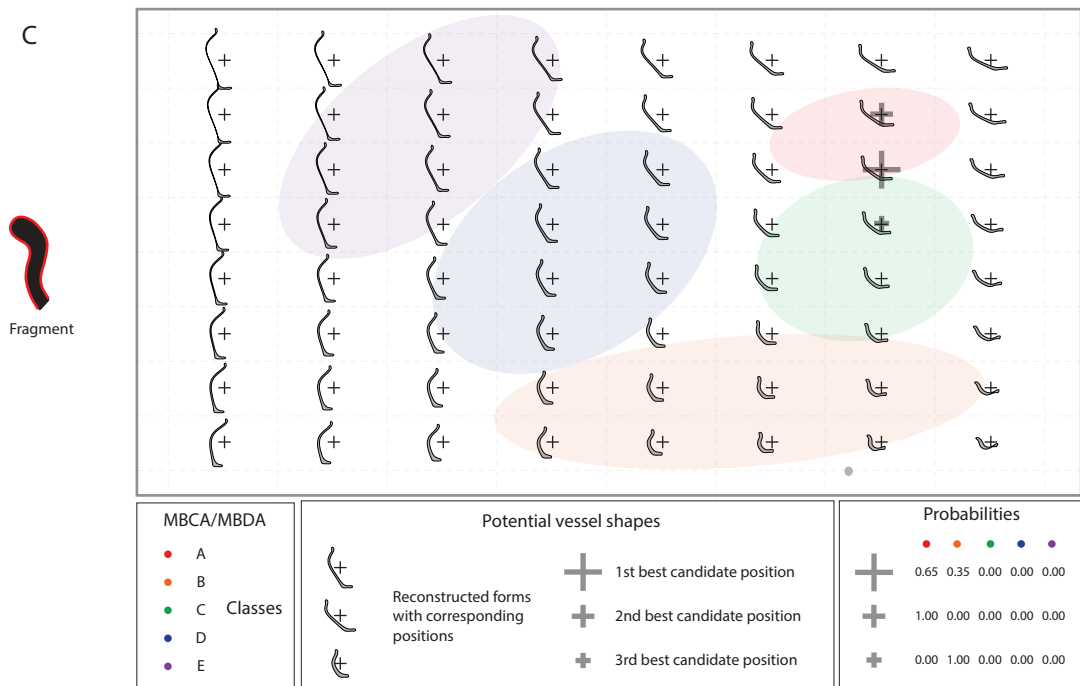
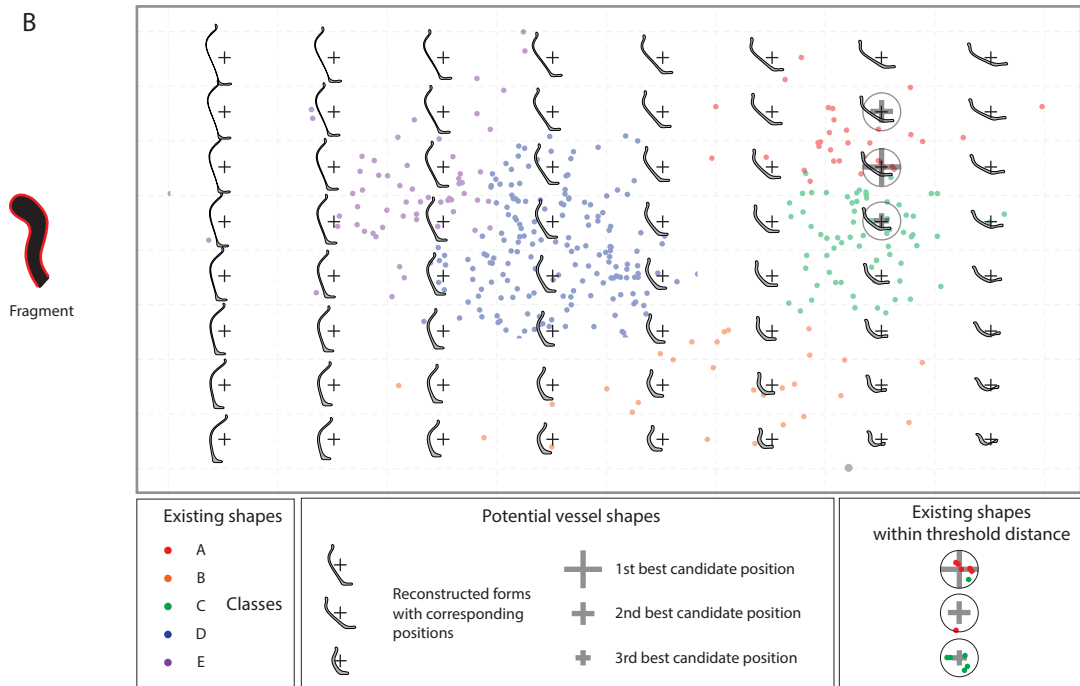
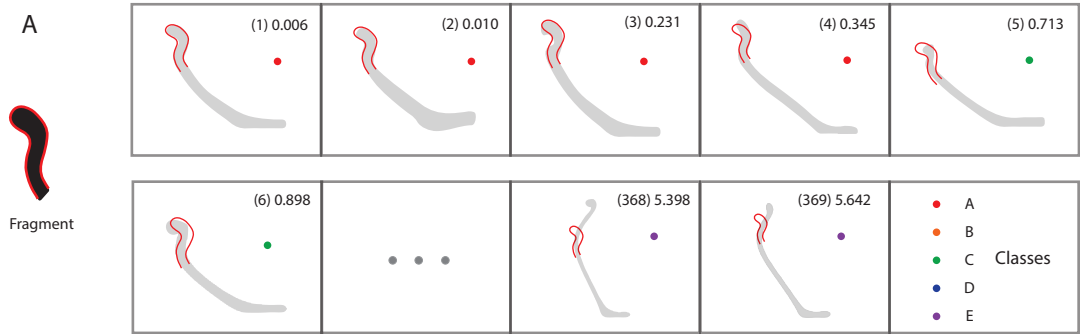


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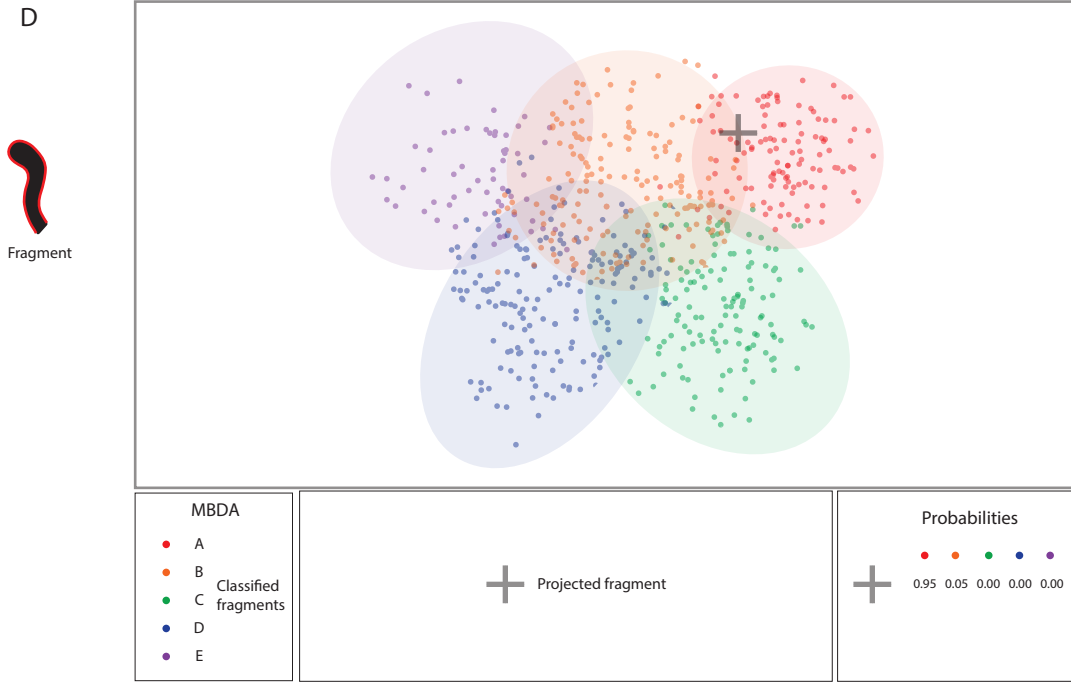


Fig 1. Three proposed workflows for fragment classification. A) Workflow 1. Classification according to the most similar vessel in the database. Numbers in rectangles correspond to the similarity (i.e. distance) between fragment and vessel. B-C) Workflow 2. Classification according to potential vessel shapes (see text for explanation). D) Workflow 3. Classification by projecting the fragment into a PCA-based morphospace of all possible fragments, obtained by mathematically fragmenting all previously classified complete vessel shapes in the database.

All ceramic profile images available in the literature (see [Supplementary materials S1](#)) were scanned at 600 dpi resolution ([Fig. 2:A](#)). Neighbouring image pixels with the same colour intensity served to trace the x and y coordinates of profile outlines ([Fig. 2:B](#); e.g. [Freeman, 1974](#); [Madhvanath et al., 1999](#); [Szeliski, 2011](#)). Each profile outline was scaled to its original size, inverted, and positioned so that the rotational axis coincided with $x=0$, and the rim with $y=0$ ([Fig. 2:C](#)). The profile outline was then reduced to 400 points, with the starting point at the rim, and the inner and outer profiles were traced ([Fig. 2:D](#)). Profile outlines were used in workflows 1 and 2, while fragments were used in workflow 3. These fragments were open outlines, expressed as percentages (e.g. 10%, 15%, etc.), based on points equally spaced along the profile ([Fig. 2:E](#)).

2.2. Identifying the most similar analogy

The best analogy in workflows 1 and 2 is identified by a modified version of an Iterative Closest Point (ICP) algorithm. The ICP algorithm is widely applied in computer vision for 3D model alignment (e.g. [Besl and McKay, 1992](#); [Turk and Levoy, 1994](#); [Zhang, 1994](#); [Fitzgibbon, 2003](#); [Pomerleau et al., 2013](#)). A complete profile (target), is positioned in the coordinate system, and the fragment to be matched (source), is iteratively translated, scaled, and rotated, to minimise the distance between target and source ([Fig. 3](#)). The general cost function to align target and source rz projection (where r corresponds to a radius, and z to the rotational axis) can be written as ([eq. 1](#)):

$$\min_{a,b,\varphi,S} \left(\sum_{i=1}^M d^2(ps_i, pr) \right) \quad [eq. 1]$$

where $d^2(ps_i, pr)$ is the sum of the squared distances between each of the M points of the source, noted ps_i , and their corresponding closest points on the target, pr . The source is translated (a, b) along the r and z axis, rotated (φ), and sized by the scalar (S).

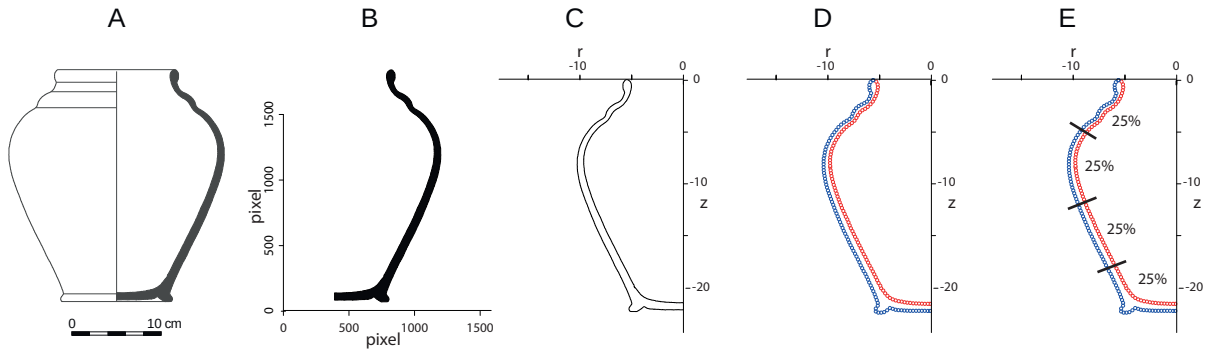


Fig. 2. Data acquisition. A) Example of an original drawing from the literature (here [Čizmařová, 2005, Fig. 65:8](#)), scanned at resolution 600 dpi. B) Extraction of the x and y coordinates of points along the profile outline. C) Scaling the outline to the original size of the vessel, mirroring it, and repositioning it, according to (i) the axis of symmetry (i.e. z axis), and (ii) r axis. D) Sampling of 200 equally spaced points along the profile outline, and extraction of the outer (blue points) and inner (red points) profiles. E) Extraction of open outlines of fragments, expressed as percentages (here 25%), based on points equally spaced along the profile outline.

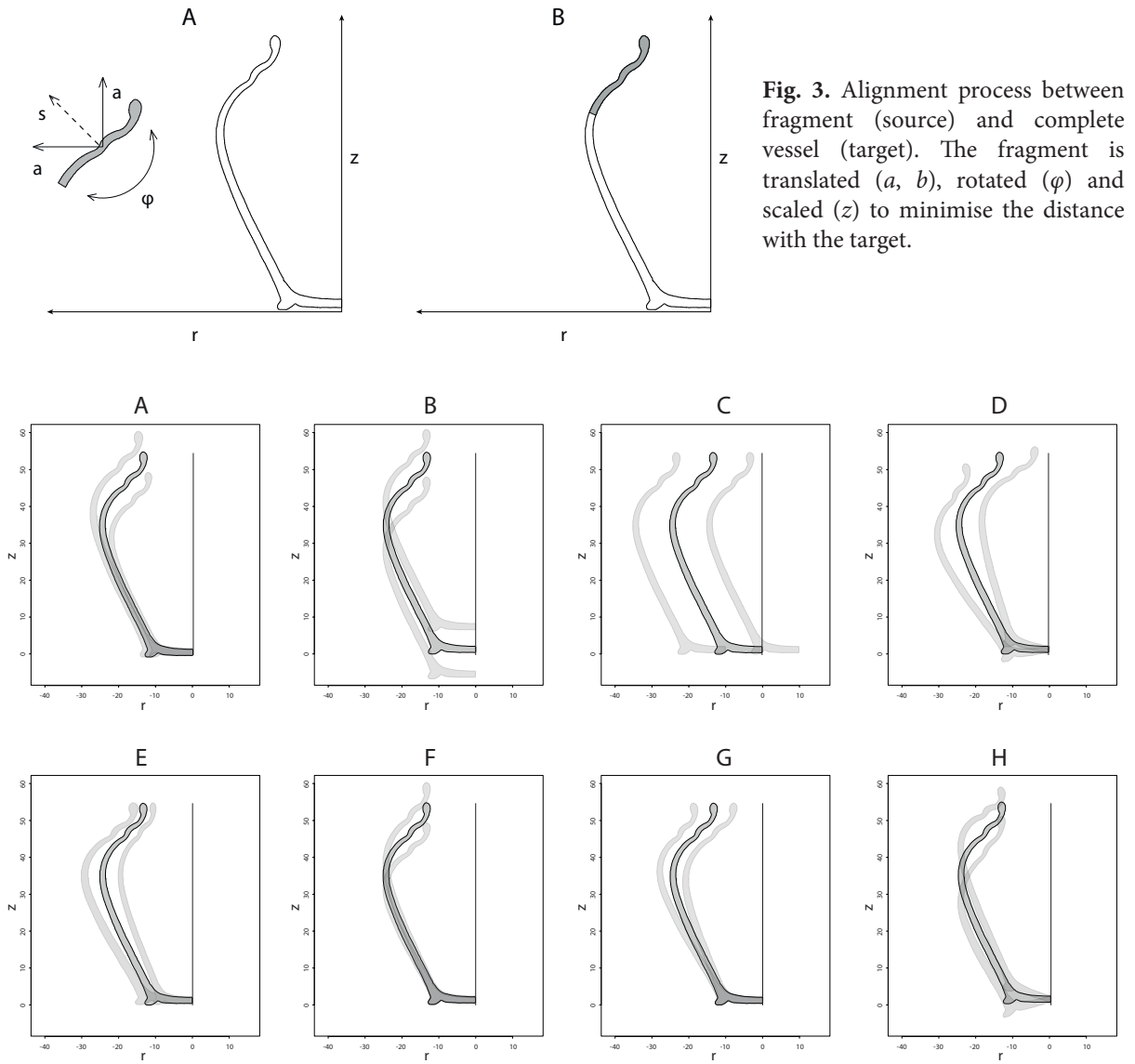


Fig. 3. Alignment process between fragment (source) and complete vessel (target). The fragment is translated (a , b), rotated (ϕ) and scaled (z) to minimise the distance with the target.

Fig. 4. Basic uniform transformations (i.e. transformations that leave parallel lines parallel) of the complete pottery profile. Original (dark grey polygon) and transformed (light grey polygon) profiles. A) Uniform scaling. B) Translation along the z axis. C) Translation along the r axis. D) Rotation around the profile centroid. E) Dilation along the r axis. F) Dilation along the z axis. G) Shearing along the r axis. H) Shearing along the z axis.

It is important to define the criteria used to identify shape similarity in pottery vessels and fragments. Shape in morphometrics is usually defined as “all the geometric information that remains when location, scale and rotational effects are filtered out from an object” (Zelditch et al. 2004, p. 76; Kendall, 1977). Figure 4 shows that, in morphometrics, scale and translation along the z axis (Fig. 4:A,B) are the only transformations that do not affect the original shape of the vessel.

2.3. PCA-based morphospaces

2.3.1. Discrete Cosine Transform and Elliptic Fourier Analysis

The original Cartesian coordinates were processed by Discrete Cosine Transform (DCT) for open outlines (fragments), and by Elliptic Fourier Analysis (EFA) for closed outlines (complete vessel profiles). Both methods transform the original Cartesian coordinates into a set of harmonics, represented by coefficients in pairs (for DCT) and quaternaries (for EFA), commonly known as shape variables (Kuhl and Giardina, 1982; Rohlf and Archie, 1984; Lestrel, 1989; Navarro et al., 2004; Dommergues et al., 2007). To retain shape information, the number of harmonics was set to the maximum defined by the Nyquist criterion (Lestrel et al. 1989, 2004a, 2004b, 2005). Because pottery size (related to vessel volume) is important in traditional morphological ceramic classifications, coefficients in this study were not size-standardised.

2.3.2. Morphospaces of complete vessels and fragments

The morphospace of complete vessel shapes (closed outlines, used in workflow 2) was created by Principal Component Analysis (PCA; Jolliffe, 2002), on a variance-covariance matrix, calculated from EFA coefficients. Two currently popular probabilistic methods were then used for classification. As a first step, Model-Based Cluster Analysis (MBCA) was used to classify the closed profile outlines in an unsupervised standardised way (Dempster et al., 1977, Fraley and Raftery, 1998, 2002; McLachlan and Peel, 2000). Then Model-Based Discriminant Analysis (MBDA) was used to better discriminate the clusters formed by MBCA.

The potential vessel shapes were calculated from equidistant positions in the 3D morphospace by inverse Fourier function (Claude, 2008). An ellipsoid encompassing 95% of the shape variability confidence interval was used to avoid creating impossible shapes.

The morphospace of all possible fragments (open outlines, used in workflow 3) was formed by a PCA of the variance-covariance matrix calculated from DCT coefficients. As the class membership of fragments must be known in advance, the MBDA was used for fragment class discrimination.

The R software (R core team, 2017) was used for all calculations.

3. Results and Discussion

3.1. Workflow 1

3.1.1. Testing process

The matching algorithm was tested on the database of profile outlines, drawn from 47 complete vessels (Fig. 5). To examine the effect of fragment size on attribution, all vessels in the target database were segmented into four size-classes (Fig. 5:A): large fragments (4 breaks), medium fragments (6 breaks), small fragments (8 breaks), and tiny fragments (10 breaks). All 1,316 fragments thus created (source database) were then transformed, to account for any possible imprecision produced by the drawing process (Fig. 5:B), and matched to all 47 complete vessels (target database, Fig. 5:C). Optimisation of the general cost function was calculated by the Simulating Annealing method (Hajek, 1988; Bélisle, 1992; Cortez, 2014).

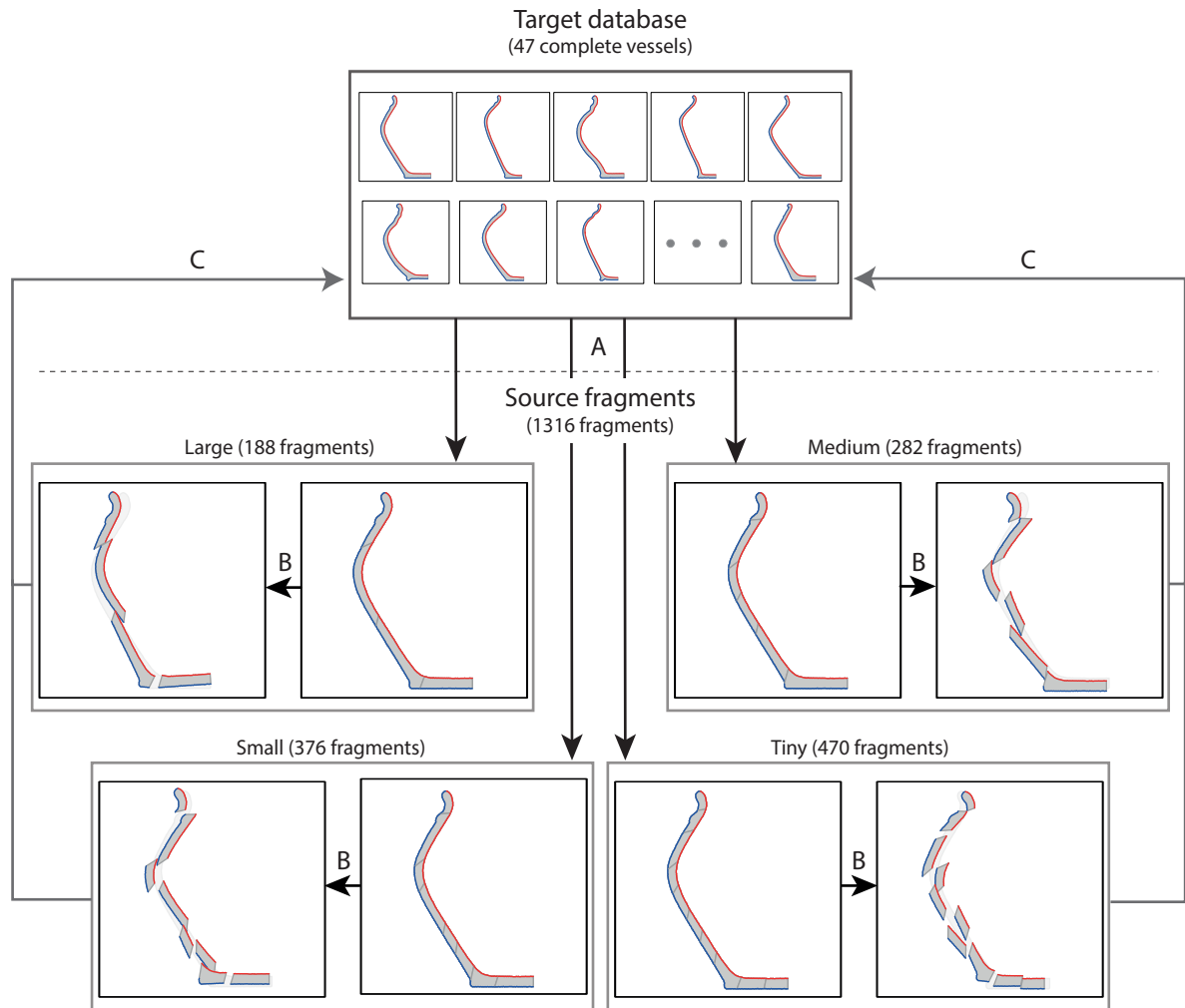


Fig. 5. Fragment identification. A) A set of 47 complete vessels (target database) is segmented into four size-classes. B) Fragments are transformed, to account for any possible imprecision produced by the drawing process. C) Fragments are matched to the target database.

3.1.2. Results

The original vessels were identified as the best candidate for 74% of the large fragments and 50% of the tiny fragments (88% for large and 68% for tiny when the first two best candidates were considered; [Table 1](#)).

	Large (4 breaks)	Medium (6 breaks)	Small (8 breaks)	Tiny (10 breaks)
1st Best candidate	139	179	200	239
2nd Best candidate	27	45	62	80
3rd Best candidate	12	17	34	36
The rest	10	41	80	115
Total	188	282	376	470

Table 1. Results of testing process.

It could be tempting to conclude that fragment size affects attribution results. However, closer examination of the results ([Figure 6](#)) reveals that fragment position, rather than fragment size, seems to play a key role in attribution. The best outcomes were observed for rim fragments, which were correctly recognised in all cases, no matter what the size of the fragment. Adequate

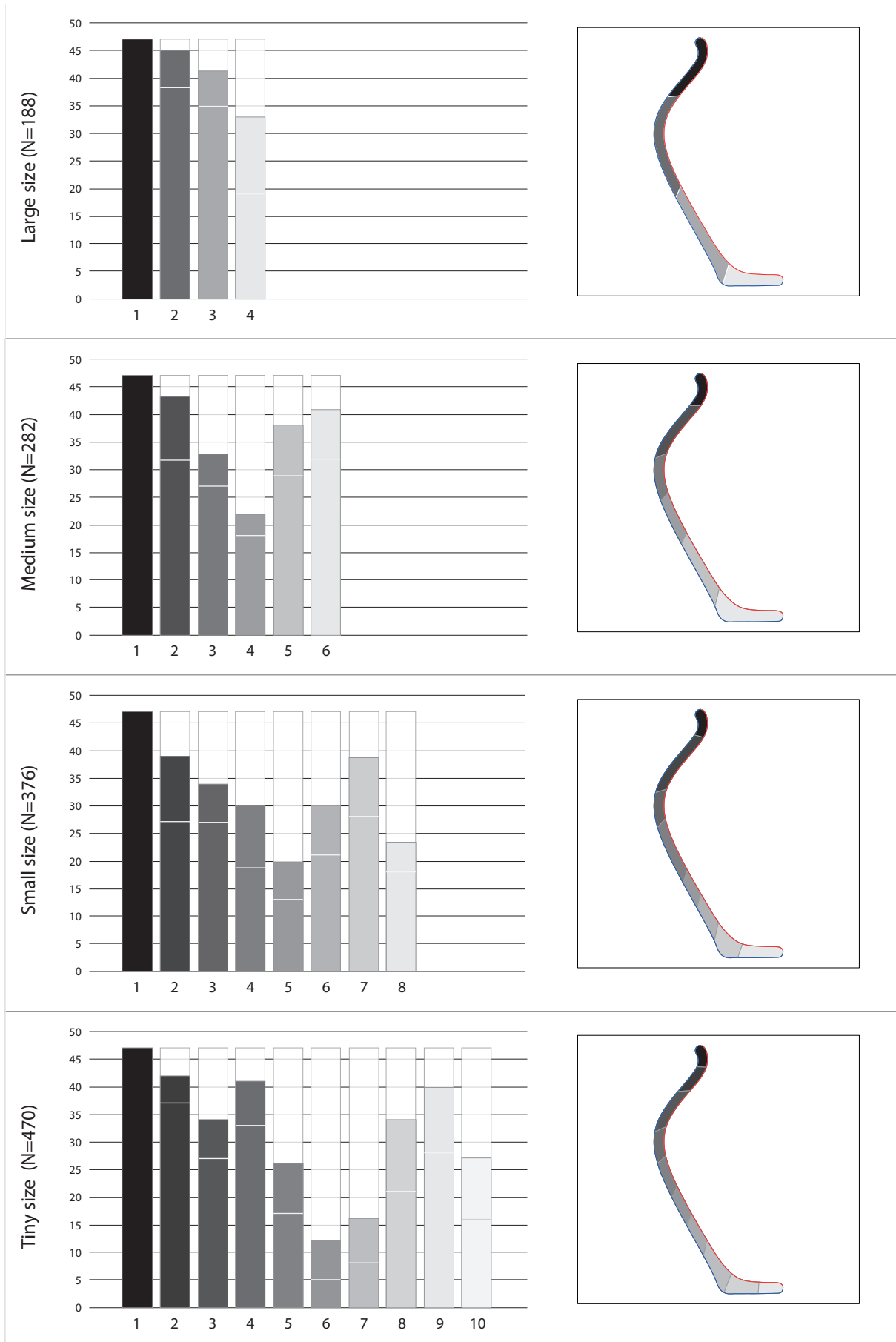


Fig. 6. Results for workflow 1. Bars in barplots are divided by a white line into two segments; the lower segment indicates the total for 1st best candidates, the higher segment for 2nd best candidates. Greyscale in barplots corresponds to the position of the fragment on the original vessel depicted on the right.

results were also obtained for fragments from vessels whose shape varies considerably along the profile (shoulder and corner-base fragments). Other vessel parts (lower body and flat-base fragments) did not give satisfactory results because: (i) they are very similar for all individuals, and (ii) they do not vary in shape, but rather in size.

3.1.3. Discussion

Attribution can only be successful if the target database contains a suitable analogy for the fragment. Attribution quality depends on (i) fragment size, (ii) fragment position on the original vessel, (iii) shape variability in the target database, and (iv) boundary values for search parameters. Although these factors may seem self-evident, their effects do not seem to have been examined on archaeological pottery. The effect on attribution of fragment size and position can now be used to answer two important questions about archaeological pottery: “Is this fragment too small to be identified?”; and “Is it possible to identify non-rim fragments?”.

Results for the test database showed that fragment attribution can be successful for rim and upper body, and the transition between lower body and base. Even some of the smallest fragments may still be suitable for attribution (i) if both source fragment and target vessel are more complex in shape, and (ii) if this shape complexity is a discriminating factor for the vessels in the database.

The matching algorithm can also provide deeper insight into some aspects of ceramic function and the fabrication process, in terms of technology, technique, gesture and know-how. For example, by allowing the rotational parameter used in optimisation to vary more freely, shape will be affected, but the “hand signature” of the pottery manufacturer becomes more easily identifiable, even in very different pottery vessels (Fig. 7:A). Aligning a series of complete vessels confirms that rims, which are supposed to be very significant for typological classification, are sometimes far less important than the remainder of the vessel (Fig. 7:B). This difficulty arises because rim shape variability, reflecting chronological, spatial and technological differences, may be much greater than the global shape variability of the vessel, which is generally related to its function.

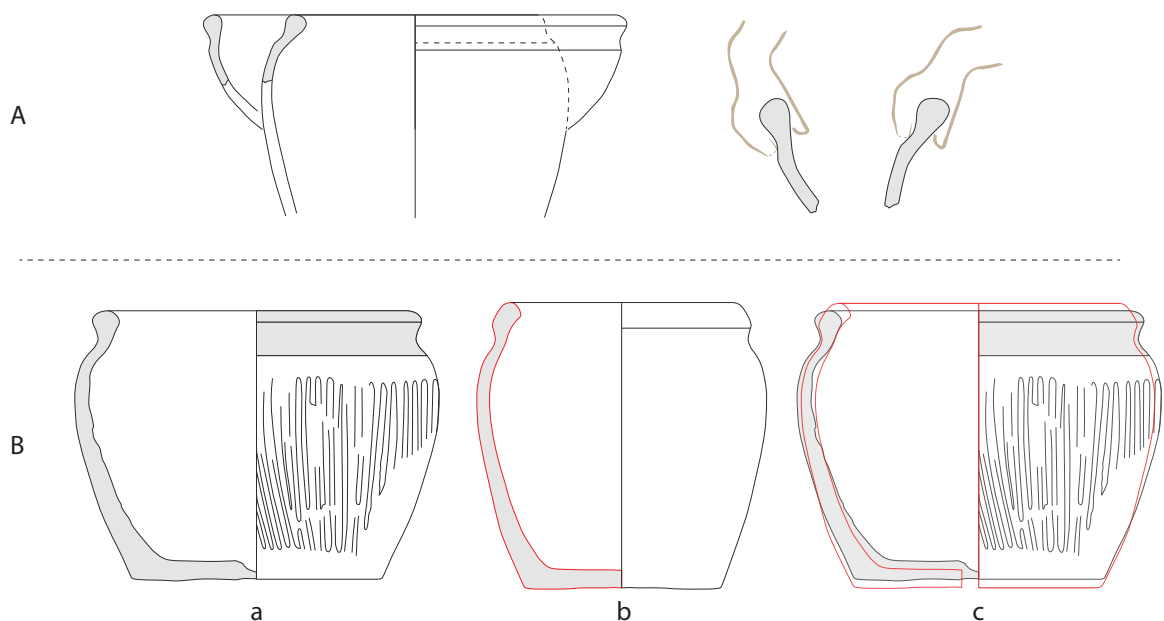


Fig. 7. Two examples of further applications of the alignment algorithm. A) Identification of the same gesture used for the fabrication of two different vessel shapes. B) Matching of two complete vessels: a) target vessel, b) source vessel, c) superimposed target and source vessels.

3.2. Workflow 2

3.2.1. Testing process

The original database containing 47 vessels was divided into two subsets (Fig. 8): 44 target vessels representing the corpus of vessels previously known to archaeology, and 3 source vessels representing newly discovered vessels, to test the accuracy of the attribution process. Each source vessel was therefore fragmented into 4 sections of different lengths (12 source fragments).

Target vessels were projected into the PCA-morphospace (target morphospace), computed on all harmonics obtained by Elliptic Fourier Analysis (Fig. 8:B). Model-based cluster analysis was used to identify potential classes, based on the first three principal components. The optimal classification model, EEI (equal volume and shape, and identical orientation of clusters), with three classes, was selected using the Bayesian information criterion (see [Supplementary materials S2](#)).

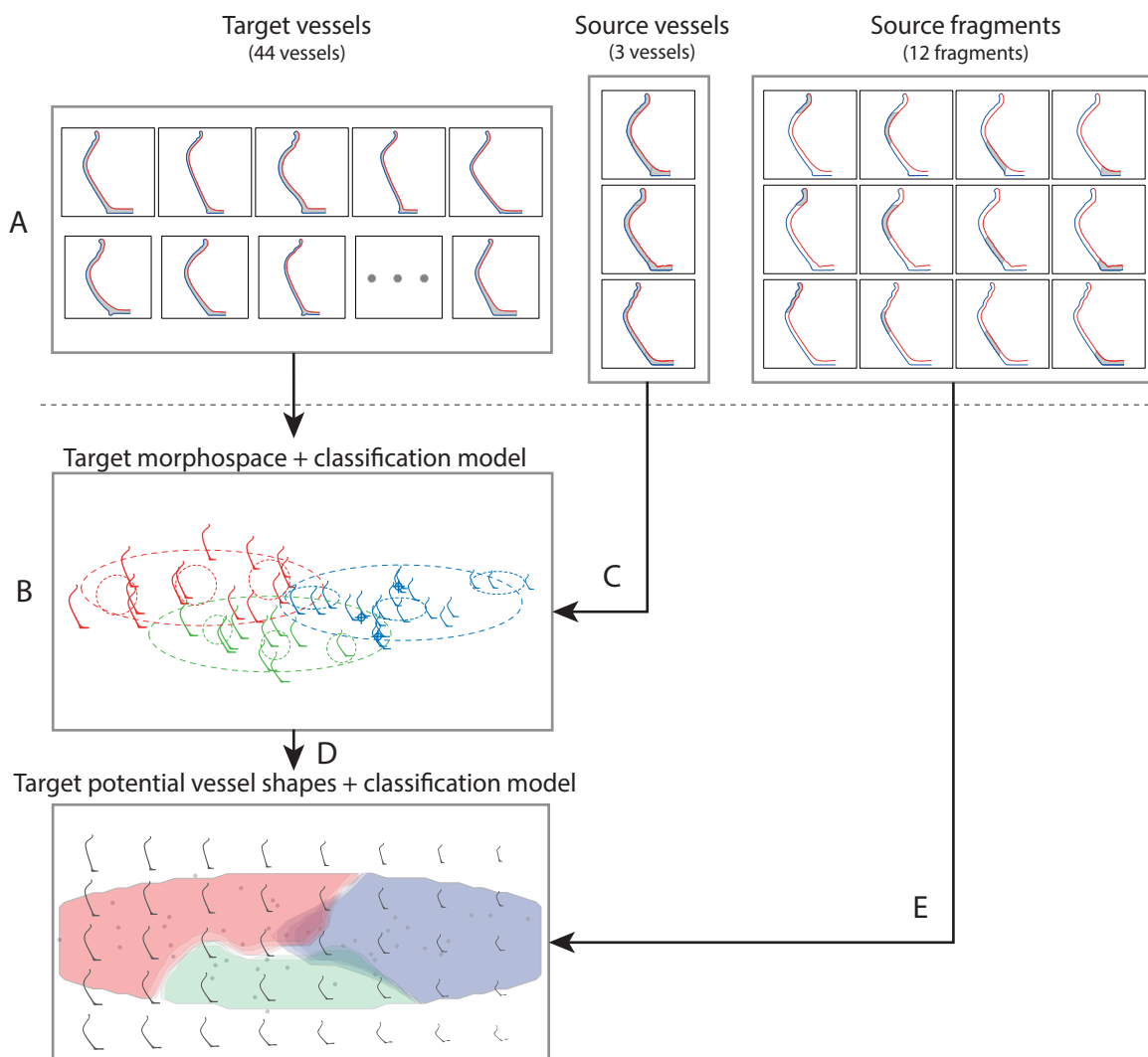


Fig. 8. Fragment classification workflow based on potential vessel shapes. A) The original corpus is divided into target vessels and source vessels. Source vessels are segmented into 12 fragments (source fragments). B) Target space is created by PCA from target vessels. Target vessels are classified by MBCA and the target classification model is calculated by MBDA. C) Source vessels are projected on to the target space and classified by the target classification model. D) Based on the target space, potential vessel shapes are reconstructed and typologically classified by the target classification model. E) Source fragments are matched to these reconstructed vessel shapes.

Such a classification yields acceptable visual results (Fig. 9). Small, globular, rather closed forms (G1; in blue) are concentrated on the right, while tall, situla-shaped vessels (G2; in red) occupy the upper-left area of the PC2 vs PC1 space. Globular vessels, with a slightly open rim (G3; in green), are located in the lower part of the PC2 vs PC1 space.

To further improve this classification, the MBDA was calculated from the data (see [Supplementary materials S3](#) for histogram of probabilities, and [S4](#) to perceive class limits in the morphospace). This target classification model was used to classify the three source vessels, projected into the morphospace. All three vessels were attributed to the tall situla-shaped class (G1; see blue crosses in Fig. 9).

The number of principal components necessary to reconstruct the original shape of all vessels was assessed by the improvement in shape quality as the number of principal axes increased. Thus, even a relatively small number of principal axes (3-5) is sufficient to retrieve the global shape of all vessels (see [Supplementary materials S5](#) for animated reconstruction of six vessels).

The effect of the number of principal components on vessel shape reconstruction is demonstrated in [Figure 10](#). Profile colours correspond to the maximum squared distances between the original 44 shapes and their reconstructions. Shape improvement with the lower principal components (from 3 to 10) is strongly related to the rim and base. This is not surprising, as rims are highly variable and often possess decorative features, while bases often possess omphali, concentric rills, or abruptly curved legs.

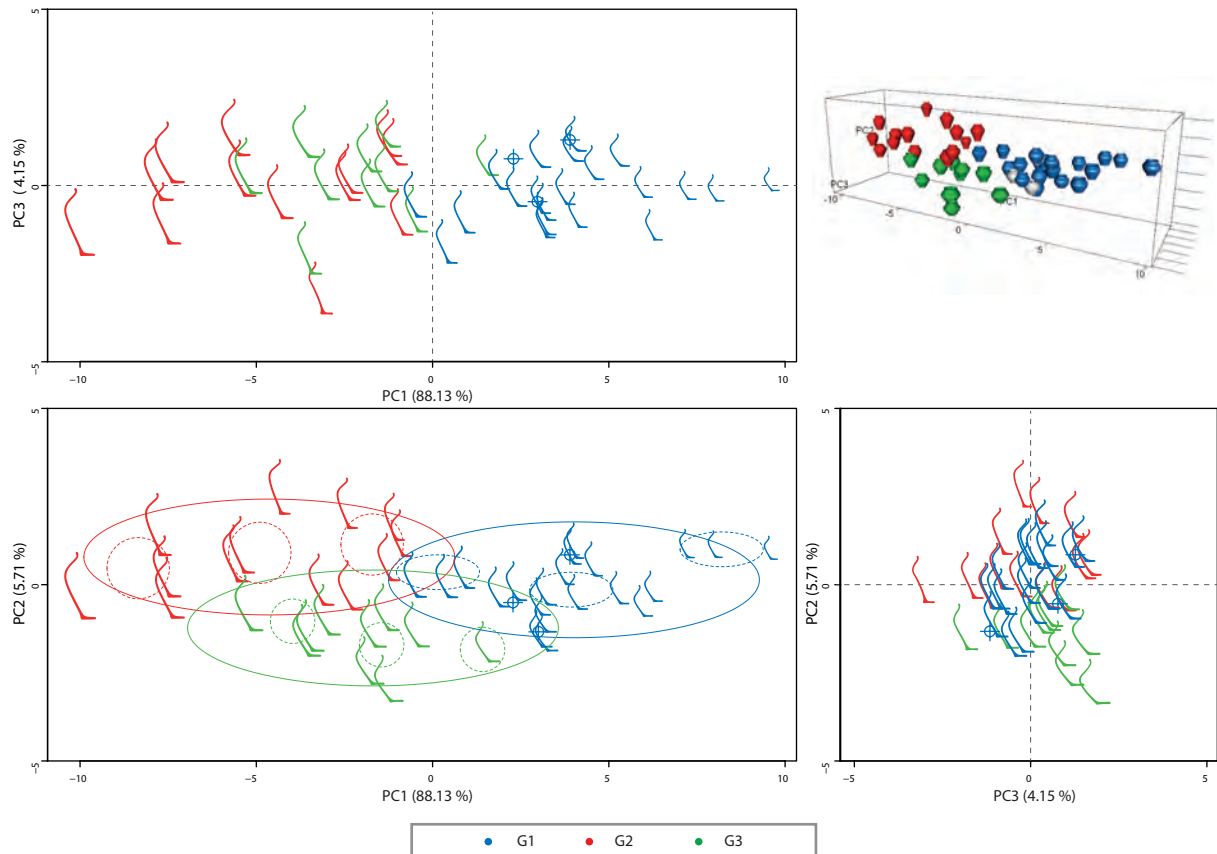


Fig. 9. Principal Component Analysis. Projection of 44 target vessels (coloured profiles) and three source vessels (crosses) into a PC3 vs PC2 vs PC1 morphospace (target space). Target vessels are coloured according to MBDA classification, source vessels according to MBDA discrimination. Solid line ellipses delimit MBDA clusters; dashed line ellipses delimit MBDA clusters (for 3D visualisation of the classification space, see [Supplementary materials SM5](#)).

Optimal attribution would require a highly dimensional space (at least 20 principal components; Fig. 10, see also [Supplementary materials S6](#)). For testing purposes, however, only three principal components were retained. Within the morphospace of these first three PCA dimensions, a 3D ellipsoid was created to encompass 95% of corpus variance. Within this ellipsoid, a set of 49,195 equidistant (0.1 standard deviation of PCA) potential vessel shapes was created (i.e. new target vessels). The probability of each potential vessel being a member of one of the three classes (G1, G2, and G3) was then calculated by MBDA.

The probabilistic source fragment attribution process was tested on the three source vessels, for which the typological class and position in the morphospace were known. Each of the 12 source fragments was matched to the 49,195 new target vessels. Only one parameter (fragment translation along the z axis) needed to be optimised, because coefficients were not size-standardised in the new target vessel morphospace.

Results for matching the 12 source fragments were then checked by (i) superimposing each source fragment on its best candidate for visual inspection, and by (ii) comparing the position of the best candidate to the location of the original source vessel in the target morphospace.

3.2.2. Results

[Figure 11:A](#) shows that fragment outlines generally fit well to the corresponding best candidates. For each source vessel, one fragment is less well aligned (for S1, F2; for S2, F1; for S3, F1). In each case, the least well-aligned fragment is the furthest from the source vessel in the morphospace ([Fig. 11:B](#)).

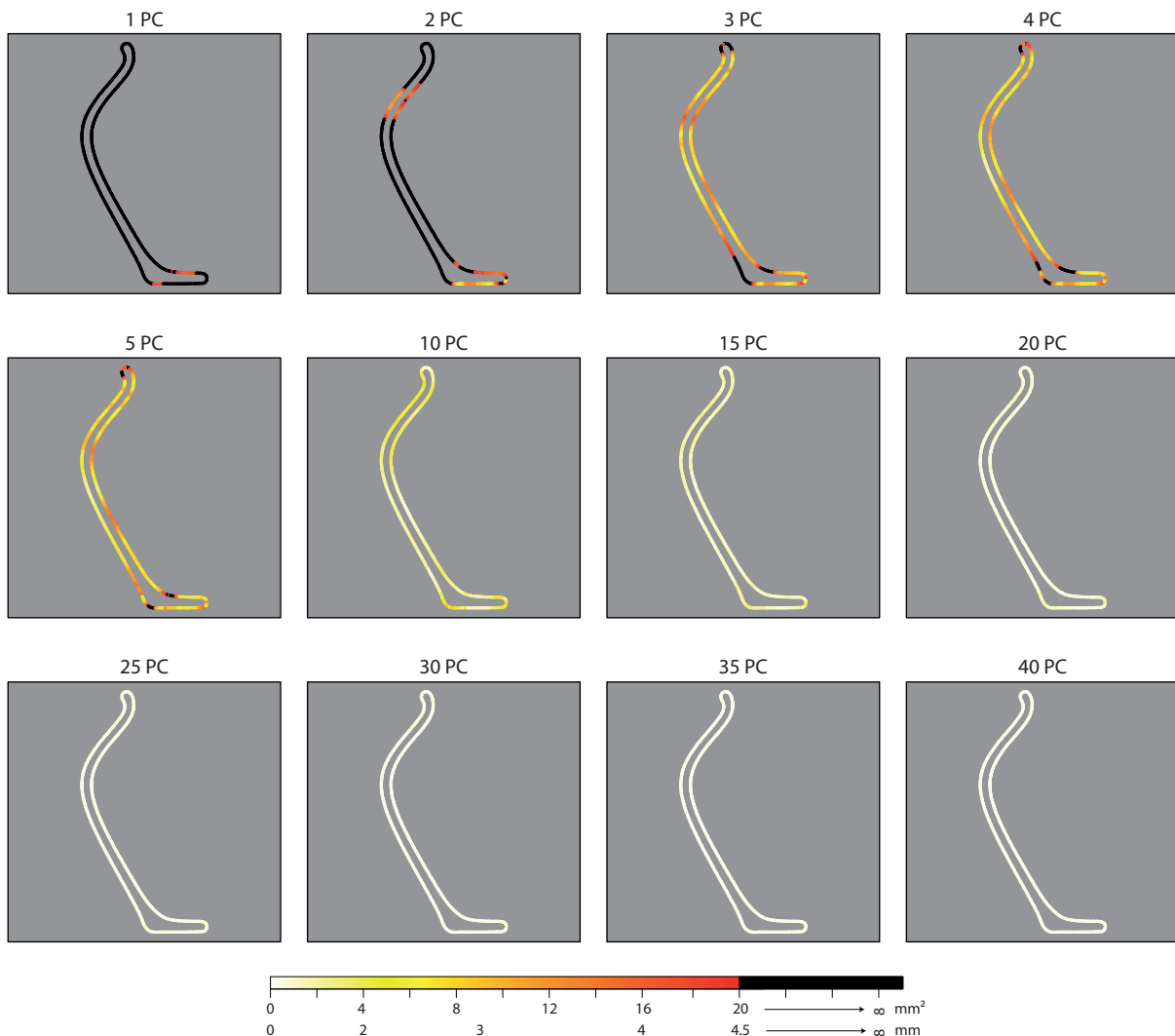


Fig. 10. The effect of the number of principal components on vessel shape reconstruction.

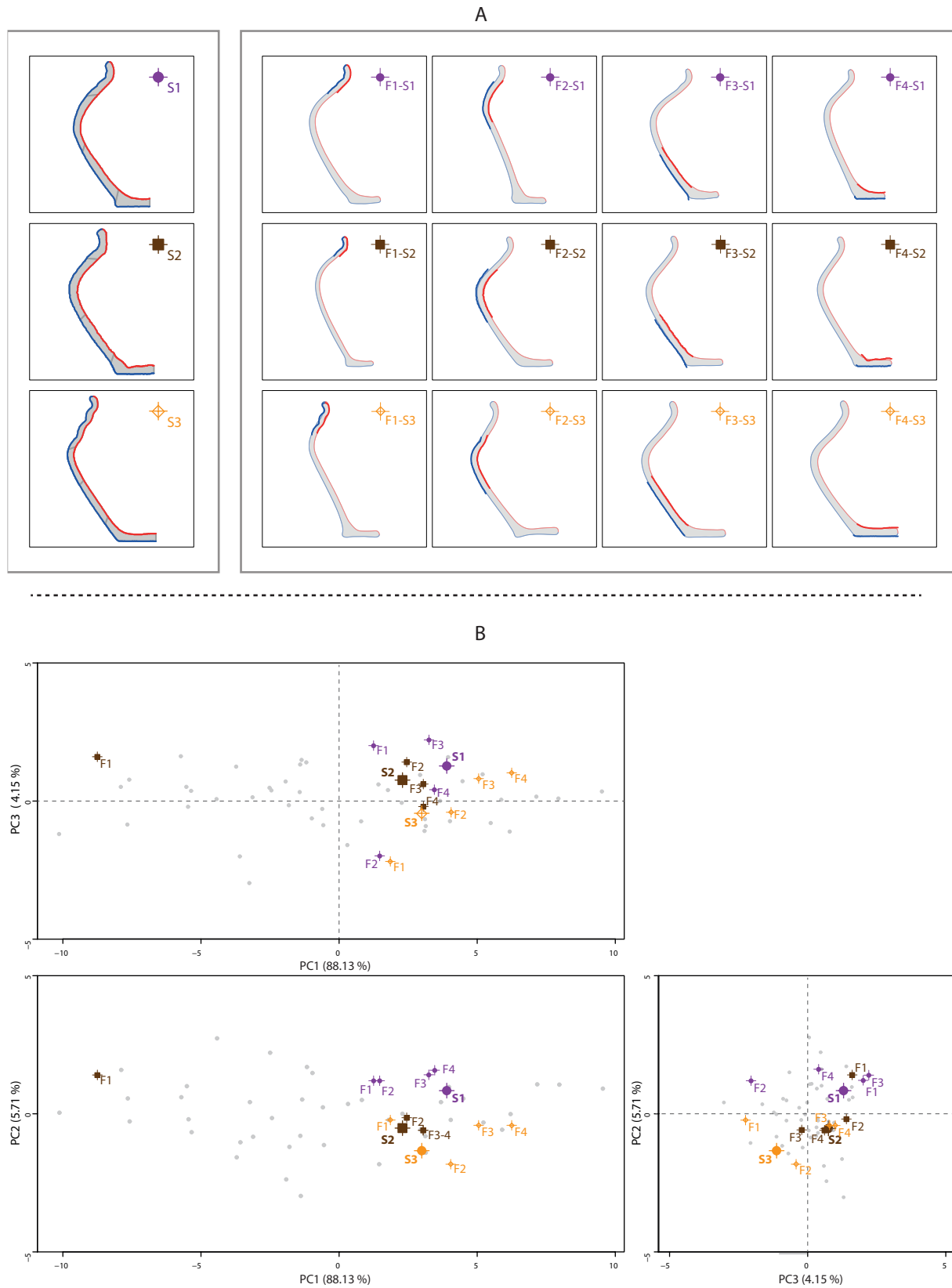


Fig. 11. Result of the fragment classification workflow based on potential vessel shapes. A) Visualisation of source fragments with their corresponding best candidates. B) Principal Component Analysis. Projection of 44 target vessels (grey dots), three source vessels (large crosses), and best candidates for source fragments (small crosses) in a morphospace (target space). Source vessels and best candidates are coloured as in [Fig. 11:A](#).

	Number of best candidates used for classification of 12 fragments											
	100			500			1000			10000		
	G1	G2	G3	G1	G2	G3	G1	G2	G3	G1	G2	G3
F1-S1	27	73	0	267	233	0	514	486	0	6648	3352	0
F2-S1	95	5	0	405	95	0	731	269	0	7710	2290	0
F3-S1	100	0	0	500	0	0	1000	0	0	9884	115	1
F4-S1	100	0	0	500	0	0	1000	0	0	9906	31	63
F1-S2	0	100	0	2	468	30	89	808	103	5112	4320	568
F2-S2	86	14	0	368	132	0	705	295	0	5734	3750	516
F3-S2	100	0	0	498	0	2	979	0	21	9596	192	212
F4-S2	99	0	1	492	0	8	986	0	14	9339	0	661
F1-S3	100	0	0	483	0	17	850	1	149	4743	2702	2555
F2-S3	100	0	0	496	4	0	940	57	3	5543	3914	543
F3-S3	100	0	0	500	0	0	1000	0	0	9675	176	149
F4-S3	100	0	0	500	0	0	1000	0	0	9191	0	809

Table 2. Classifications of 12 fragments (originally members of class G1) according to different numbers of candidates (100, 500, 1000, and 10000). Each best candidate was classified as G1, G2 or G3 based on the highest probability. Classes which significantly differ from others are highlighted in bold (differences were calculated by a Multiple Chi Square post-hoc test). Coherent classes are highlighted in blue, non-coherent in red.

On visualising alphashapes of the 1000 best candidates, however, even the worst-fitting best candidate falls very close to, and generally within, the appropriate class ([Supplementary materials S7](#)). Alphashape size indicates the variability of potential shapes: the smaller the alphashape, the better the attribution results. Better results may therefore be expected for bases and lower parts (e.g. [Supplementary materials S7:F4-S1, F3-S2, F4-S2](#)), indicating that even non-rim vessel parts can be useful for typological classification.

Classifications based on different numbers of candidates are provided in [Table 2](#). Source fragments were classified based on the 500 best candidates, approximately equivalent to 1% of all possible results ([Table 2](#)). Of the 12 fragments, only one rim part (F1-S2) was misclassified as G2 instead of G1. Another rim fragment (F1-S1), although correctly classified as G1, cannot be attributed to this class without ambiguity, as both the graphic and the Chi-square post-hoc test reveal that class G2 is not significantly different from G1 ($p < 0.128$).

[Figure 12](#) shows that the alphashapes for all four fragments converge at the location of the source vessel (see [Supplementary materials S8](#) for all source vessels). Therefore, identifying a greater number of locations with low precision, rather than the best matches themselves, may give better information about the position (i.e. shape) of the source vessel itself. This hypothesis was further tested on the three source vessels, each randomly broken into 10 fragments. Each set of fragments was then matched to the 49,195 new target vessels. For each fragment, a large number of best candidates (10,000) was selected and marked as a hit. The number of hits for each target vessel was then counted. The new target vessels with the greatest number of hits were encompassed in a 3D alphashape for each set of fragments.

The results obtained support the initial hypothesis that the source vessels are close to or even within the alphashape ([Fig. 13](#); see also [Supplementary materials S9](#) for all source vessels). These results are reproducible even when threshold values are modified (i.e. number of best candidates and of hits). An additional test successfully reconstructed from its fragments a potential vessel randomly positioned in the 3D morphospace ([Supplementary materials S9:S4](#)).

3.2.3. Discussion

Workflow 2 initially identified three classes of pottery vessels in the database. The significance of these classes for archaeological interpretation (e.g. chronological, economic or social role in society) was not among the aims of this study. Further investigation of these classes would

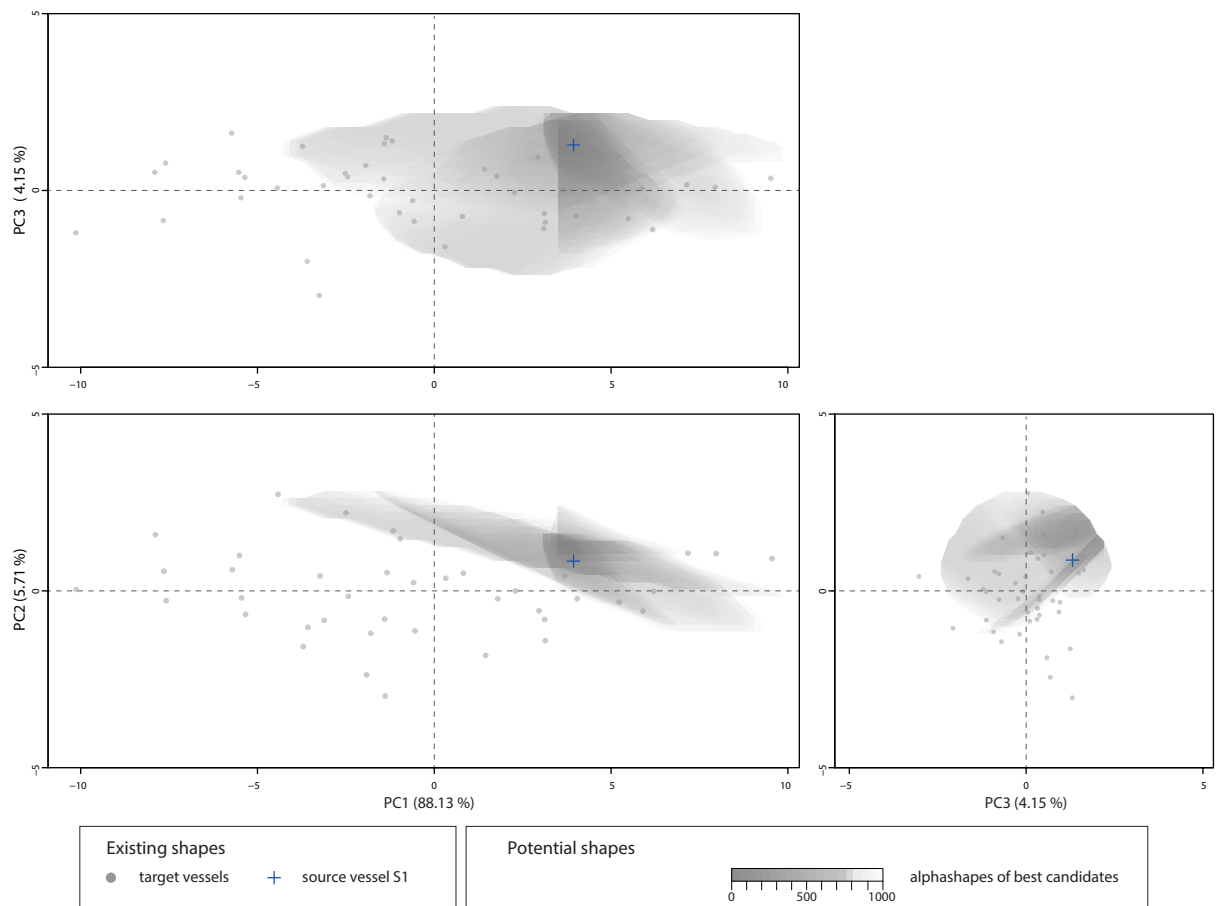


Fig. 12. Alphashapes highlight distribution in the morphospace of the 1000 best candidates for each of the four fragments of source vessel S1 (identified by a blue cross). See [Supplementary materials S8](#) for all source vessels.

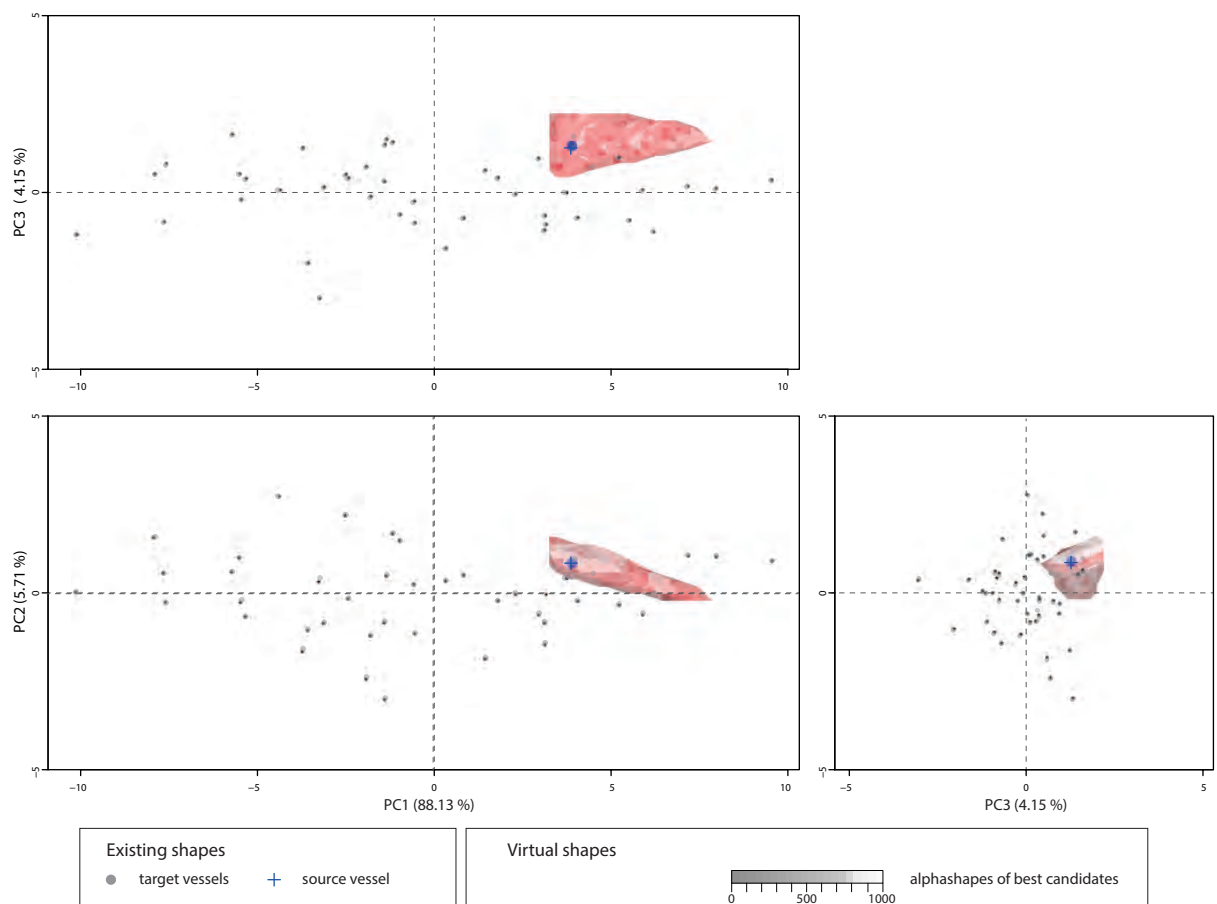


Fig. 13. A 3D alphashape highlights target vessels possessing the greatest number of hits (source vessel S1 broken into 10 fragments). See [Supplementary materials S9](#) for all source vessels.

require other evidence, such as spatial distribution, gender association for graves, technological or petrographic analyses, etc.

The shape variability of known (excavated) artefacts served to create a vast number of potential vessel shapes in the morphospace, thus increasing the quantity of investigable data available. The greater the number of existing vessels used in morphospace creation, the more precise the shape reconstruction. In theory, the spectrum of reconstructed potential vessels should correspond to shapes that could have been created by ancient manufacturers. Morphometrics could thus help to identify certain conceptualisations existing among ancient populations.

The quality of attribution is guided by aspects inherent to the initial database (profile outline quality, shape variability, etc.), but also by space dimensionality and the resolution used for the creation of potential vessels. The attribution procedure is here demonstrated only in a three-dimensional space. Although the morphospace retained 98% of overall shape variability, the profiles of the original vessels, notably rims and bases, could not be retrieved in precise detail. Despite this fact, the procedure was successfully used to reconstruct the global shape of the source vessels, and to correctly classify most of the source fragments (9 out of 12 fragments were correctly reconstructed, and 10 out of 11 were well classified). For practical purposes, the probability score for the best potential candidate or candidates provides archaeologists with helpful clues as to whether and where to classify the fragment.

The procedures presented here may help in the treatment of incomplete or unclassified vessels, but also help in solving the pottery puzzle problem (see e.g. [Kleber and Sablatnig, 2009a, 2000b](#), [Sonet et al., 2013](#)). The process follows the logic of unsupervised machine learning. The morphospace of potential vessels shows the machine what the desired output should look like. The superimposition of the alphashapes for a set of fragments will highlight the vessel shape or shapes most typical for that set ([Fig. 12-13](#); [Supplementary materials S8, S9](#)).

3.3. Workflow 3

3.3.1. Testing process

Fragment attribution was tested on the 44 target vessels, previously classified into 3 classes (G1, G2, and G3; [Fig. 14:A](#); see section 3.2.1 for classification). To test the limits of the classification process, only small fragments were processed, varying in size between 5% and 15% of the original vessel. A 200-point open outline was produced for each of the 174,240 small fragments thus obtained ([Fig. 14:B](#)). These outlines were then standardised so that their maximum z coordinates coincided with the z axis. Outline coordinates were then transformed by Discrete Cosine Transform. To retain as much shape information as possible, all DCT harmonics were used to produce the PCA-morphospace (target fragment morphospace) where fragments were then projected ([Fig. 14:C](#)).

3.3.2. Results

The first three principal axes of the target fragment space account for 98.6% of the total variance. Fragments in the morphospace are distributed like “pearls on strings” ([Fig. 15](#)). Each pearl corresponds to a fragment, while each string represents one of the 11 differently sized sets of fragments produced for each target vessel. The fragments on each string are always naturally ordered from rim to base (see [Supplementary materials S10](#)).

Fragments in the morphospace seem to be clustered by target vessel class (G1, G2, and G3), with MANOVA-attested statistical significance ($p < 2.2e-16$). To further improve the typological fragment class discrimination model, the MBDA was calculated using the first three PCA scores of each fragment (i.e. target fragment classification model).

The power of the target fragment classification model can be observed using an arbitrary threshold (minimum *a posteriori* probability) to classify fragments. This threshold value should

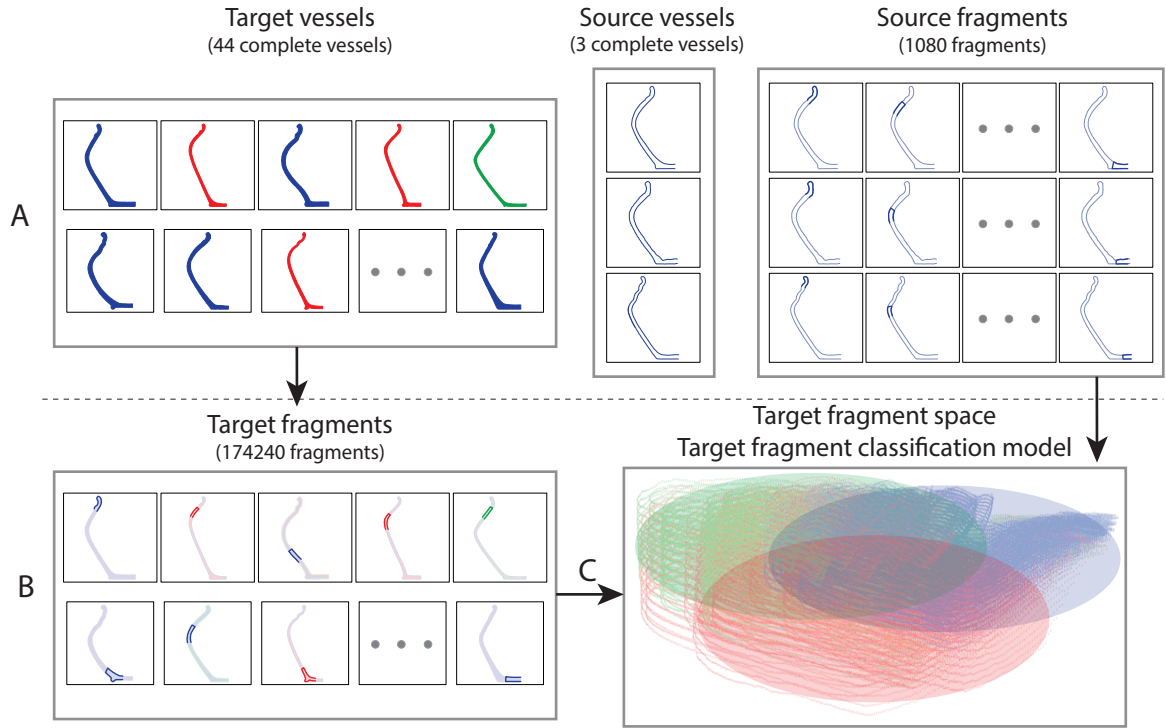


Fig. 14. Fragment classification workflow based on projection into the fragment morphospace. A) Original database of 47 vessels is divided into 44 target vessels and 3 source vessels. Each source vessel, previously classified as G1, G2 or G3, is segmented into 360 fragments (source fragments). B) Target vessels are segmented into 174,240 small fragments (5 to 15 %). C) Target fragments are used to create a PCA-based target fragment space. Target fragments are used to calculate the target fragment classification model. D) Source vessels are projected into the target fragment space and classified by target fragment classification model.

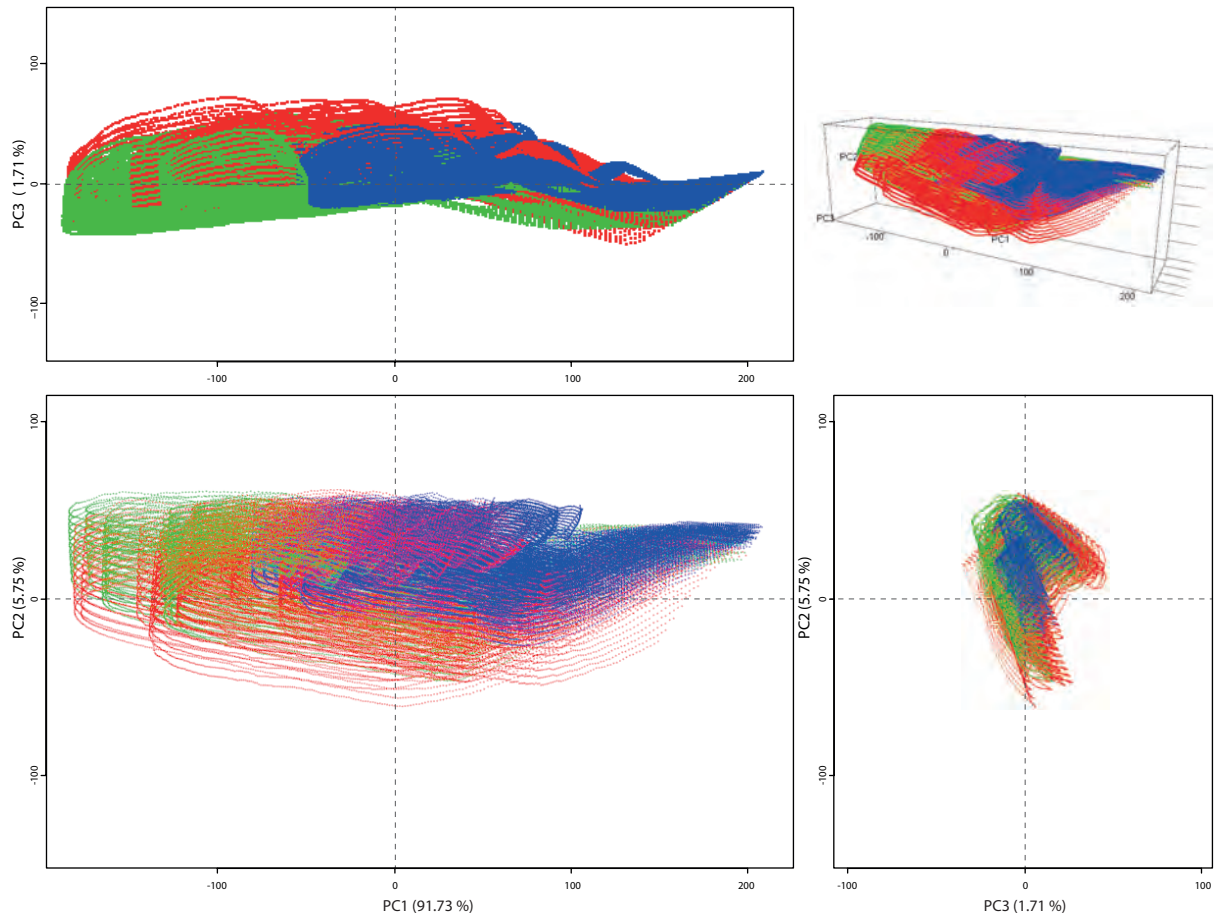


Fig. 15. Projection of 174,240 target fragments in the PCA-based target fragment space. Fragments are coloured according to the class of the original vessel.

be chosen by the archaeologist, and can then be assessed by three different parameters: (i) percentage of fragments classified, (ii) percentage of fragments well classified, and (iii) correct position on target vessel of well-classified fragments.

The evolution of these three parameters for a given threshold is shown in Figure 16. The maximum threshold to classify 100% of the fragments is approximately 0.4, but only 60% will then be correctly classified (Fig. 16:A). As the threshold is raised, the number of fragments that can be classified falls dramatically, but the number of fragments correctly classified rises considerably. With a threshold of 0.91, for example, only 1% of fragments can be classified, but around 98% of these 1,191 fragments are correctly classified (Fig. 16:B).

To examine the effect of size and position on fragment attribution (i) the percentage of well-classified target fragments, and (ii) the precise location of these well-classified target fragments were both plotted for each fragment size, from 7 to 14% (Fig. 17). For higher probability values (threshold at 0.80), little difference was observable between classification success rates, whatever the fragment size (Fig. 17:A). Similarly, for the same threshold value, the correctly classified fragments were distributed all along the profile (Fig. 17:B). However, at 0.85, the smaller fragments were correctly classified only in the upper part of the profile, whereas the larger fragments remained correctly classified all along the profile. With the threshold at 0.90, no fragments from 7 to 9% were classifiable, while from 10 to 13% the correctly classified fragments gradually spread along the profile from the uppermost part downward. At 14%, correctly classified fragments could be found all along the profile.

Source fragment classification example

In a real-life situation, where fragment classification is initially unknown, the archaeologist should select the optimal threshold for correct classification, previously identified for this corpus as 0.8 (Fig. 17). With this threshold, the expected success rate should be around 95% of classifiable fragments correctly classified. For the first source vessel (S1), all the classifiable fragments are correctly classified, all 24 are located on the rim (Fig. 18:S1). For source vessel S2, a similar result is obtained, with all 9 fragments correctly classified, and again all located on the rim (Fig. 18:S2). For source vessel S3, a threshold value of 0.8 is in fact too high, as no fragments are classifiable at that level (Fig. 18:S3).

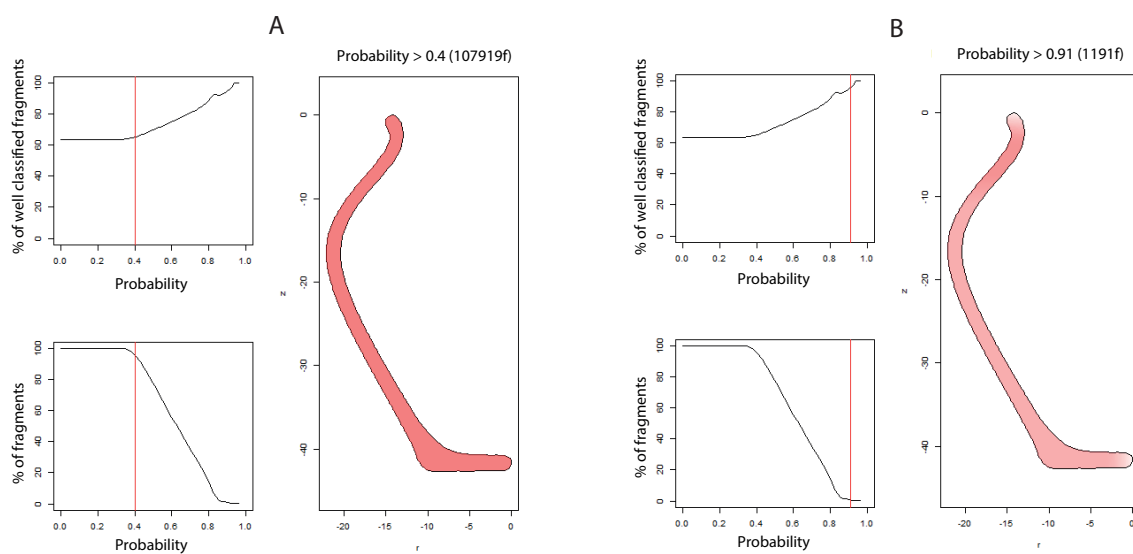


Fig. 16. The power of the target fragment classification model with two threshold values (0.4 and 0.91). Three parameters can be inspected: (i) proportion of fragments, (ii) proportion of well classified fragments, and (iii) position of well-classified fragments on vessels.

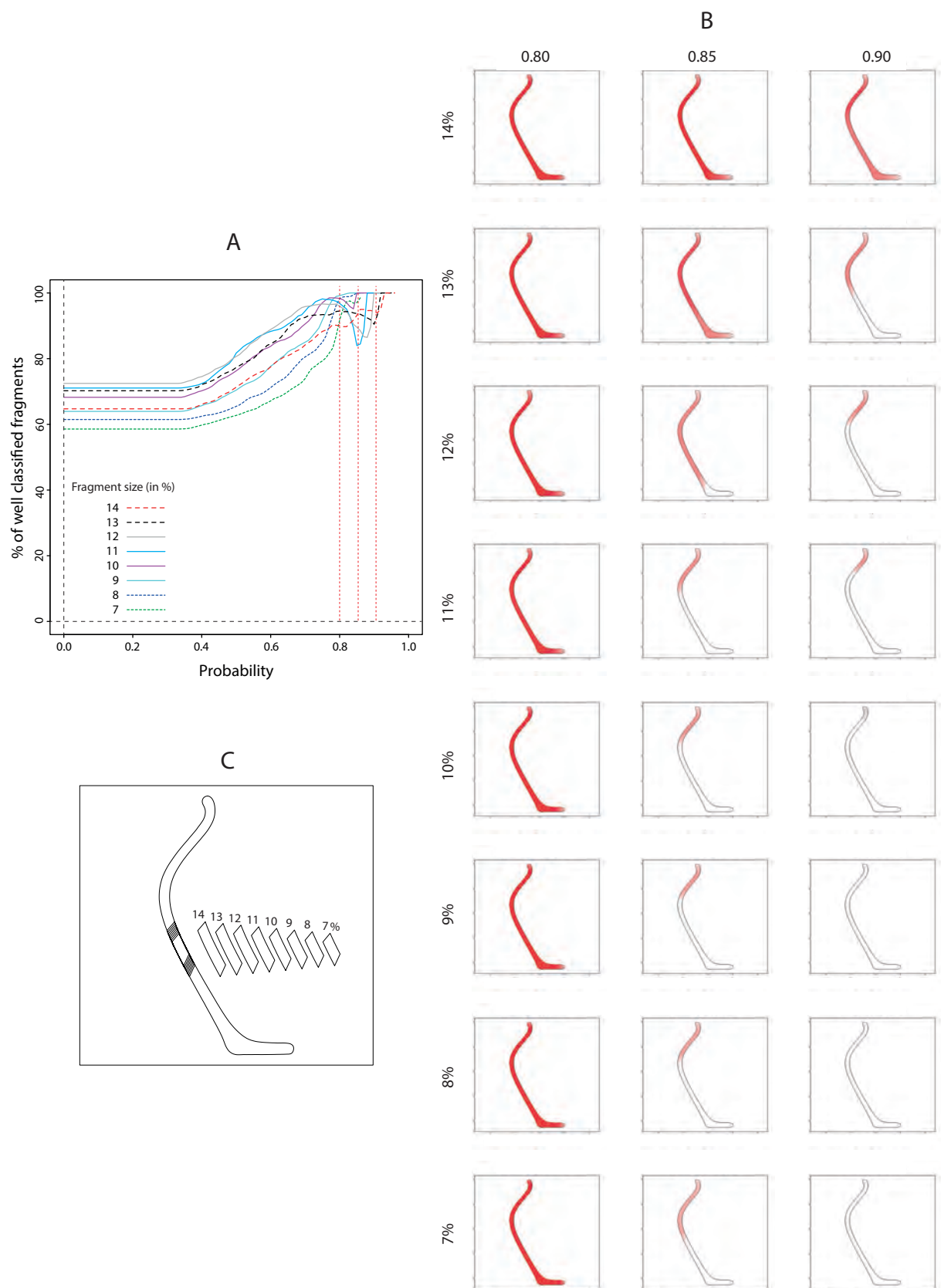


Fig. 17. The effect of size and position of fragment attribution. Three probability values are highlighted (0.8, 0.85, and 0.9). A) Proportion of well-classified fragments. B) Position of well-classified fragments on vessel. C) Illustration of fragment sizes.

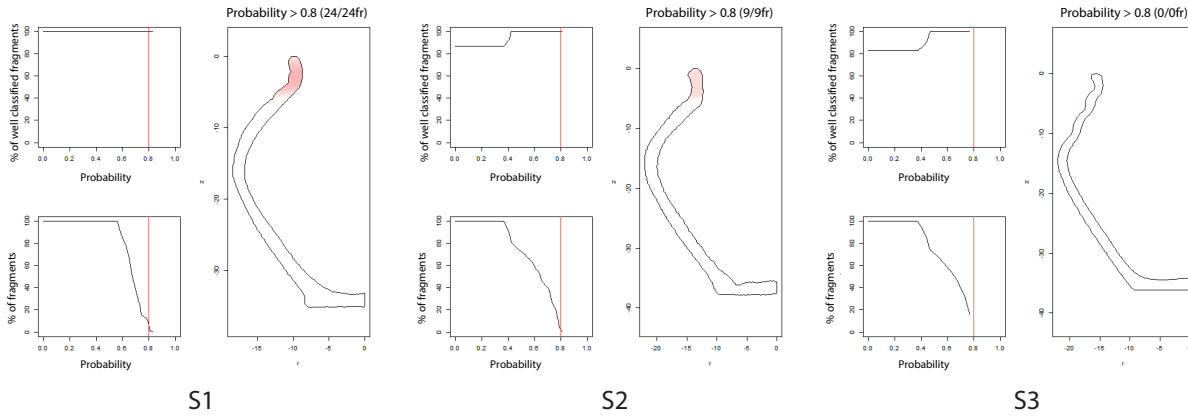


Fig. 18. Classification of source fragments.

3.3.3. Discussion

Workflow 3 demonstrated that any fragment can be classified by simple projection into a morphospace of previously classified fragments. The classification process can be controlled by the operator, who selects the minimal probability value to be used for classification. By arbitrarily fixing this threshold value (e.g. at 0.90 or 0.95), the classification process can be fully automated. The advantage of this process is that it can classify fragments of any size, located anywhere on the original vessel. The only two assumptions are (i) that the fragment is of approximately the same size as the fragments used to create the fragment morphospace, and (ii) that the morphospace contains a suitable analogy for the fragment. The results of the classification process therefore depend on (i) the quality of the database, (ii) the classification model, (iii) the number of fragments in the morphospace, and (iv) the range of fragment sizes used to create the fragment morphospace.

Results showed that even some small fragments (5-15% for the test database) can be correctly classified. As previously observed for fragment matching (see section 3.1), the best outcomes were observed for vessel rims and upper parts.

4. Conclusion

The goal of this paper was to present three workflows for attribution and/or classification of archaeological pottery. Results show that the best analogy for a given fragment can be identified in a referential database containing both (i) existing vessels and (ii) potential vessel shapes, defined by the overall shape variability of existing complete pottery vessels.

The advantage of the minimisation algorithm is that it does not necessarily require profile segmentation. Defining one or more landmarks, and completing shape information with other observable descriptors (e.g. clay, decoration, or fabrication technique) may increase the accuracy and rapidity of the matching process. The alignment process presented here allows full control of the matching criteria. By setting these criteria, the process can be fully automated. Visual inspection, with minimal residual values and best result classes, will allow the operator to make the most important decisions: whether the fragment is classifiable, and which class it could belong to. The matching process presented here is therefore semi-automatic, exploratory, and complementary to traditional approaches, and other existing computer-aided classification systems.

Matching fragments to potential vessel shapes, reconstructed based on all the shape information in the database, offers a way of overcoming the limitations of post-depositional processes. The experiments showed that vessel reconstruction in full detail requires a high-dimension

morphospace (above 20 dimensions). However, although only three principal axes were used here, the global shape of the three source vessels was correctly reconstructed. Extending computer calculation capacity and adapting optimisation algorithms should considerably improve the results obtained here.

Based on the reconstructed vessels, it was possible to correctly classify most of the fragments. Superimposing the alphashapes of the best candidates for a set of fragments highlighted the position, i.e. shape, of the most probable reconstructed vessels. This feature could be extremely useful when reassembling complete objects from fragments, and could therefore be applied in many domains where incomplete elements are studied (e.g. palaeontology, archaeozoology, prosthetics(?), etc.).

The spectrum of reconstructed potential vessels in the morphospace corresponds to shapes that could realistically have been created by ancient manufacturers. Morphometrics may therefore largely contribute to the identification of the ideological and cognitive conceptions of ancient populations, based on visual representations, which may overcome both regional and linguistic limitations.

The procedure, based on projecting fragments into the morphospace of previously classified fragments, facilitates probabilistic classification, for fragments varying in size and initial position.

Results show that even some small non-rim fragments may be suitable for attribution or classification. The procedures proposed here can thus considerably increase the quantity of data available for the study of a wide range of questions related to the dynamics of ancient populations. All the algorithms used here can save a considerable amount of time, while increasing precision and reproducibility in classification.

All procedures were programmed in the open-source R program, and can thus be made accessible to the wider scientific community to be tested, discussed, and further improved.

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Chapter 5

Morphometrics of Second Iron Age ceramics strengths, weaknesses, and comparison with traditional typology

This chapter presents the first case-study published within the framework of this doctoral project. The main goal of this paper was to ensure that geometric morphometric methods, rarely applied in archaeology at that time, were appropriate for typological goals. The efficacy of these methods were evaluated in comparison with traditionally produced typologies.



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Morphometrics of Second Iron Age ceramics – strengths, weaknesses, and comparison with traditional typology



J. Wilczek ^{a,b,*}, F. Monna ^a, P. Barral ^c, L. Burlet ^a, C. Chateau ^d, N. Navarro ^e

^a ARTÉHIS, UMR 6298 CNRS-Université de Bourgogne, ARTÉHIS, Bat. Gabriel, F-21000 Dijon, France

^b Ústav archeologie a muzeologie, Masarykova univerzita, Arna Nováka 1, 602 00 Brno, Czech Republic

^c Laboratoire Chrono-Environnement, UMR 6249 CNRS, Université de Franche-Comté, UFR Sciences et Techniques, 16 route de Gray, F-25030 Besançon Cedex, France

^d Université de Bourgogne, UFR SVTE, 6 bd Gabriel, F-21000 Dijon, France

^e Laboratoire PALEVO, Ecole Pratique des Hautes Etudes, UMR uB/CNRS 6282-Biogéosciences, Université de Bourgogne, 6 Boulevard Gabriel, F-21000 Dijon, France

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ABSTRACT

Although the potential of geometric morphometrics for the study of archaeological artefacts is recognised, quantitative evaluations of the concordance between such methods and traditional typology are rare. The present work seeks to fill this gap, using as a case study a corpus of 154 complete ceramic vessels from the Bibracte *oppidum* (France), the capital of the Celtic tribe Aedui from the Second Iron Age. Two outline-based approaches were selected: the Elliptic Fourier Analysis and the Discrete Cosine Transform. They were combined with numerous methods of standardisation/normalisation. Although standardisations may use either perimeter or surface, the resulting morphospaces remain comparable, and, interestingly, are also comparable with the morphospace built from traditional typology. Geometric morphometrics also present the advantage of being easily implemented and automated for large sets of artefacts. The method is reproducible and provides quantitative estimates, such as mean shape, and shape diversity of ceramic assemblages, allowing objective inferences to be statistically tested. The approach can easily be generalised and adopted for other kinds of artefacts, to study the level of production standardisation and the evolution of shape over space and time, and to provide information about material and cultural exchanges.

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1. Introduction

Ceramics are the most abundant and well-documented archaeological artefacts from the Second Iron Age. Traditionally, ceramics are used to study a wide range of issues: from establishing chronological sequences and cultural entities, through the definition of social, economic and cultural relationships and organisation, to the understanding of how archaeological complexes were formed. Archaeologists seek to answer these questions through the creation and application of ceramic typologies (e.g. Orton et al., 1992; Shepard, 1985).

Typology aims at assigning ceramic fragments to predefined types. Classification uses one or several discriminant criteria, such

as the overall form, the shape of a particular ceramic part (body, rim, foot...), the decoration, or the clay used. Although the method has been proved to be effective for treating huge ceramic assemblages, it may be affected by operator subjectivity, ensuing from specialisation, personal skills and professional experience. This has been demonstrated by the pioneer work of Hodson et al. (1966) on Iron Age brooches. To improve objectivity, at least in data treatment, over the past few decades, archaeologists have introduced statistical techniques, such as the popular seriation (Kendall, 1969, 1971) and cluster analysis (Hodson, 1970), occasionally supplemented by other methods, such as non-metric multidimensional scaling (Drennan, 1976), correspondence analysis (Duff, 1996), principal component analysis (Hodson, 1969) or principal coordinate analysis (Camiz et al., 2003). However, these approaches cannot overcome drawbacks related to the choice of relevant descriptors, and also to difficulties which may arise in coding the state of categorical variables without any ambiguity.

Recent developments in image, video and audio processing have led to remarkable advances in geometric morphometrics, used

* Corresponding author. ARTÉHIS, UMR 6298 CNRS-Université de Bourgogne, ARTÉHIS, Bat. Gabriel, F-21000 Dijon, France. Tel.: +33 420 549 497 782.

E-mail addresses: wilczek@mail.muni.cz, josef.wilczek@hotmail.com (J. Wilczek).

nowadays to study shapes in a broad range of fields. Many studies by anthropologists and evolutionary biologists are now based on these techniques (e.g. Slice, 2005; Zelditch et al., 2004), while their application in archaeology, except for biological remains (physical anthropology, archaeozoology, archaeobotany, and palynology; e.g. Ottoni et al., 2013; Terral et al., 2010; Vigne et al., 2007) is rarer and more recent (e.g. Karasik and Smilansky, 2008, 2011; Saragusti et al., 2005). Nonetheless, their potential for expressing forms unambiguously has been successfully exploited to study lithic artefacts (Brande and Saragusti, 1996; Buchanan, 2006; Buchanan and Collard, 2007, 2010; Lycett, 2009 and citations therein), sculptures (Buxeda i Garrigós and Gordaliza, 2011; Urbanová et al., 2011) and Bronze Age axes (Forel et al., 2009; Monna et al., 2013). Up to now, morphometric applications for ceramic taxonomy are scarce, but very promising: Gilboa et al. (2004) used computational typology to classify ceramics, while other authors have examined the potential of morphometrics for predicting the complete form from shards that include the rim or the bottom of the vessel (Karasik and Smilansky, 2011; Martínez-Carrillo et al., 2009, 2010). In practice, several parameters of geometric morphometrics can be fine-tuned, from sampling resolution and size normalisation (i.e. the acquisition process and shape definition), to the computation of shape variables.

The corpus used here was discovered at the *oppidum* of Bibracte, Burgundy, France. During the late Second Iron Age, Bibracte was the capital of a Celtic tribe, the *Aedui*. Archaeological excavations, undertaken each year over several decades, have unearthed a wide variety of ceramics, from plates to bottles (Barral et al., 1995; Paunier and Luginbühl, 2004; Paunier et al., 1994). If properly applied, computer-intensive techniques can do far more than merely categorise objects that can be readily differentiated by the naked eye. The present work provides an assessment of various operational combinations, considering either open contours treated by Discrete Cosine Transform (DCT) or closed contours treated by Elliptic Fourier Analysis (EFA), where the effects of size and orientation have been removed with different normalisation techniques. These approaches correspond to different ways of apprehending the ceramic shape. All these combinations produce morphospaces, which were all compared with each other and with the morphospace produced by more traditional descriptive methods (Vaginay and Guichard, 1988). Thus, it is possible to evaluate i) the pertinence of the traditional typology using several different quantitative estimates of shapes, and ii) the strengths and weaknesses of these various definitions of ceramic shapes in an operational context.

Morphometrics allows the calculation of an average shape, an idealised centroid shape, which takes into account all the individuals in the group, and not merely a single element, supposed to be the best representation, as often used in typology. Another parameter of interest is the diversity of shape within a group, which is at least as important as the mean shape, and may reveal the level of product standardisation of a given ceramic type. These two parameters are also explored here.

2. Material and methods

2.1. Corpus

Data were acquired from a corpus composed of ceramics discovered at Bibracte, all dating from the 2nd century B.C. to the beginning of the 1st century A.D. Only items presenting a complete, well-preserved cross-section, for which drawings are available in the literature, were retained for the present study (literature details are available in Supplementary materials S1). Although the “complete, well-preserved cross-section” constraint considerably

downsized the number of analysable individuals, drawings of 154 ceramics were processed. Some may not be entirely reliable because of the errors inherent in any manual representation (lack of precision, possible interpretation in the drawing process, etc.). However, the disparity of the corpus is large enough to render such uncertainties negligible.

Using the well-established typological system of Bibracte (Barral et al., 1995), the 154 ceramics studied were attributed to 8 main functions on the basis of their shape: 35 plates, 20 dishes, 24 bowls, 18 cups, 14 goblets, 27 pots, 9 vases and 7 bottles.

2.2. Numerical analysis of classical typology

2.2.1. Character coding

Structured descriptive typology elaborated by Vaginay and Guichard (1988) for a contemporary site (Feurs, Rhône-Alpes, France) has been selected to describe our ceramics. Such a system allows the shape similarity between items to be quantified using a combination of several observed features, categorised metric indexes and angles. Only descriptors directly related to the morphology of vessels were taken into account. For some descriptors, additional levels were added to allow a full description of items from Bibracte (see Supplementary materials S2 for details). The final descriptive system consists of 15 categorical descriptors, each with between 2 and 10 levels.

2.2.2. Similarity computation between ceramics

The degree of resemblance between objects was achieved by computing Gower's coefficient (Gower, 1971; Legendre and Legendre, 1998) from the 15 categorical descriptors mentioned above. This similarity coefficient, S_{ij} , between two individuals (i and j) for p variables, was computed following:

$$S_{ij} = \frac{\sum_{k=1}^p w_{i,j,k} g_{i,j,k}}{\sum_{k=1}^p w_{i,j,k}}, \quad (1)$$

where all variables were treated as qualitative. Therefore $g_{i,j,k} = 1$ when both character states agree, and 0 otherwise. Kronecker's delta $w_{i,j,k}$ is a weight equal to 0 when the information is missing, and which evolves between 0 and 1 otherwise, allowing more weight to be ascribed to the most important descriptors for typological distinction. An arbitrary triple weight was attributed to certain descriptors characterising the whole form, because they are fundamental in traditional typology to identify the main types (see Supplementary materials S2 for details). The absence of certain signs for two items, for example the absence of a foot, was considered as an indicator of similarity, so that $g_{i,j,k} = 1$ in such cases. A Principal Coordinate Analysis (PCoA), also called metric multidimensional scaling, was computed to visualise the level of similarity of items. In practice, this analysis used a dissimilarity matrix D . The transformation $D = \sqrt{1 - S}$ was preferred for computation because it is less likely to generate negative eigenvalues, synonymous with non-Euclidean dimensions (Legendre and Legendre, 1998).

2.3. Morphometrics

2.3.1. Sampling outlines

Drawings were scanned in greyscale at a resolution of 1200 dpi and stored in TIF format. All images were oriented on the natural axis on which the artefacts originally stood. After possible noise had been eliminated from the scans, the half cross-section was extracted, in the R environment (R Core Team, 2012), using a modified version of the function “conte” (Claude, 2008). Sampling was at

equally spaced points, providing a dataset of 154 (number of individuals) $\times$ n (number of points per individual) $\times$ 2 (planar x- y-coordinates).

The optimal number of points, n , to be sampled to produce a good representation of the original contour was investigated using a representation error percentage, $E\%$, computed on the basis of the original perimeter, P (Sheets et al., 2006):

$$E\% = 100 \times \left(1 - \frac{\sum d_{ij}}{P} \right), \quad (2)$$

where d_{ij} is the distance between two consecutive points on the reconstructed outline. By varying n , the optimum was chosen when $E\%$ stabilised around low values. This choice will be exposed in the “Results and Discussion” section. Symmetric contours were obtained by mirroring the coordinates of the external half-section drawings (Fig. 1).

2.3.2. Shape variables

Computation of shape variables is known to be very sensitive to the orientation and size of individuals, as well as to the position of the starting point (Tatsuta et al., 2004). Different approaches normalising size and orientation were therefore tested: several types of baseline registration (Bookstein, 1991), and normalisation to perimeter (Schmittbuhl et al., 2003) or surface (Navarro et al., 2004), which are currently and frequently used in the literature (see Fig. 2 for a visual summary of the different methods of normalisation). They refer to different views of the object size. It is

noteworthy that orientation and size are frequently normalised with the major axis of the first ellipse (Kuhl and Giardina, 1982). This procedure was not followed here, as it is not adapted to the present corpus: the major axis does not always correspond to the plane on which artefacts originally stood (imagine a bottle and a plate), affecting the homology between objects, in terms of orientation.

Shape variables characterising ceramics were obtained by Discrete Cosine Transform (DCT) for open outlines, and by Elliptic Fourier Analysis (EFA) for closed ones. In both these Fourier-type methods, the outlines are decomposed into an infinite series of repeating trigonometric functions called harmonics. Each harmonic is expressed by two Fourier coefficients, in the case of DCT (Dommergues et al., 2007; see also the appendix of Forel et al., 2009), and four for EFA (Kuhl and Giardina, 1982; Lestrel, 1989; Lestrel et al., 2010; Rohlf and Archie, 1984). The contribution of harmonics is additive for the reconstruction quality of the outlines, meaning that more harmonics reflect more precisely the form studied. The DCT was computed using an adaptation for R of the MATLAB toolkit CDFT (Dommergues et al., 2007). The EFA was processed using a modified version of the “efourier” function, as reported in Claude (2008).

It is noteworthy that techniques mixing landmarks and sliding semilandmarks on outlines (Bookstein, 1997) were not explored. The interest of the semi-landmark approach to describe outline, with respect to Fourier methods, resides in its potential to segment the whole shape into smaller pieces, comparable among individuals, using precisely defined landmarks. Although extremely

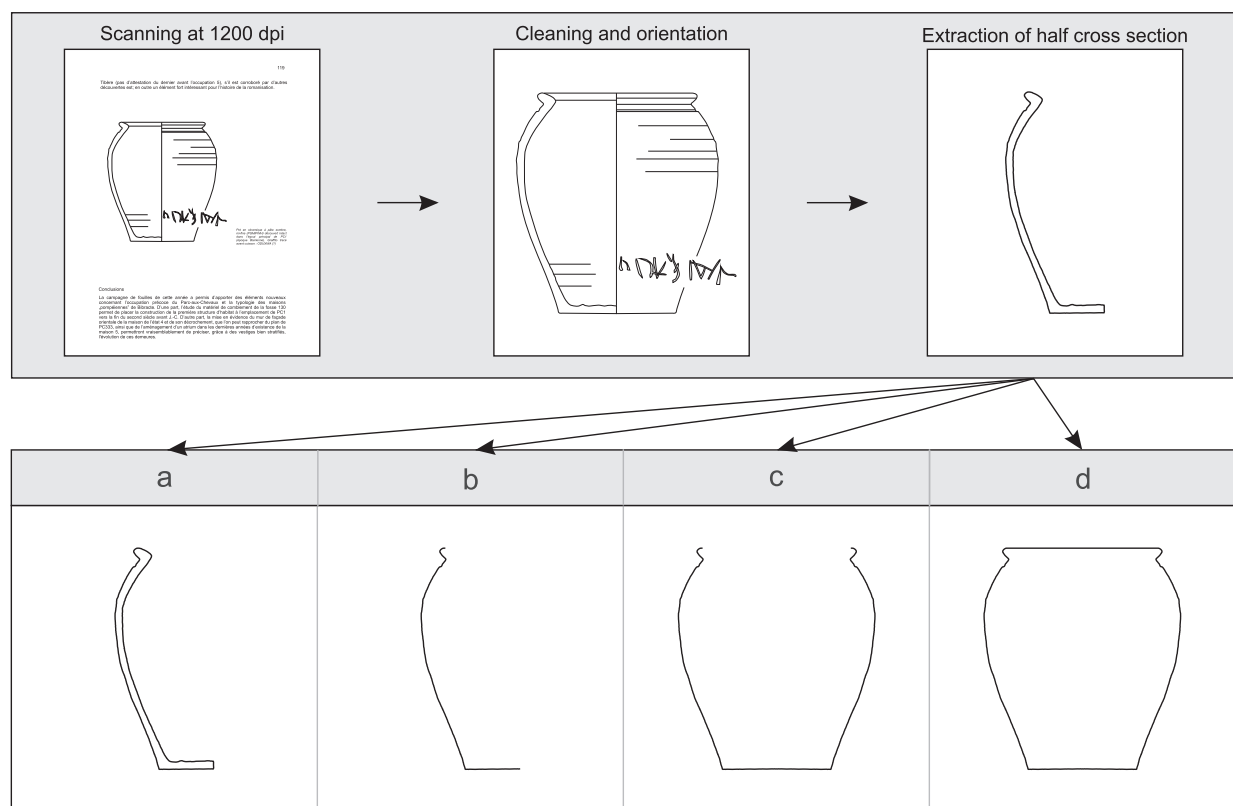


Fig. 1. Data processing and contour representations of the ceramic vessels. a) closed and b) open contour representing half of the complete cross-section; c) open contour of the outer shape of the ceramic vessel without the rim, and d) closed contour of the outer shape of the complete ceramic vessel.

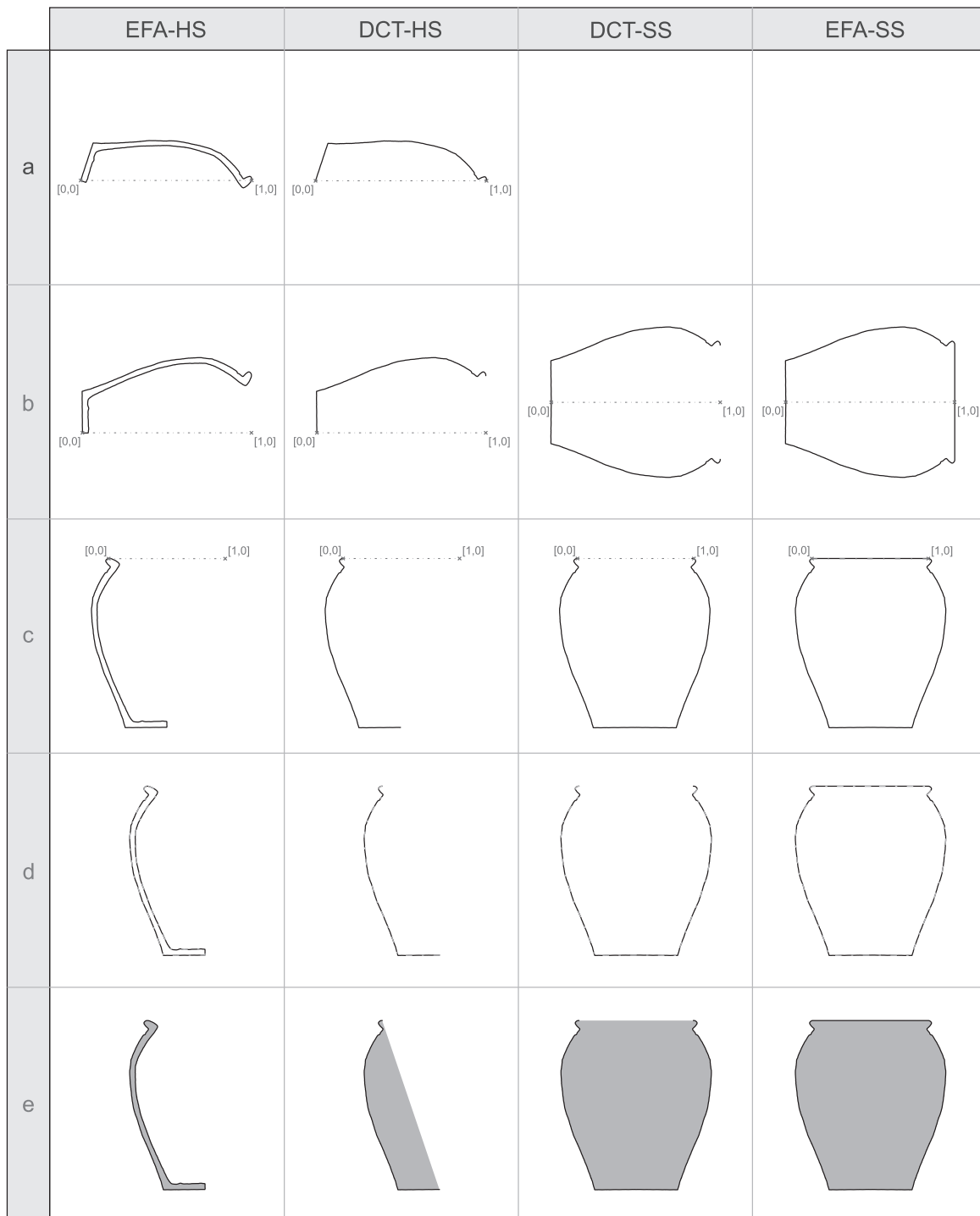


Fig. 2. Normalisation of size and orientation. EFA and DCT; Elliptic Fourier Analysis and Discrete Cosine Transform for closed and open contours, respectively. HS and SS for half section and symmetrised section. a–c) Baseline registration (Bookstein, 1991) sends two specific points to the respective coordinates (0,0) and (1,0), rotating, translating and scaling all contours; the coordinates are no longer dependent on size, position or orientation; d) size normalisation relative to the perimeter (Schmittbuhl et al., 2003) and e) size normalisation relative to the square root of the area bounded by the polygon contour (Sundberg, 1996; Hurth et al., 2003; Navarro et al., 2004); in cases d and e, the orientation was standardised to the natural axis on which the ceramic vessels originally stood.

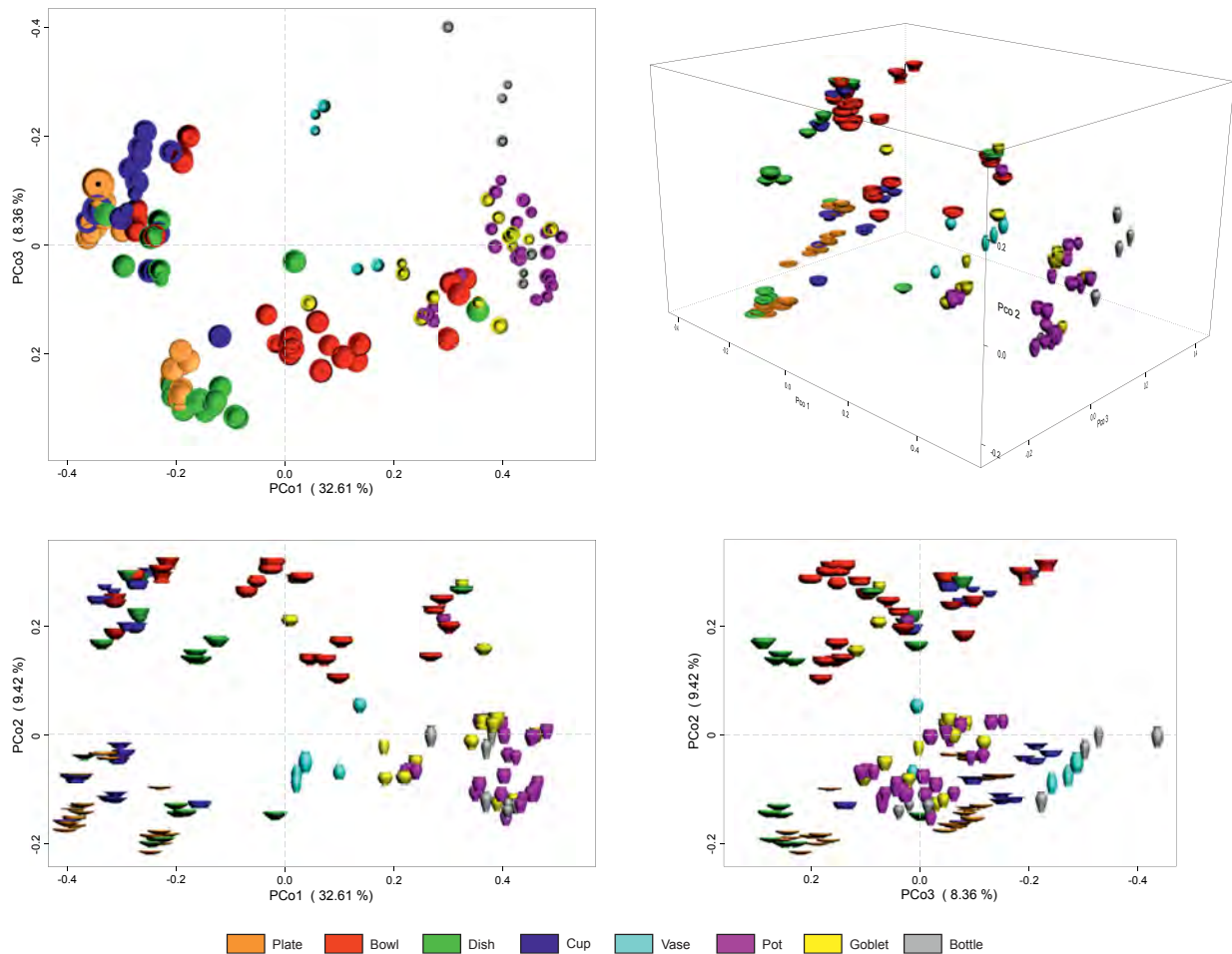


Fig. 3. Principal Coordinate Analysis. Projection of the 154 items in a PCo3 vs PCo2 vs PCo1 morphospace. These three axes (out of a total 154 axes – number of items) account for more than 50% of the total variance (see [Supplementary materials S3](#) for an animation depicting the projection of items in a 3D morphospace). Items were attributed to ceramic types following the typological system of [Barral et al. \(1995\)](#). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

powerful, they were not applied here because a sufficient set of true landmarks could not be straightforwardly identified across our heterogeneous corpus.

Principal component analyses (PCA) were then performed on the covariance matrices of the obtained Fourier coefficients, producing a series of morphospaces. All individuals can be plotted in each morphospace, to allow comparison in a visually friendly form.

2.4. Concordance between morphospaces

The agreement between the typological-based morphospace (PCoA) and each geometric morphometric-based morphospace (PCA) was checked using PROTEST, a Procrustean randomisation Test, which appears to be more powerful than the traditional Mantel test ([Jackson, 1995](#); [Peres-Neto and Jackson, 2001](#)). Basically, the PROTEST approach compares multivariate datasets: the first matrix is translated, reflected, rotated and dilated to minimise the sum of the squared residual deviations between the elements of the first (original) and the second (target) matrices ([Jackson, 1995](#)). A PCoA space can be constructed, allowing a visual inspection of morphospace similarities, while the statistical significance can be

obtained by permuting elements (here $n = 9999$) within the matrices under comparison ([Jackson, 1995](#); [Peres-Neto and Jackson, 2001](#)).

2.5. Unsupervised model-based clustering

Multivariate Gaussian mixture models were used for unsupervised classification ([Fraley and Raftery, 2007](#)) of ceramics using the R package mclust ([Fraley et al., 2012](#)). The number of clusters was explored from 2 to 12 groups for different modelling of the variance-covariance matrices. The optimal model for classification (number of clusters and matrix modelling) was determined from the Bayesian Information Criterion, BIC ([Fraley and Raftery, 1998](#); [Schwarz, 1978](#)), according to the value of the Δ -BIC (i.e., the gain in information when an additional cluster is envisaged).

2.6. Mean shapes and morphological diversities

The mean shape for each ceramic group was reconstructed by inverse Fourier functions. Palaeontologists have long developed various metrics to analyse diversity of shape (disparity), where

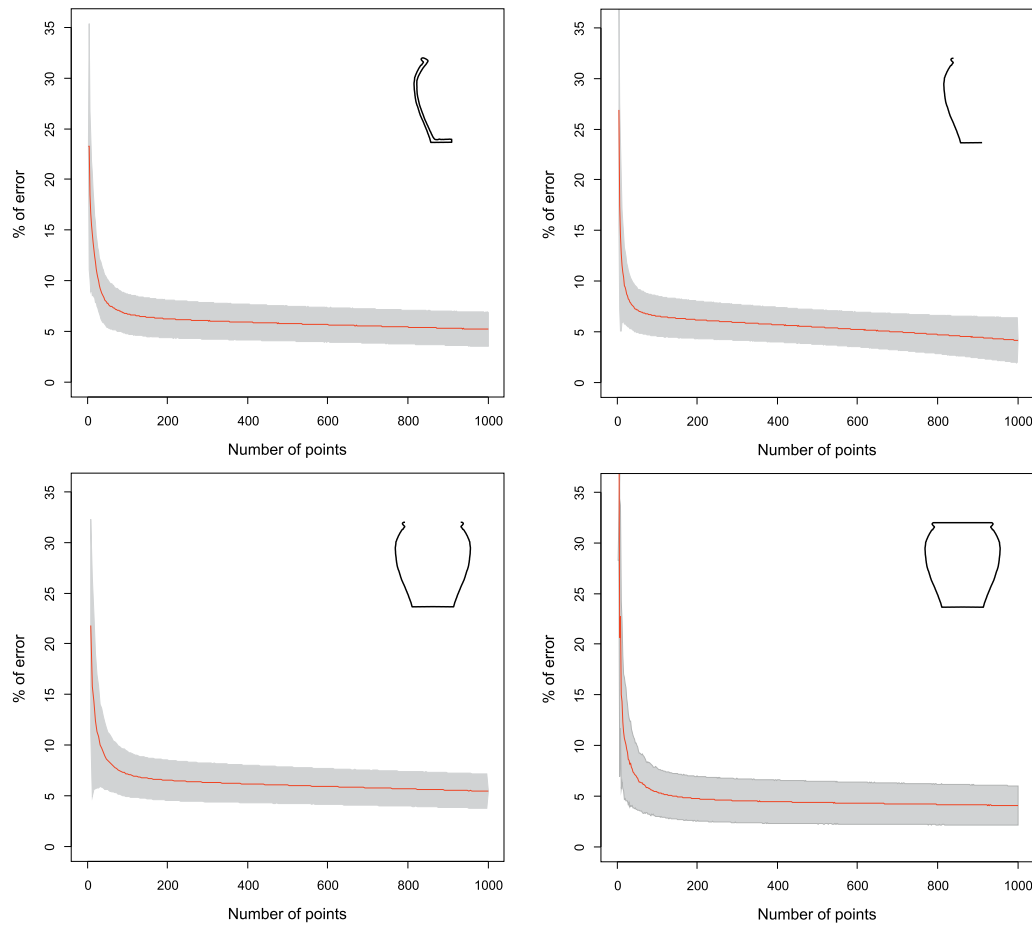


Fig. 4. Effect of the number of points sampled along the outline on contour quality for each of the four representations. The red line represents the mean value of root-mean-square deviation, and the grey surface the 95% confidence region. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

emphasis is laid on dissimilarity (e.g. total variance) or spatial distribution within the morphospace (e.g. convex hull, clustering vs scattering; Navarro, 2003). Among all possible metrics, the area of the convex hull on the first two principal components was privileged as a surrogate of morphological diversity within each ceramic group.

3. Results and discussion

3.1. Typological-based morphospace

Computation of PCoA using the dissimilarity matrix does not produce any negative eigenvalues, so the space is Euclidian and the distances between items are meaningful. The first three principal coordinate axes account for more than 50% of the variation. Fig. 3 illustrates the shape variation of ceramics in the PCoA-based morphospace, although the underlying parameters are categorical (see Supplementary materials S3 for an interactive projection of items in a 3D morphospace). Note that in some cases, certain individuals (e.g. vases), which slightly differ in terms of morphology, occupy exactly the same position in the graph. Even a typology based on a large number of categorical descriptors cannot produce a morphospace able to express the entire shape variability of ceramics. Despite this disadvantage,

the Bibracte types are reasonably well clustered: negative values of both PCo1 and PCo2 characterise flat forms, such as *plates*, while *bottles* and *pots* are clustered in the bottom right-hand corner. Other types are more broadly scattered (i.e. *bowls* and *cups*), illustrating the great variation in shape that exists within these functional groups.

3.2. Morphometric-based morphospaces

In Fig. 4, the evolution of the error percentage according to the number of points sampled exhibits a stabilisation around 100 points, with a steady representation error at about 5–7%. Nevertheless, as a precaution, 200 points were sampled, regularly spaced along the outlines.

The effect of the number of harmonics on the quality of representation was also studied. The original form was first obtained by a 200-point sampling, and then compared to the shapes reconstructed by increasing the harmonic number from 1 to 100; the second value corresponding to the Nyquist frequency (Lestrel, 1989; Lestrel et al., 2004a,b, 2005, 2010). The reconstruction quality criterion is the root-mean-square deviation, RMSD, between the 200 points composing the original outline, and their equivalent on the reconstructed outline. Although the representation becomes steady with a low number of harmonics (ca. 10),

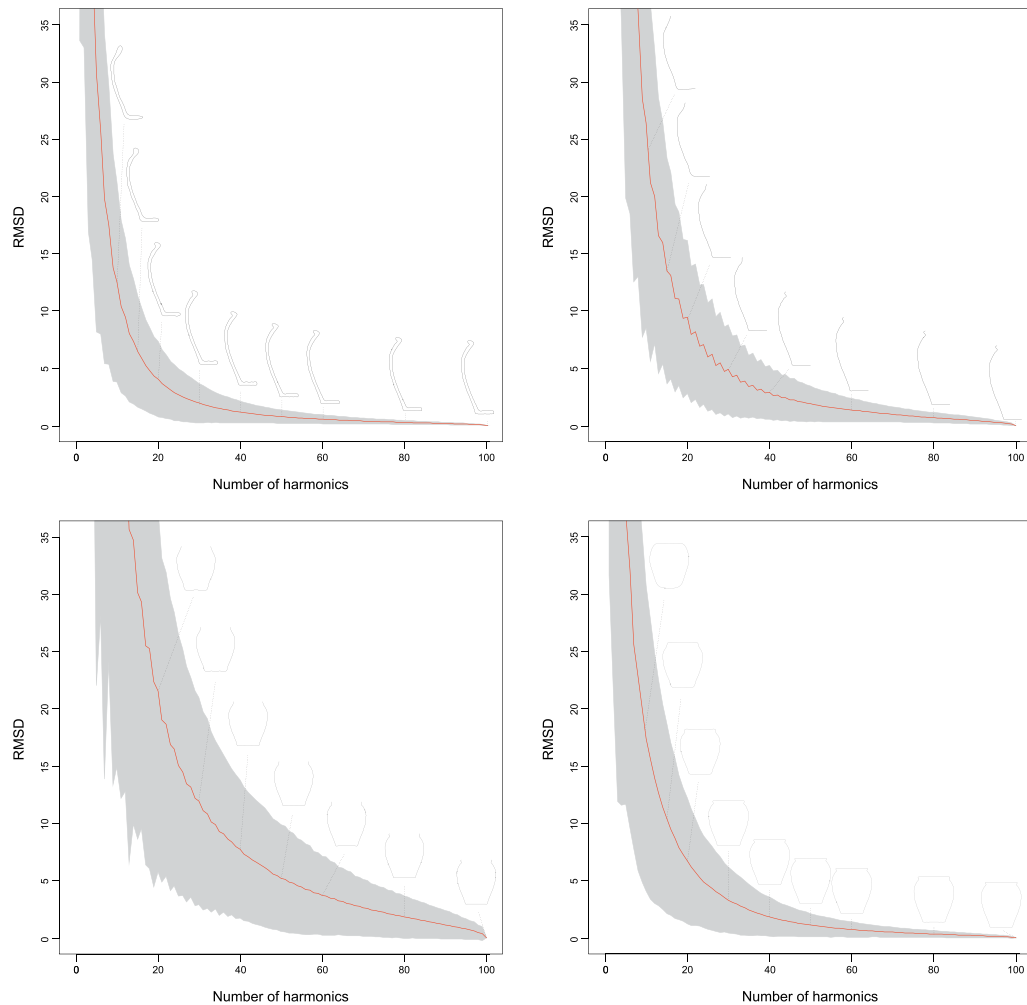


Fig. 5. Effect of the number of harmonics on the contour quality for each of the four representations. The red line represents the mean value of root-mean-square deviation, and the grey surface the 95% confidence region. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

additional details are recovered with more coefficients (Fig. 5). That is why, for the following analyses, 20 harmonics were retained in both DCT and EFA.

Principal components were computed for each of the 18 approaches depicted in Fig. 2 (four of them are depicted in Fig. 6). In all cases, the total variance explained by the two first principal components ranges between 80% and 99%, depending on the method used. The elongation of the artefacts is mainly carried by PC1. This feature clearly dominates the analyses as, similarly to the PCoA-based morphospace, the *bottles* and *plates* are located at opposite sides of PC1, while the other items are found in a more central position.

3.3. Evaluating the similarities between the different approaches

A total of 18 distance matrices based on the different morphometric approaches and normalisation (Fig. 2) and one distance matrix based on typology were processed. The agreement between all pairs of matrices appears to be significant using PROTEST (Table 1). However, despite this overall resemblance, the degree of similarity varies for each pair considered (Fig. 7). The procedures

based on perimeter (d) and surface (e) normalisations occupy the same position in Fig. 7, while the others appear as separate clusters (a, b, c). It is not really surprising to find the perimeter (d) and surface (e) normalisations clustered in a central position, between the normalisations based on height (b) and width (c), because the former are somehow a combination of the latter. In any case, normalisation to the surface or perimeter should be preferred because they are closely related to volume, which is as fundamental as shape in the purpose of ceramics. The morphospace using standardisation by chord (a) lies very close to the traditional typology. Nevertheless, because of the nature of this Bookstein registration using chord, two half cross-sections cannot be merged to reconstruct the original shape, as information about opening is lost. If the aim of the study is related to the overall aspect of artefacts, then losing a feature, such as the opening angle, is prohibitive. The chord is thus no better adapted for standardisation than the major axis of the first ellipse. It may sometimes be preferable to emphasise local features of the outline, containing relevant information about the artefact, such as its symbolic or social role, and the know-how of craftsmen, which could have been masked by treating only the overall aspect.

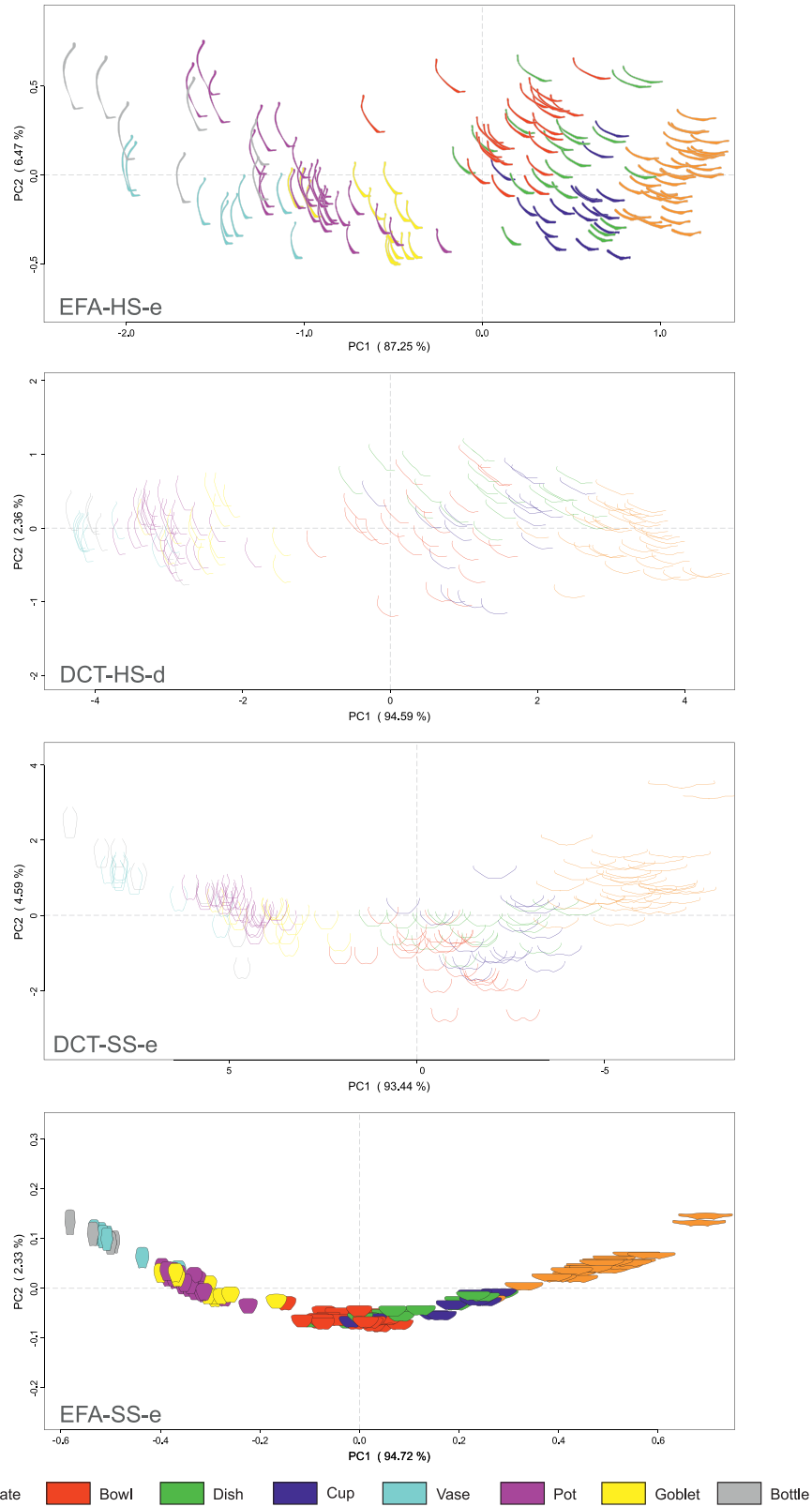


Fig. 6. Principal Component Analysis. Projection of the 154 items in a PC2 vs PC1 morphospace. The two axes always account for more than 93% of the total variance. The figure code helps to find the corresponding approach in Fig. 2. Items were attributed to ceramic types following the typological system of Barral et al. (1995). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1Similarity matrix for the different approaches. Lower half matrix: PROTEST statistic. $p < 0.05$ in all cases. See Fig. 2 caption for acronym definition.

	TYPO	EFA-HS					DCT-HS					DCT-SS				EFA-SS			
		a	b	c	d	e	a	b	c	d	e	b	c	d	e	b	c	d	e
TYPO		1.00																	
EFA-HS	a	0.59	1.00																
	b	0.53	0.64	1.00															
	c	0.47	0.49	0.68	1.00														
	d	0.62	0.71	0.91	0.84	1.00													
	e	0.59	0.67	0.85	0.88	0.95	1.00												
DCT-HS	a	0.55	0.84	0.67	0.49	0.68	0.62	1.00											
	b	0.50	0.57	0.99	0.68	0.89	0.83	0.65	1.00										
	c	0.44	0.43	0.67	0.99	0.82	0.85	0.47	0.68	1.00									
	d	0.58	0.61	0.89	0.84	0.98	0.93	0.63	0.89	0.83	1.00								
	e	0.58	0.66	0.91	0.80	0.95	0.91	0.71	0.90	0.79	0.96	1.00							
DCT-SS	b	0.50	0.57	0.99	0.68	0.89	0.83	0.65	1.00	0.68	0.89	0.90	1.00						
	c	0.44	0.43	0.67	0.99	0.82	0.85	0.07	0.68	1.00	0.83	0.79	0.68	1.00					
	d	0.59	0.62	0.88	0.86	0.98	0.94	0.62	0.87	0.84	0.98	0.94	0.87	0.84	1.00				
	e	0.57	0.62	0.92	0.86	0.97	0.94	0.66	0.92	0.85	0.97	0.96	0.92	0.85	0.97	1.00			
	b	0.50	0.56	0.99	0.67	0.88	0.82	0.63	1.00	0.67	0.88	0.90	1.00	0.67	0.87	0.91	1.00		
EFA-SS	c	0.47	0.45	0.68	0.99	0.83	0.86	0.48	0.68	1.00	0.84	0.80	0.68	1.00	0.86	0.86	0.68	1.00	
	d	0.61	0.64	0.88	0.87	0.98	0.94	0.65	0.87	0.86	0.98	0.95	0.87	0.86	0.98	0.97	0.87	0.88	1.00
	e	0.59	0.63	0.95	0.81	0.97	0.92	0.68	0.95	0.81	0.96	0.97	0.95	0.81	0.95	0.98	0.95	0.82	0.97

Classical typology can be viewed in essence as a degeneration of morphometrics, because it takes into account only some shape criteria, while morphometrics contains all shape information. Although typology does not exactly match any of the morphometric morphospaces (Fig. 7), there is good statistical agreement between the two approaches (Table 1). This concordance clearly demonstrates the value of the classification proposed by Vaginay

and Guichard (1988). Morphometrics, however, has greater potential, more particularly in terms of quantitative evaluation, which is not readily accessible with the traditional descriptive approach.

Constraining the outline approach by landmarks may improve point correspondence because the homology between elements on the outline is increased (e.g. McCane, 2013). Here ceramics were constrained with only two landmarks (see Fig. 2), because it was

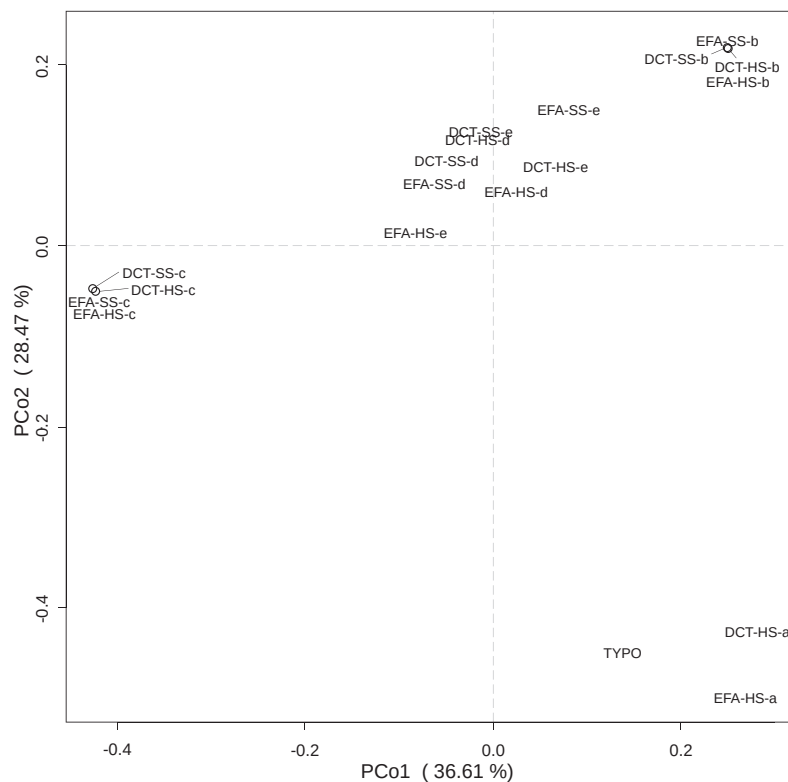


Fig. 7. Principal Coordinate Analysis. Projection of the typological morphospace (TYPO) and the 18 different morphospaces of shape variables in a PCo2 vs PCo1 space based on PROTEST statistics. The two axes account for more than 65% of total variance. See Fig. 2 for acronym definition.

Table 2

Concordance between the traditional typological classification of Barral et al. (1995) and unsupervised model-based clustering based on the EFA-SS-e morphospace.

Traditional typology		Model-based groups						
		1	2	3	4	5	6	7
Plates		29			6			
Dishes		6	13		1			
Bowls			6	15	2	1		
Cups		1	2		15			
Goblets						9	5	
Pots						11	16	
Vases							2	7
Bottles					1		2	4

not possible to define additional landmarks easily and unambiguously across the corpus. As a consequence, pairwise point matching is probably not fully optimal and could be improved. That would be problematic if precise description of curve inflexion, such as that at the neck, was required. Here, losing such precision is not as important, given the disparity of ceramic shapes and the embedding of broad homology within the complete outline.

3.4. Unsupervised model-based clustering

Unsupervised model-based clustering was performed from the “EFA-SS-e” harmonic coefficients. An optimal model with seven clusters and a VEI modelling (i.e. diagonal, varying volume, equal shape) of the variance-covariance matrices was chosen in the light of the Δ -BIC evolution with an increasing number of groups (see [Supplementary materials S4](#)). These seven clusters match well the traditional typology of Barral et al. (1995), with each cluster corresponding mainly to one single type ([Table 2](#) and [Supplementary materials S5](#)), except when the size drives the difference (e.g., goblets and pots). This is additional evidence that the traditional typology is still compatible with a more recent quantitative expression of shape. Morphological typology built over years, although founded on empirical observations, may therefore produce very reliable systems. Modern morphometrics followed by

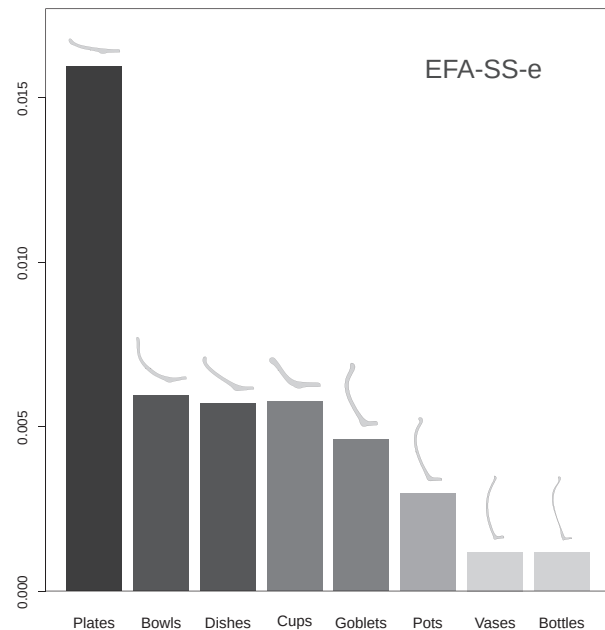


Fig. 9. Shape diversity of the eight ceramic groups using the “EFA-SS-e” morphospace ([Fig. 6](#)), computed as the surface of the convex hull of PC2 vs PC1 (the two axes account for more than 97% of the total variance).

unsupervised classification nevertheless provides a quantitative tool to build these typologies numerically, independently of the operator.

3.5. Reference shape and diversity estimation

The mean shapes of the 8 known ceramic types from Bibracte are displayed in [Fig. 8](#). Dispersion of each ceramic group was computed as the area of its convex hull in the “EFA-SS-e”

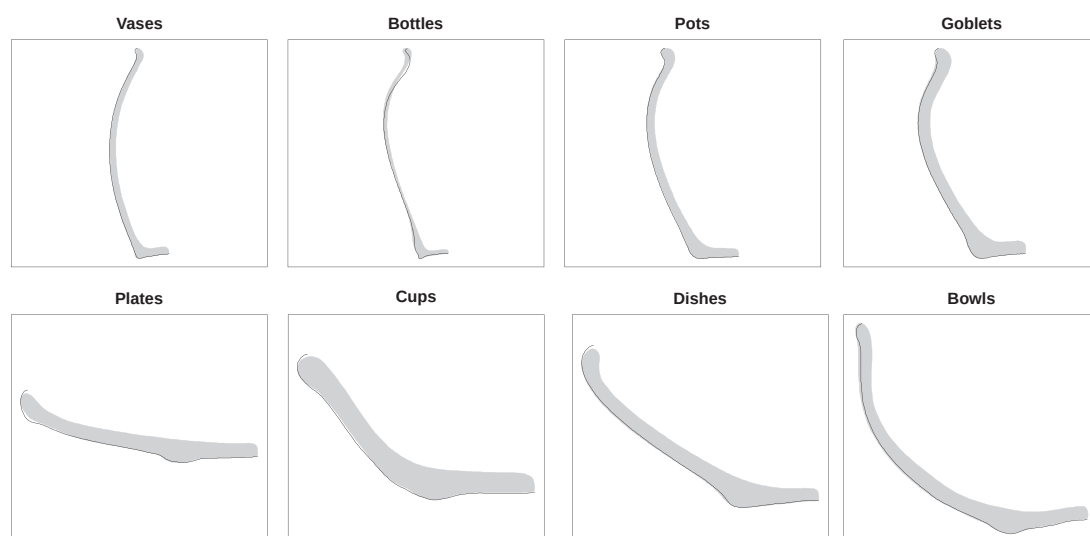


Fig. 8. Mean shapes of main ceramic groups from Bibracte following the typological system of Barral et al. (1995). Grey polygons represent inverse transform for EFA-SS-e, while solid black lines are the inverse transform for DCT-HS-d.

morphospace, using PC1–PC2 (Fig. 9). The diversity of ceramics used for presenting food (*plates*) is twice that of the other functional groups. Ceramics used for consuming food (*bowls, dishes*) and liquids (*cups, goblets*) show similar dispersion, followed by the vessels for cooking and storage (*pots*). The most homogenous groups were related to ceramic types dedicated to storing and presenting liquids (*vases and bottles*). Interestingly, group disparity may be related to functional and technical constraints, but also to the ostentatious nature of the vessels (Barral, 2002, 2005; Barral and Huet, 2006; Paunier and Luginbühl, 2004; Paunier et al., 1994).

4. Conclusion

Shape corresponds to the variation that remains, once position, size and orientation are removed. The definition of size (e.g. volume, height, etc.) may be governed by functional considerations. It has profound implications for the resulting shape, so that this choice must be made by the archaeologist in relation to the intended goal. Here, both approaches, DCT and EFA, combined with perimeter or surface standardisation can be efficiently used. As a very heterogeneous and diversified assemblage, consisting of specimens with low and high shapes (e.g. *plates and barrels*) has been taken into account, morphospaces are mainly driven by elongation. This is probably why morphometrics matches traditional ceramic typology so well. This finding could be perceived as trivial, but in fact the same approach can be applied to a group of individuals presenting far more resemblance. Shape diversity is expressed in simple, readable graphics, allowing comparisons and intuitive understanding of the entire assemblage. Within a given type, the amount of shape variation quantifies the structural constraints together with the level of standardisation of the products. Morphospaces are continuous by nature and can accept forms that have not yet been discovered. Mean shape can easily be reconstructed and is a better reference form than an isolated individual, supposed to reflect some typical standard.

The entire morphometrics pipeline (acquisition, standardisation, and computation of shape variables) can be fully automated, ensuring rapid processing and reproducibility.

Acknowledgements

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jas.2014.05.033>.

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Chapter 6

Unsupervised model-based clustering for typological classification of Middle Bronze Age flanged axes

This chapter presents the second paper published within the framework of this doctoral project. Several morphometric (EFA) and statistical methods (MBCA) were applied to create and validate a new unsupervised typology for Middle Bronze Age axes and ingots found west of the Alps.



Unsupervised model-based clustering for typological classification of Middle Bronze Age flanged axes



J. Wilczek^{a,b,*}, F. Monna^a, M. Gabillot^a, N. Navarro^c, L. Rusch^a, C. Chateau^d

^a ArTeHiS, UMR CNRS 6298, Université Bourgogne Franche-Comté, 21000 Dijon, France

^b Ústav archeologie a muzeologie, Masarykova univerzita, 602 00 Brno, Czech Republic

^c Biogéosciences, UMR CNRS 6282, Ecole Pratique des Hautes Etudes, Université Bourgogne Franche-Comté, 21000 Dijon, France

^d UFR SVTE, Université Bourgogne Franche-Comté, 21000 Dijon, France

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ABSTRACT

The classification of Western European flanged axes dating to the Middle Bronze Age (1650–1350 BC) is very complex. Many types of axe have been identified, some of which have numerous variant forms. In the current French terminology, all axes are divided into two generic groups: namely “Atlantic” (*Atlantique*) and “Eastern” (*Orientale*). Each of these generic groups, however, is highly polymorphic, so that it is often very difficult for the operator to classify individual axes with absolute confidence and certainty. In order to overcome such problems, a new shape classification is proposed, using morphometric analysis (Elliptic Fourier Analysis) followed by unsupervised model-based clustering and discriminant analysis, both based on Gaussian mixture modelling. Together, these methods produce a clearer pattern, which is independently validated by the spatial distribution of the findings, and multinomial scan statistics. This approach is fast, reproducible, and operator-independent, allowing artefacts of unknown membership to be classified rapidly. The method is designed to be amendable by the introduction of new artefacts, in the light of future discoveries. This method can be adapted to suit many other archaeological artefacts, providing information about the material, social and cultural relations of ancient populations.

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1. Introduction

Many types of flanged axes produced in Western Europe during the Middle Bronze Age (1650–1350 BC) have been recognised by archaeologists (e.g. Abels, 1972; Briard and Verron, 1976; Gomez de Soto, 1980; Kibbert, 1980; David-Elbiali, 2000; Gabillot, 2003; Michler, 2013). Most types have numerous variants, so that fine typological classification on the sole basis of their shape is generally problematic. The situation is even more complex because typologies generally combine several criteria, such as edge height, the possible presence of ornaments, and the total size of the object, but do not always take all of them into account. These descriptive criteria are not always given the same weight in type definition. Briard and Verron (1976) merged axe types into two generic groups: namely “Atlantic” (*Atlantique*) and “Eastern” (*Orientale*), broadly following the location of the find: closer to the Atlantic coast, or closer to the Alps. Nevertheless, this distinction no longer seems completely adequate to differentiate rapidly between axes of each generic group. For instance, the shapes of concave-blade flanged axes (Atlantic group) and those of the Neyruz type (Eastern

group), which each have several variant forms, are at first glance very similar (Fig. 1a:2, 4). Since the 1970s, specific studies on axes, and regional syntheses (Butler, 1995/1996; David-Elbiali, 2000; Gabillot, 2003; Michler, 2013) on metallic artefacts dating from the Bronze Age have refined the previous classification presented by Briard and Verron (1976), but they have not really called into question this early work. Without a precise location for the find, it is impossible to attribute a flanged axe to a group, except for some specific types, such as Roseaux-Morges, Möhlin, or the large cutting blade type (Fig. 1b:12–14; Abels, 1972; Briard and Verron, 1976). In any case, a typological system based on the location of the find, which may seem convenient, would not be appropriate to tackle archaeological questions relating to the quality of exchanges or potential stylistic and technological influences between cultural entities.

During the Middle Bronze Age, in addition to flanged axes, another category of object, the so-called axe-ingots, was also produced (Fig. 1c). Their shape is quite similar to common flanged axes, but they are almost exclusively composed of copper (e.g. Rychner and Kläntschi, 1995), and do not seem to have been used after casting (e.g. Nicolardot and Verger, 1998). The casting cone and burrs on the edges are still present on axe-ingots, unlike functional axes. Two main hypotheses concerning their function have been formulated: they could have been designed as copper ingots for future casting operations, or they

* Corresponding author at: ArTeHiS, UMR CNRS 6298, Université Bourgogne Franche-Comté, 21000 Dijon, France.

E-mail address: wilczek@mail.muni.cz (J. Wilczek).

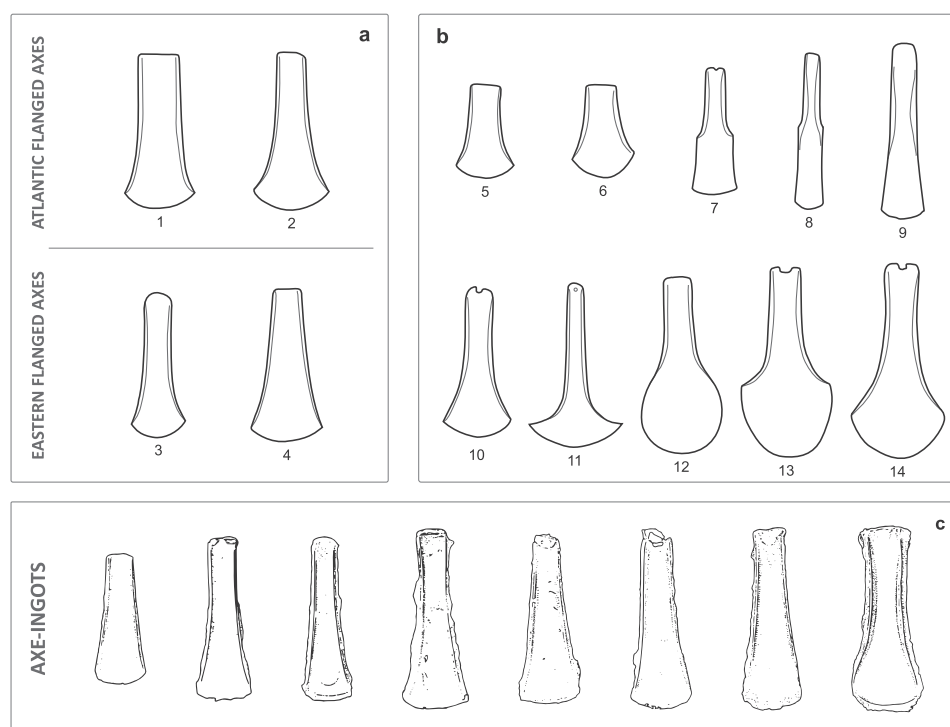


Fig. 1. Typological classification of Middle Bronze Age flanged axes, based on Briard and Verron (1976). a) Atlantic and Eastern types integrated into the corpus, b) morphologically specific flanged axe types not included in the corpus, c) examples of axe-ingots found in several eastern French sites. 1) Narrow-blade flanged axes, 2) Concave-blade flanged axes, 3) Salez type, 4) Neyruz type, 5) Low flanged axes, 6) Languedoc types, 7) Shoulder type, 8) Baraque type, 9) Ricardelle type, 10) Porcieu-Amblagnieu type, 11) Large cutting blade type, 12) Roseaux-Morges type, 13) Möhlin type, and 14) Langquaid type.

may have served as a means of exchange. Their potential for use as genuine axes cannot be excluded (Nicolardot and Verger, 1998; Delrieu et al., 2015).

The present study aims at systematising the typological classification of these flanged axes. Our approach is based exclusively on object shapes and their treatment by objective statistical techniques, reproducible in time and space by any operator. Since the 1960s, many morphometric methods have been developed. They are based on linear and angular measurements of objects (e.g. Roe, 1968; Hodson, 1971; Barker, 1975; for Bronze Age axes see Lull, 1983), sometimes simplified by deduced categorical variables (e.g. Hodson et al., 1966; Sackett, 1966; Vaginay and Guichard, 1988), and they have proved their worth in archaeological classification. More recently, morphometrics applied to archaeology has evolved into more complex methods including more information (e.g. Brande and Saragusti, 1996; Gilboa et al., 2004; Lycett, 2009; Karasik and Smilansky, 2008, 2011). These methods are known to allow a better description of the entire shape and a separation of shape and size. They provide a continuous morphospace allowing more complex statistical analyses, including the reconstruction of the mean shape and shape diversity within the group of interest (Adams et al., 2004; Navarro, 2003; Zelditch et al., 2004; Slice, 2005; Wilczek et al., 2014). Two recent studies undertaken on Bronze Age palstaves (Forel et al., 2009; Monna et al., 2013) have already demonstrated that combining geometric morphometrics with spatial analyses can be very effective for the better understanding of artefact production and use.

Our first goal was to apply these techniques to closed contours obtained from a corpus of 247 axes (all available as drawings, either in published literature or in personal collections), in an area circumscribed by the French Atlantic coast, the Rhine valley and Switzerland. A new classification approach, based on shape similarities, unsupervised clustering with Gaussian mixture modelling, and discriminant methods,

was then developed. The performance of this model was spatially checked using multinomial scan statistics and compared to classifications currently used in the study area. Finally, 21 axe-ingots were introduced into the typological model, for attribution to one of the newly established groups.

2. Material and methods

2.1. Corpus

The choice of the corpus was guided by several constraints: (i) the objects had to be intact and undamaged by use or corrosion, and (ii) their silhouette must not have been drastically reworked after they came out of the mould. Axe preservation was estimated visually from available items, or obtained from the literature (Bocquet, 1970; Abels, 1972; Gomez de Soto, 1980; Kibbert, 1980; Gabillot, 1997, 2003; Nicolardot and Verger, 1998; Mélin, 2012; Gabillot et al., 2014; Thevenot, unpublished). Although the above-mentioned constraints considerably reduced the number of individuals available (approximately 50–60% of available items were kept for further analysis), this selection process is expected to produce robust results. The final corpus consists of 247 reasonably contemporaneous flanged axes (126 from the Atlantic group, and 121 from the Eastern group), discovered in 132 sites, located in what is now France, Switzerland and Germany. Other more specific types, visually very different from the corpus of interest (Fig. 1b), or simply very scarce (e.g. type Strasbourg, Herbrechtingen, Luzern, Riquewihl), were not integrated into the present study. Finally, four generic groups of axes (concave and narrow blades for the Atlantic generic group and Salez and Neyruz types for the Eastern generic group) were retained in the study. The spatial distribution of these axes (Fig. 2) is marked by a clear gap between the two groups, possibly due to the relative absence of archaeological

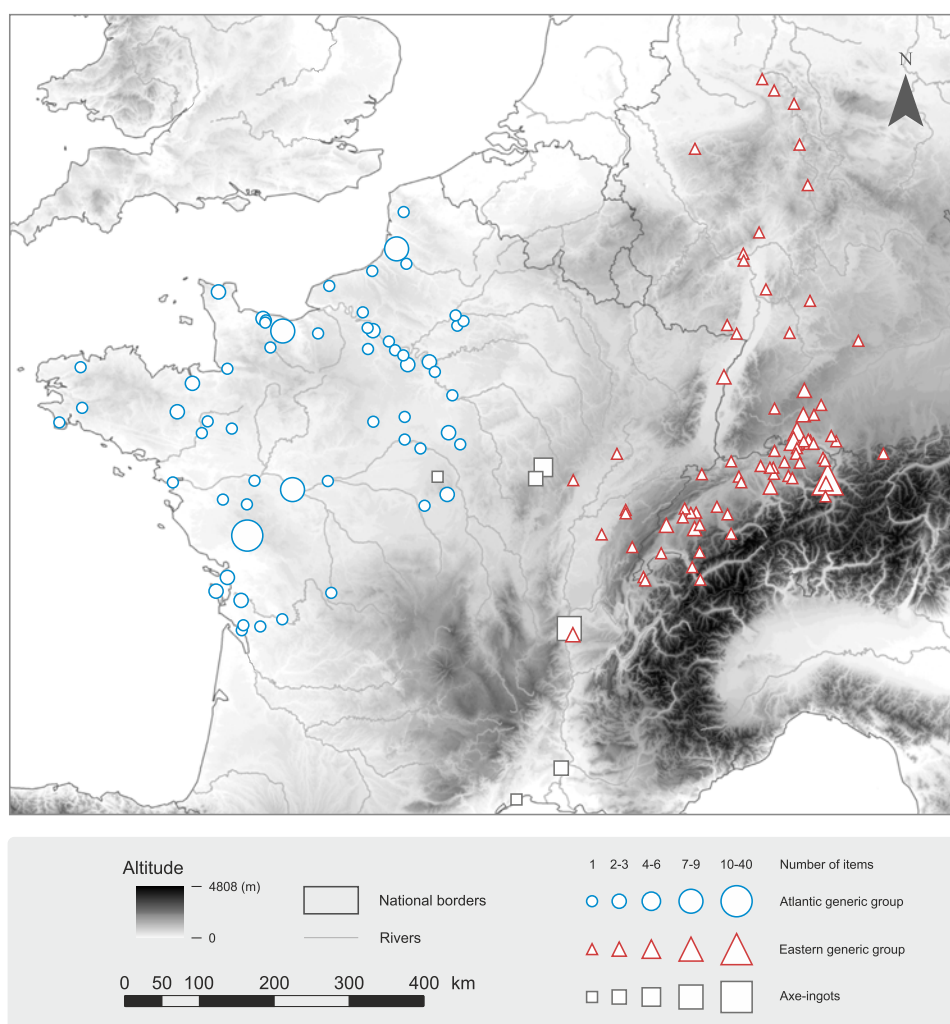


Fig. 2. Distribution map of flanged axes and axe-ingots.

exploration in that area, but which could also reflect an archaeological reality. This corpus was supplemented by 21 “axe-ingots” (e.g. Nicolardot and Verger, 1998), found in hoards located in the area roughly separating the zones of Atlantic and Eastern flanged axes.

2.2. Morphometrics

2.2.1. Data acquisition, drawings, and outlines

Acquisition follows the procedure described in Forel et al. (2009), Monna et al. (2013) and Wilczek et al. (2014). Briefly, all published images obtained from the available documentation (Supplementary materials S1, Fig. 3ab) were redrawn on tracing paper by a single operator (Fig. 3c). These drawings were then scanned at 300 dpi, and each silhouette was orientated vertically, using the first eigenvector computed on the x -, y -coordinates (Fig. 3d). After this step, each silhouette was sub-sampled, using 200 equally spaced points, starting from the point possessing the minimum y -coordinate value (Fig. 3e). Original drawings were hand-produced by the same operator, but may differ to some extent from the original object. However, given the number of artefacts processed, it is reasonable to think that any possible awkwardness in

the drawings of some specimens will be insignificant for the final results.

2.2.2. Extraction of morphological data

The contour of the axe is taken to be a parametric equation defined as $x(t)$ and $y(t)$, where t is the displacement step along the outline, $t \in [1, \dots, 200]$. Decomposition is then performed by Elliptic Fourier Analysis (EFA), producing four coefficients by harmonics i (i.e. A_i , B_i , C_i , and D_i). The greater the number of harmonics, the better the reconstruction of the original contour. These coefficients are commonly used as new variables to describe the shape (Kuhl and Giardina, 1982; Lestrel, 1989; Navarro et al., 2004). Normalisation of flanged axes was performed by the major axis of the first harmonic (Kuhl and Giardina, 1982; Rohlf and Archie, 1984; Furuta et al., 1995; Zhan and Wang, 2012), and coefficients were size-normalised, using the square root of the harmonic amplitudes. The first three coefficients of the first harmonic (A_1 , B_1 and C_1) become constants and can be ignored in further calculations. Note that the fourth coefficient (D_1) of the first harmonic is retained, as it represents the minor axis of the first ellipse, and contains

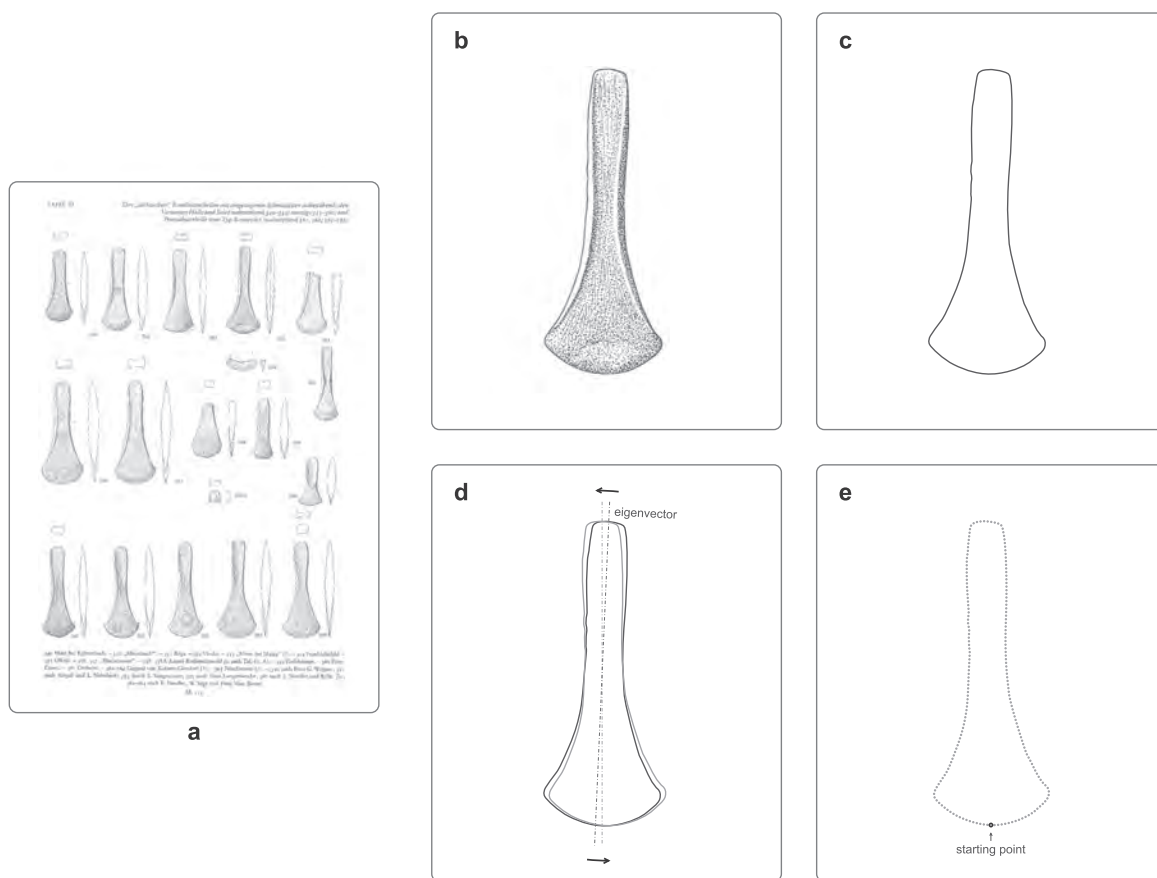


Fig. 3. The data acquisition pipeline: a) example of original drawings accessible in the literature (here Kibbert, 1980), b) original drawing of a flanged axe, c) axe outline redrawn by the operator and scanned at a resolution of 300 dpi, d) vertical orientation of the axe, using the first eigenvector computed on the x-, y-coordinates, and e) sampling of 200 equally spaced points along the outline.

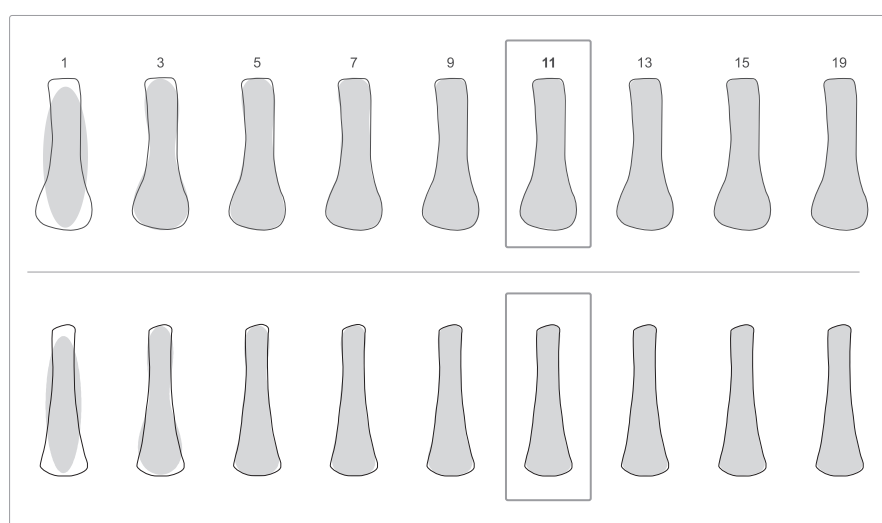


Fig. 4. The quality reconstruction of flanged axes, based on an increasing number of harmonics. The original silhouette corresponds to the black line while the shape reconstructed by a given number of harmonics is expressed as a grey polygon. Further calculations were performed using 11 harmonics (see grey boxes).

information about elongation (Iwata et al., 1998; Michaux et al., 2007; Helvacı et al., 2012).

The minimum number of harmonics needed to properly reconstruct shape was investigated by the harmonic power (Lestrel, 1997). First, the power carried by the harmonics i was defined as P_i :

$$P_i = \frac{A_i^2 + B_i^2 + C_i^2 + D_i^2}{2}. \quad (1)$$

The reconstruction quality obtained for a given number of harmonics, say n , was estimated by calculating the relative cumulative power of the first n harmonics, expressed as a percentage of the total power, i.e. the sum power carried by all harmonics (Renaud et al., 1999; Helvacı et al., 2012). The relative cumulative power stabilises at a value close to 100%, on average, from 7 harmonics (Supplementary materials S2), but 11 harmonics (corresponding to 41 Fourier coefficients) were finally retained, to ensure a finer reconstruction of artefact shapes in all cases (Fig. 4).

2.2.3. Statistical treatment of morphological data

Fourier coefficients were treated by Principal Component Analysis, PCA (Jolliffe, 2002), a procedure which maximises the variance carried by the first axis. This analysis produces a projection of the artefacts into a low-dimensional, visually friendly, Euclidian space, or morphospace, in morphometric terms. However, one major drawback with PCA is that the representation is strongly affected by the most outlying artefacts, i.e. those contributing most to the variance. This implies that any possible structure within the set of the most common artefacts, that is to say those projecting close to the centre of the PCA space, will be blurred. To circumvent this issue, self-organizing maps (SOMs), a class of neural-network algorithms (Kohonen, 2001), were also computed. These techniques are now increasingly used for data visualisation, clustering and classification of large datasets (Yin, 2008). They allow representation of a multidimensional dataset by nonlinear projection of artefacts in a lower dimension space, usually represented by discrete locations in a regular 2D lattice. Despite the loss of linearity in the output space, the topological relationships between objects (the order of the distances) are preserved (Liu and Weisberg, 2005). In our case, the use of SOMs can provide a first, exploratory step for further clustering of large datasets (Yin, 2008). Artefacts are assigned to the most similar prototype (also called codebook) vectors, which represent a set of locations summarizing the original data. Interestingly, the density of codebook vectors increases with the density of artefacts, making possible (unlike PCA) a high level of detail in the structure of the data where the artefacts are the most numerous. Practically, the codebook vectors are initiated at random. They are then progressively displaced by an iterative process, following an algorithm reasonably similar to the one applied for the widely used k-means clustering (more details about the procedure can be found in Wehrens and Buydens, 2007; Wehrens, 2011; Kung, 2014). It should be noticed however that different tunings (number of codebook vectors, type of topology used, etc.) will produce different outputs. Several tries are recommended. If a relevant structure exists within the data, the maps produced will tend to be similar.

The final determination of groups was then performed using the increasingly popular Gaussian mixture models (McLachlan and Peel, 2000), where membership association of individuals is probabilistic (Wehrens, 2011). The clusters are assumed to follow multivariate normal distributions here. The point is to find the density of each cluster, their covariance matrix and mean, as well as the conditional probability of membership. This is achieved using a two-step iterative Expectation–Maximisation (EM) algorithm (Dempster et al., 1977), where the conditional probabilities are estimated by the Expectation step and the features of the cluster by the Maximisation step. The optimal number of clusters for classification is determined by likelihood. However, the likelihood is expected to increase with the number of clusters, so that the final decision should be taken by examining a measure penalized by

the number of parameters sought. The choice of the number of clusters was made using the Bayesian Information Criterion, or BIC (Schwarz, 1978; Baudry et al., 2010; Wehrens, 2011), and more particularly by close examination of the function:

$$\Delta BIC = 2(\log L_i - \log L_j) + (p_j - p_i) \log n \quad (2)$$

where j is the number of groups considered and i is this number of groups minus 1, L corresponds to the likelihood, p to the number of free parameters in the model, and n to the sample size. In other words, ΔBIC expresses the gain in information when an additional group is considered. The posterior membership attribution, computed using an increasing number of clusters, was also examined. If the model is relevant and artefacts well classified in the clusters thus created, the posterior probabilities should most of the time be close to 1. Discriminant Analysis based on Gaussian finite mixture modelling was then applied to establish group membership for unknown artefacts (here, axe-ingsots).

2.2.4. Geographical treatment of morphological data

Axe distributions were mapped by applying a Gaussian kernel function to the location of artefact finds. Map smoothness depends greatly on bandwidth, a parameter of the kernel function: the higher the bandwidth, the smoother the kernel surface (Baxter et al., 1997). Optimal values may be obtained by cross-validation, as recommended by Wand and Jones (1994), but this algorithm tends to create roughness in the estimate. To reduce the risk of overinterpretation, a bandwidth of 65 km was applied, following the procedure of Stevens et al. (2009). Multinomial scan statistics (Jung et al., 2007) were used to complement the geographical mapping of different groups of axes. The aim of this method is to identify non-random spatial patterns (i.e. clusters) in the geographical space. The null hypothesis assumes that the probability of belonging to a given type is the same everywhere. The alternative hypothesis states that, at least for one group, membership probability is not uniform in all parts of the area. Basically, a zone of interest is progressively scanned using a scanning window of increasing diameter, and the number of items in each category is counted, inside and outside the scanning window (Jung et al., 2007). Significant geographical clusters, where type distributions are different from the rest of the map, are identified using likelihood ratio tests. A Monte Carlo procedure allows p -values to be calculated (Jung et al., 2010).

In practice, acquisition and statistical treatments were performed using the set of functions found in Claude (2008), together with the MASS (Venables and Ripley, 2002), mclust (Fraley and Raftery, 2002, 2007; Fraley et al., 2012), momocs (Bonhomme et al., 2014), kohonen (Wehrens and Buydens, 2007), and ks (Duong, 2014) packages, all written for the free R software (R Core Team, 2014). Multinomial scan statistics and geographical mapping were performed using SatScan v9.3.1 (<http://www.satscan.org/>; Kulldorff et al., 1998) and Quantum GIS 2.6.1 (<http://www.qgis.org/>; QGIS Development Team, 2015), both of which are freely available software programs.

3. Results and discussion

3.1. Evaluating current typology (Atlantic vs Eastern generic groups)

Axes belonging to the Atlantic and Eastern groups were projected onto a PCA-based morphospace, in which the first two components account for almost 86% of the total variance (Fig. 5). The first principal component (PC1) is characterised by an overall enlargement of flanged axes. The second, PC2, depicts the ratio between cutting edge length and body width. Atlantic and Eastern axes occupy approximately the same area in the PCA-based morphospace, so that neither generic group can be clearly distinguished by this analysis. It should, however, be recalled that PCA is not specifically designed to test the presence of possible

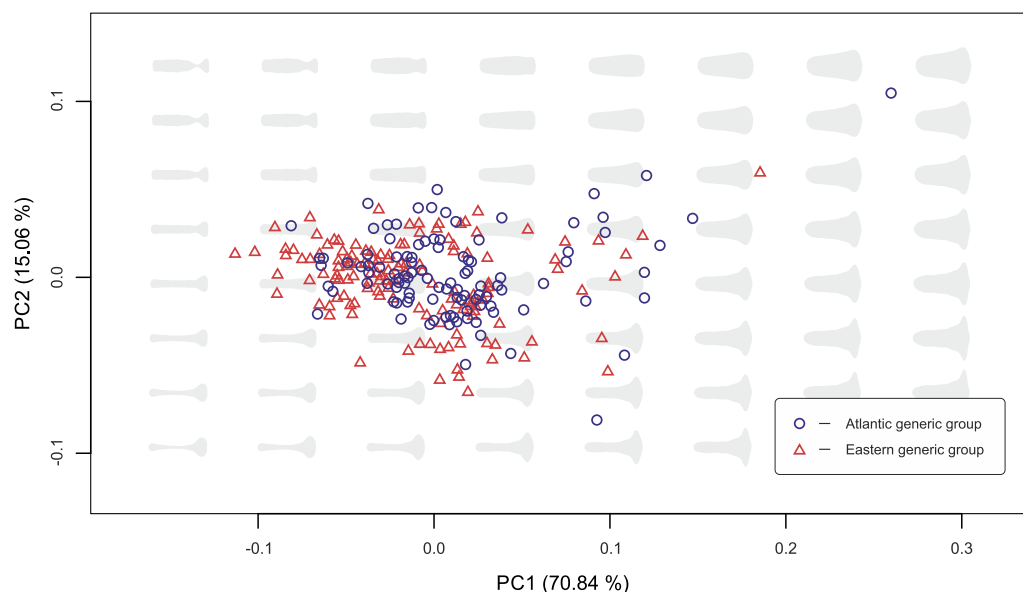


Fig. 5. Principal Component Analysis. Projection of the 247 items in a PC2 vs PC1 morphospace. The first two principal components together account for almost 86% of the total variance. Grey shapes, reconstructed by the inverse Fourier transform, represent 64 virtual axes.

groups, which is why a multivariate analysis of variance (MANOVA) was computed on the Fourier coefficients. The difference between the two generic groups, Atlantic and Eastern, is highly significant ($p < 10^{-15}$). A significant percentage of specimens are *a posteriori* well classified, when linear discriminant analyses (LDA) are applied, supplemented by either leave-one-out (85%) or two-fold cross-validation (88%). Provided that both generic groups, previously determined by archaeologists, are to some extent validated in terms of statistics, it could be tempting to cease our investigations here. Finer structuration of axe shapes was nonetheless explored, using self-organizing maps, SOMs (Fig. 6), as this procedure may allow local features to be better recognised than by PCA (Wehrens, 2011). The SOM is broadly coherent with the results of the MANOVA and LDA, as there is no major overlap between Atlantic and Eastern axes (Fig. 6a). However, the Atlantic axes are projected on two opposite regions of the SOM, separated by

the Eastern group, which does not fit the original interpretation, with only two groups.

3.2. Exploring a new typological system

The evidence from the SOM clearly suggests the possible existence of finer structuration within the two existing generic groups, at least for the Atlantic corpus. Identifying this internal structure deserves more attention, and this hypothesis was therefore explored with Gaussian mixture models. A VEM model (varying volumes, equal shapes, and identity orientation of clusters) was selected by applying a Δ BIC procedure. It appears that, after 6 clusters, there is no notable improvement in terms of information (Supplementary materials S3b), so that further modelling was performed using this value. For each flanged axe, the probability of belonging to each of the six clusters formed was also

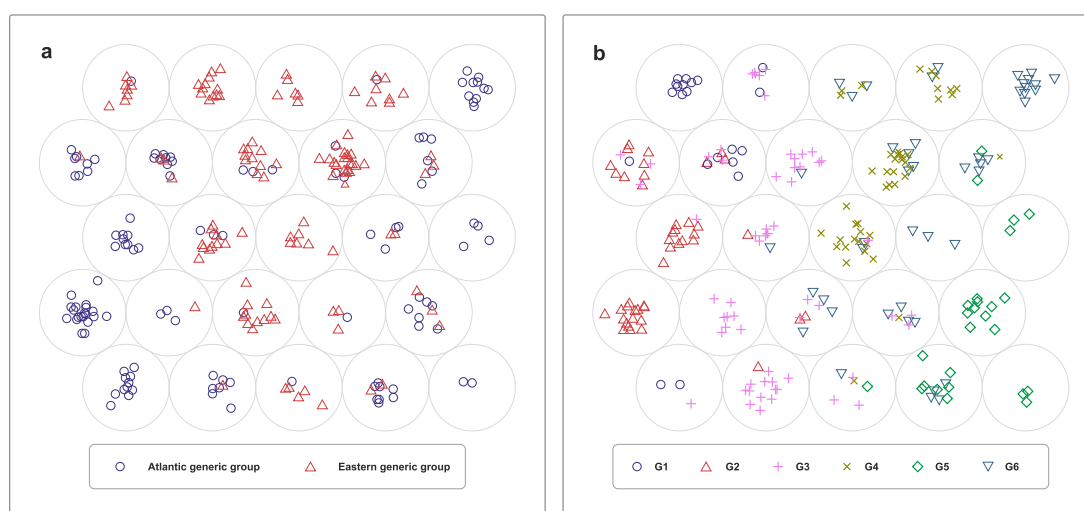


Fig. 6. Projection of 247 flanged axes on a self-organizing map. Artefacts belong to a) the Atlantic generic group (blue dots) or the Eastern generic group (red triangles); and b) one of the six groups obtained by unsupervised model-based clustering. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

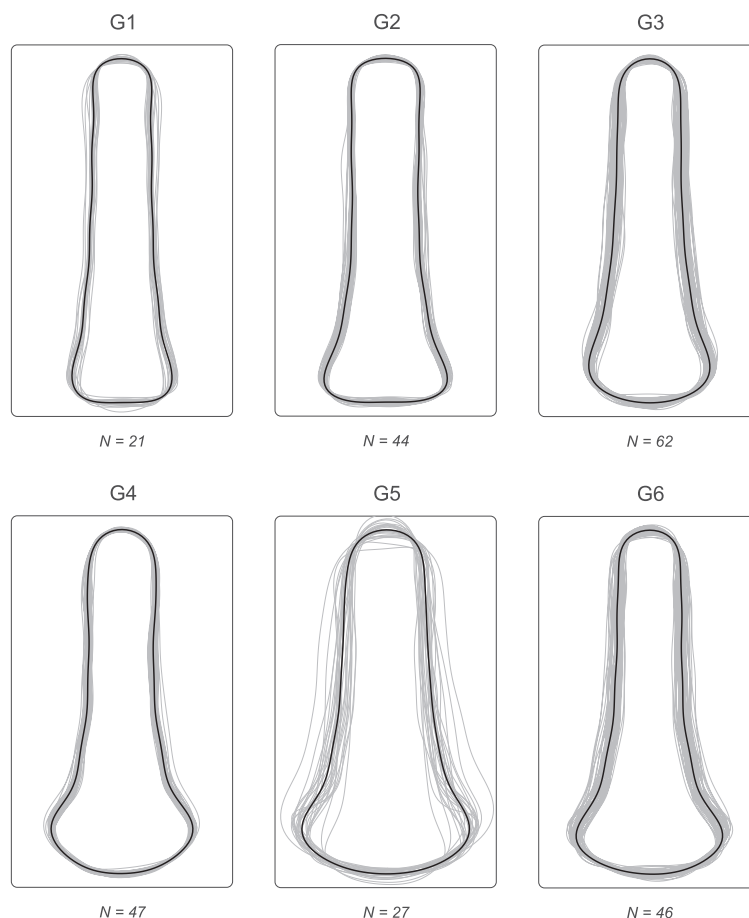


Fig. 7. Visual representation of the six groups obtained by model-based clustering. Black lines represent the mean shapes, while grey lines correspond to all group members.

computed. If the clusters are properly defined, the probabilistic attribution of items to groups should not be ambiguous. As a result, the maximum membership probability should be close to 1. Out of a total of 247 items, 224 have a maximum probability above 0.99, with 240 achieving a probability of 0.90 (Supplementary materials S3a). Such results suggest that the six groups constructed do not strongly overlap in terms of shape. This result is confirmed by a model-based DA, with leave-one-out cross-validation, in which none of the specimens was erroneously classified. Interestingly, with six groups, the SOM also depicts clearly separated items (Fig. 6b). To reinforce cluster validation, two complementary analyses were performed for each group: (i) artefacts were plotted on the same graph, together with the mean shape, resulting in clearly visible differences between the six groups (Fig. 7), and (ii) the location where each item was found was mapped, to investigate possible geographical clusters (Fig. 8). The multinomial scan statistics show non-random spatial distribution. Four circular zones (clusters) were detected, in which the distribution of items (in terms of representation of groups) was significantly different from the rest of the map ($p < 0.001$). Zone A corresponds to a greater relative abundance of the flanged axes belonging to G1 and G2, zone B has a greater number of axes from G3, zone C contains more axes from G3 and G4, while D, the final zone, encompasses most of the items from G5 and G6 (Fig. 8, Table 1).

The statistical analysis therefore confirms the existence of two traditions of flanged axes, corresponding to the Atlantic and Eastern generic groups mentioned above. Nevertheless, within this classification, a more complex internal organisation was discovered: the Atlantic axes

can be subdivided into four groups (G1, G2, G5, and G6), while the Eastern axes form two groups (G3 and G4). This result explains why the initial linear discriminant analysis was able to differentiate between Atlantic and Eastern axes so efficiently.

The first two groups (G1 and G2) are composed of similar axes, characterised by a thin, rectangular body, and a rather linear cutting edge; both forms, but more particularly G2, are very stable in terms of shape (Fig. 7). They are distributed almost exclusively in the western part of France, close to the Atlantic coast (Fig. 8, cf zone A). Given the shape homogeneity of G2, the statistical analysis tends to consider these axes as belonging to a separate group, rather than attributing them to G1. Most of these axes could result from a homogeneous unit of production, explaining why they exhibit such low shape diversity. The presence of G2 might also result from the structure of the corpus taken into account, since almost all artefacts belonging to group G2 (36 out of 44) were discovered at a single location. It is worth noting that zone A corresponds to cultural entities already recognised in central western France (i.e. the Duffaits and Vindo-Médocain groups; Gomez de Soto, 1995). The vast majority of the axes belonging to G1 and G2 were previously described as narrow-blade flanged axes (Table 2).

The axes from groups G5 and G6 exhibit the largest cutting edges and are the most heterogeneous in terms of shape (Fig. 7). They are concentrated in north-western France (more particularly in the Seine valley – zone D), and to some extent in Switzerland and southern Germany. Spatially, almost no items from G5 or G6 were discovered in zone A, while no artefacts from G1 or G2 were found in zone D

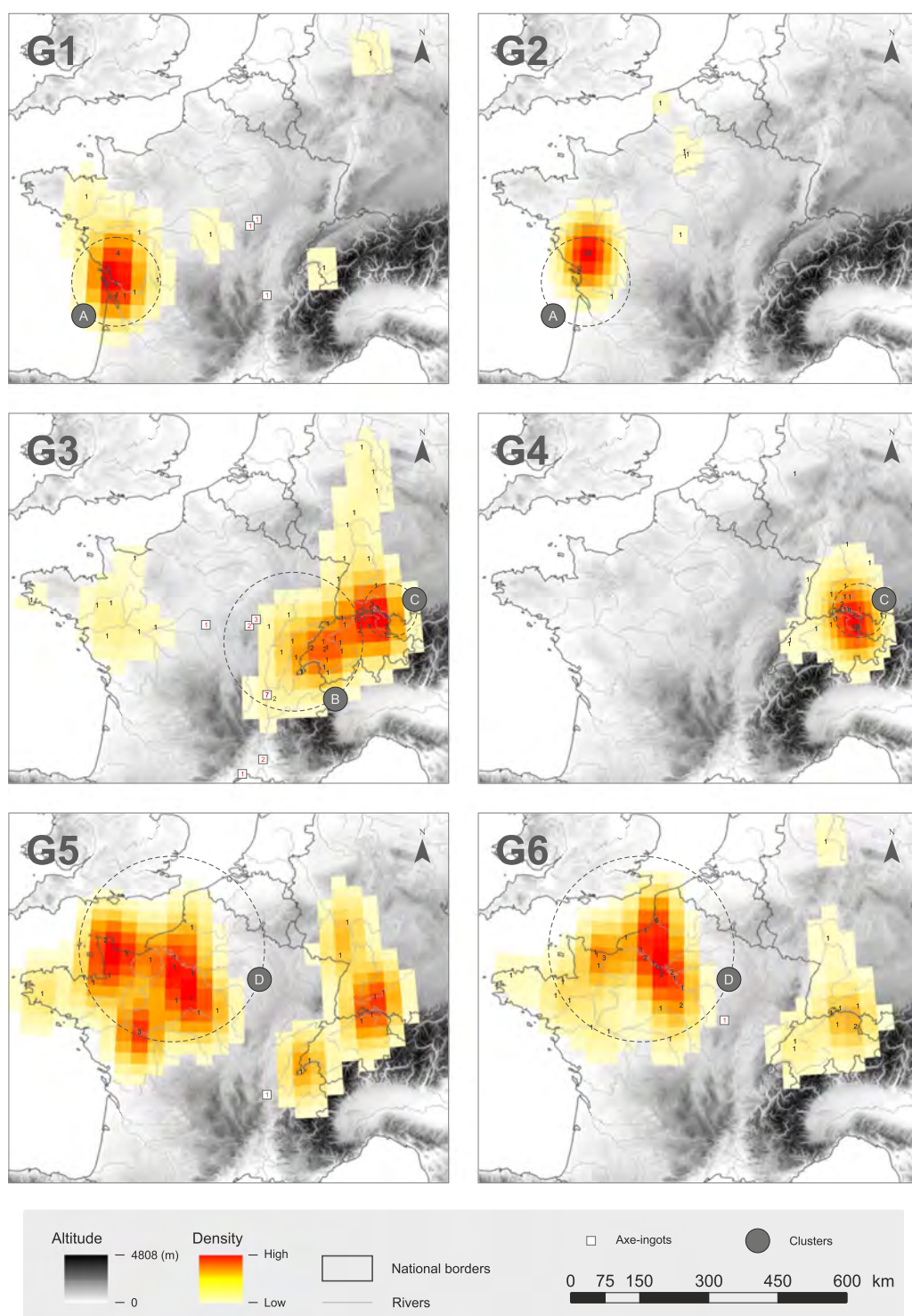


Fig. 8. Kernel density maps for the six groups of flanged axes obtained by model-based clustering. Numbers correspond to the number of axes found in one location, while dashed circles represent the clusters (A–D) provided by scan statistics (purely spatial scan statistics under a multinomial probability model, circle scanning and Monte Carlo randomisation with 999 permutations). Red numbers in squares represent the number of axe-ingots attributed by probability to one of the six clusters. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1

Concordance between model-based clustering and clustering performed by multinomial scan statistics. Bold font highlights the concordance between both methods.

		Model-based groups					
		G1	G2	G3	G4	G5	G6
Zones (Multinomial clusters)	A	15	39	0	0	0	0
	B	1	0	29	3	2	2
	C	0	0	16	38	3	4
	D	1	4	4	0	18	33
	Not associated	4	1	13	6	4	7
Total		21	44	62	47	27	46

(Fig. 8). Morphologically, groups G1–2 and G5–6 occupy very distinct regions in the SOM (Fig. 6b). The distinction between these two zones (A and D) is therefore quite clear, suggesting that consumers in each of these regions followed their own traditions. This pattern does not perfectly fit with the previous idea of a rather homogeneous Bronze Age “Atlantic world” (Briard, 1965; Coffyn, 1985; Brun, 1991), encompassing all European regions from the Atlantic coast to the North Sea. In fact, the presence of two distinct areas: (i) zone A, with flanged axes belonging to G1 and G2, and (ii) zone D, with axes from G5 and G6, tends to confirm a finer structuration of geographical space, already evoked by Butler (1963), and supported by the study of all metallic Middle Bronze Age artefacts discovered in north-western France (Gabillot, 2003). Note that zone D also corresponds to an area presenting a high density of contemporary Norman palstaves (Forel et al., 2009; Monna et al., 2013), suggesting the existence of a specific cultural area (Gabillot, 2006; Monna et al., 2013). Comparison with existing typologies reveals that most of the artefacts from G5 and G6 are members of concave-blade flanged axes (Table 2).

The axes in groups G3 and G4 are the most abundant. Their summits, flanks and cutting edges are more rounded than those in G1 and G2; this is especially true of the G4 axes (Fig. 7). Most of these artefacts were found in an area approximately corresponding to Switzerland: over the entire territory for G3 (zone B and C), and almost exclusively in the north-eastern part for G4 (zone C). Some items attributed to G3 can also be found in what is now Germany and in the western part of France (Fig. 8). These axes (G3) may indicate long-distance contact between cultural entities, but could also be viewed as noise in the data. Table 2 shows that axes from G3 correspond to Neyruz and Salez types. Axes in G4 present good shape stability, and their distribution is limited to zone C. All artefacts from G4 belong exclusively to the Salez type. Moreover, 19 artefacts out of the 25 analysed from the eponymous site of Salez (Switzerland) are members of G4. Bronze Age communities occupying this geographical area therefore used a well-defined type of axe (G4), which is statistically different from other types, at least in shape. As G4 axes have not been discovered elsewhere (except for some rare occurrences), it is reasonable to assume that they were not massively exported.

Using the discriminant analysis based on Gaussian finite mixture modelling, each of the 21 ingots can be associated with one of the six previously defined groups. Most of the axe-ingots were attributed to G3, but two artefacts were classed as G1, while the remainder were associated to the most heterogeneous group, G5. This result is not

surprising, as most of the axe-ingots were found close to zone B, where the flanged G3 axes were the most abundant (Fig. 8).

The low shape disparity within groups of flanged axes suggests a relatively well-established production pattern. Metalworkers produced axes following set technological rules of fabrication, adapted to the social and cultural standards of Bronze Age communities (Monna et al., 2013; Gabillot et al., in press). It is tempting to think that the zones with the greatest number of axes represent the areas of indigenous production. However, at least for the corpus studied here, it cannot formally be decided whether artefacts were produced (i) in one particular place and then transported to different locations, or (ii) in different places, by different metalworkers, all following the same shape and style (Orton, 1980). Except for a few examples (e.g. in France, Cabrières, Ambert and Barge-Mahieu, 1991; Saint-Veran, Barge, 2005), workshops and mines providing raw materials have not yet been identified in the archaeological record, so that production and distribution processes are still debated. In any case, if a strong spatial structure can still be recognised today, it is because exchanges at the scale of the study area were likely to be limited, as otherwise the distribution of artefacts would have been more homogeneous.

4. Conclusion

The current classification for Atlantic and Eastern flanged axes, unsystematically mixing several criteria (overall shape, supposed area of distribution, presence of decoration, etc.) was statistically validated on the outer shape of the flanged axes alone. However, the internal structure of these two groups can be much better identified by using a combination of modern computational methods. Unlike the traditional approach, the proposed classification is statistically established from unsupervised, model-based clustering analyses of morphometric data, and then validated independently by close examination of the spatial patterns. There is no need to know a priori the proper classification model, nor the number of groups. Artefacts of unknown membership can be classified, as was the case here for axe-ingots. This approach is quick, reproducible, operator-independent, easy to implement (as the tools used are freely available), and straightforward to adapt for almost any type of object where classification on the basis of shape is sought.

Unsupervised classification methods nevertheless have several limitations. Clustering basically aims to attribute all artefacts to an optimal number of groups. The issue is that what is considered as optimal today does not necessarily reflect the reality of the field in the past. Model-based clustering approaches need to have a sufficient number of items in order to identify individual groups correctly. In archaeology, where well-preserved artefacts may be scarce, some groups may have existed, but their representatives may not have survived in sufficient quantities. Model-based clustering will identify these items as extremes, but without defining a proper group for them, whereas a skilled archaeologist might be able to individualise them, and to treat them separately. Nonetheless, identifying them as extremes may allow a shift in focus for those specific artefacts.

Here, the optimal number of groups was determined using ΔBIC : i.e. $n = 6$ groups. It is worth mentioning that the operator must pay great attention to this parameter, because, with an inappropriate number of groups, classification may produce meaningless output. For example, in our case, $n = 5$ could reasonably have been considered (Supplementary materials S3), but the output would not have been as finely differentiated, because the axes classified as G5 and G6 would simply have been merged together. Note that if the n was lower, say 4 or 3, the observed spatial patterns might have not been identified. In general, increasing n above the optimal value is therefore not as unfavourable as underestimating n . In any case, the determination of meaningful groups in the dataset is a problem shared by all classification techniques (e.g., Orton, 1980; Legendre and Legendre, 1998).

The methodology proposed here is unequivocally statistical: the more data available, the better the output, meaning that this method

Table 2

Correspondence between unsupervised typology and current archaeological classification. Bold font highlights the correspondence between both methods.

		Model-based groups					
		G1	G2	G3	G4	G5	G6
Archaeological classification	Narrow-blade flanged axes	17	41	4	0	6	4
	Concave-blade flanged axes	2	3	4	0	13	32
	Neyruz type	1	0	29	0	3	2
	Salez type	1	0	25	47	5	8
Total		21	44	62	47	27	46

is not suitable for too small a corpus. Artefact assignment is probabilistic, so that it does not formally answer the question of strict attribution to a single type, except when the highest probability is considered. When the classification is based on this highest probability, subsequent analyses take for granted this classification and therefore no longer consider the level of uncertainty associated with it. It should be mentioned that the archaeologist experimenting typological attributions based on visual observation alone, faces the same restrictions, but unfortunately without any probabilistic control.

The groups defined by this approach cannot be thought of as definitive. Any new artefact added to the dataset will marginally modify the output. By its very nature, the proposed model for creating and validating the classification is one stage in a never-ending process. Any new artefact specimen or other observed descriptor for the artefact (e.g. decoration, chemical composition, or fabrication process) can test the stability of the proposed model and may thus contribute to its improvement.

It is noteworthy that the morphometric approach proposed here takes into account only the shape of the external outline of the axe. Although this feature was probably of some significance in ancient societies, as indicated by the spatial patterning found here, modern acquisition techniques allow researchers to develop better integrated descriptions of artefact shape variation, using 3D models, and not just focusing on the a priori more informative side-view of the artefact. Such features can be used and treated in similar ways.

The combination of this approach, applied to several types of objects, with other sources of information might contribute to a better understanding of the material, social and cultural relationships in ancient populations.

Supplementary data for this article can be found online at <http://dx.doi.org/10.1016/j.jasrep.2015.06.030>.

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Chapter 7

Brooches in motion: tracing shape evolution in La Tène brooches by geometric morphometrics

Until now, applications of geometric morphometrics in archaeology have focused almost exclusively on creating shape typologies or testing shape differences between several classes. However, archaeological investigations rely to a great extent on a single variable that is continuous in nature, time.

The main goal of this chapter is to test and visualise the effect of time on shape, using geometric morphometrics, on a set of 248 La Tène brooches from Central Europe.

Abstract

Brooches are among the most suitable archaeological artefacts for the study of La Tène chronology in Central Europe. The goal of this paper is to demonstrate how landmark-based analysis of shape data can be used to trace, test and illustrate the continuous evolution of brooch shape over time. The established chrono-morpho-typological classification of La Tène brooches is statistically validated. Two computer-based chrono-morpho-typological classifications are also validated. Shape differences between the two main types of LT B1 brooches (Duchcov and Münsingen) from two different regions (Jenišov Újezd and Moravia, Czech Republic) are statistically tested, confirming some regional differentiation.

Keywords: archaeology; classification; chronology; La Tène; Iron Age; brooches; geometric morphometrics; General Procrustes Analysis; Multivariate Regression

1. Introduction

Archaeology is a science largely based on data collection, organisation, and classification. Archaeological artefacts have often been classified by shape, style, decoration, or composition. The theoretical and practical aspects of archaeological taxonomy were developed in the 1960s and 1970s (e.g. [Kendall, 1963, 1971](#); [Hodson et al., 1966, 1971, 1978](#); [Hodson, 1968, 1978](#)). This taxonomy has served to organise and structure the ever-increasing quantity of data that bears witness to the succession of civilisations that have occupied most regions of the Earth. To cope with this wealth of data, computer-driven statistics (i.e. Seriation, Cluster Analysis, Principal Component or Coordinate Analysis, Multidimensional Scaling, etc.; for a general overview, see e.g. [Orton, 1980](#); [Djindjian, 1991](#); [Baxter, 2003](#); [Barcelo and Bogdanovic, 2015](#)) are being used ever more frequently in archaeology.

Statistical methods are often applied to categorical and quantitative descriptors. One of the most quintessential descriptors of an artefact is its shape, expressed either in terms of category (i.e. rounded, triangular, etc.), or as a series of measurements. Morphometrics is the quantitative analysis of form, which is a combination of size and shape (e.g. [Bookstein, 1991](#); [Lestrel, 2000](#); [Zelditch et al., 2004](#)). Geometric morphometric methods present several advantages over conventional morphometrics, especially (i) the extraction of shape information, which is always independent of size, (ii) better identification of the site of shape variability, and (iii) alleviation of the drawbacks related to auto-correlation between measurements (when measurements have the same starting point; e.g. [Zelditch et al., 2004](#); [Urbanová and Králík, 2009](#)). Shape alteration in artefacts, which can be caused by any parameter, whether discrete (e.g. type, region, or gender attribution), or continuous (e.g. size or time), can easily be statistically tested, illustrated, and interpreted (e.g. [Loy et al., 1996](#); [Rohlf et al., 1996](#); [Drake and Klingenberg, 2008](#)).

Brooches have provided one of the most precious sources of archaeological information, expressing wealth, social status, and regional or personal identity. They are also vectors of human relations, exchanges and technological developments. Brooches and other artefacts found together in archaeological assemblages have proved vital in reconstructing La Tène chronology (LT; circa 450–1 BC), subdivided into several discrete phases (LT A – LT D).

Although the chrono-morpho-typology of La Tène brooches is now well known (e.g. [Viollier, 1916](#); [Pittioni, 1928, 1930](#); [Filip, 1956](#); [Krämer, 1961, 1962](#); [Uenze, 1964](#); [Hodson, 1968](#); [Gebhard, 1991](#); [Bujna, 2003](#)), further morphometric analysis may still bring to light new insights.

The first goal of this study is to confirm that the general chrono-morpho-typology reflects shape changes over time, potentially revealing stages in design innovation. The corpus used in this study therefore presents the entire spectrum of brooch shapes that existed during this period (LT A – LT C2; circa 450–120 BC). Brooches belonging to LT D (circa 120–1 BC, e.g. Nauheim and spoon types) were not included in the study because their method of fabrication was very different.

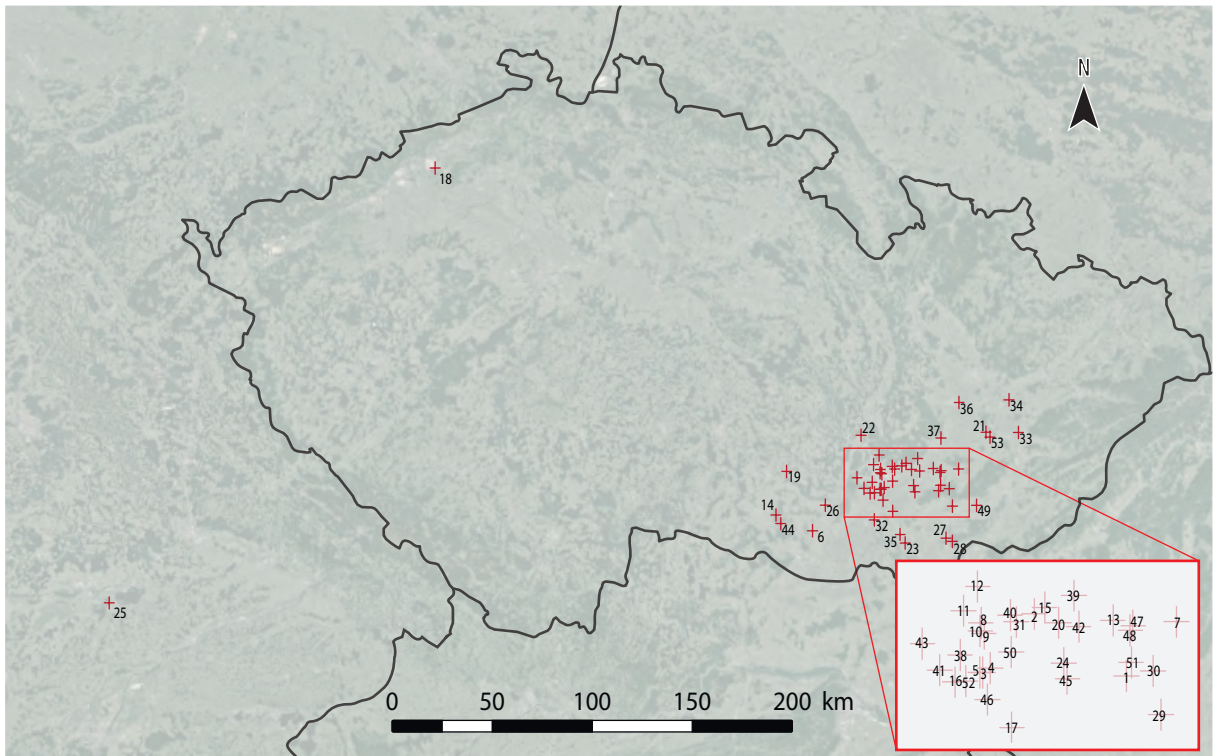


Fig. 1. Spatial distribution of brooches used in the study. Numbers correspond to site enumeration in [Supplementary materials S1](#).

The second goal is to provide visual representation of continual progressive changes in La Tène brooches over time, using Multivariate Regression techniques ([Drake and Klingenberg, 2008](#); [Klingenberg, 2013](#)).

The third goal is to illustrate these general shape changes in a specific subset of La Tène brooches from tombs in Moravia and Jenišův Újezd, and to test whether these changes are statistically significant.

The fourth goal is to test shape differences between two well-defined types of brooches from LT B1 (Duchcov and Münsingen), from two different geographic areas (Moravia and Jenišův Újezd), with multivariate statistical techniques.

Landmark-based Procrustes Analysis was used throughout the study, for all four investigations.

2. Materials and methods

2.1. Corpus

Information about La Tène brooches was collected from the available literature. Only frequently occurring types and previously classified brooches without any deformation were retained. The final corpus consists of 133 brooches, from 85 Moravian assemblages (Czech Republic), supplemented by 82 brooches from the cemetery of Jenišův Újezd (Bohemia, Czech Republic), and 33 brooches from the Manching oppidum (Bavaria, Germany; see [Fig. 1](#) for spatial distribution, and [Supplementary materials S1](#) for a detailed catalogue of brooches and [S2](#) for references used).

Precise typological description was not considered necessary for this study, so only the following broad chronological classes were retained: LT A, LT B1, LT B2, LT C1, and LT C2 ([Fig. 2](#); see [Table 1](#) for further details). However, for the fourth investigation, brooches from LT B1 were subdivided into two well-known types, Duchcov and Münsingen.

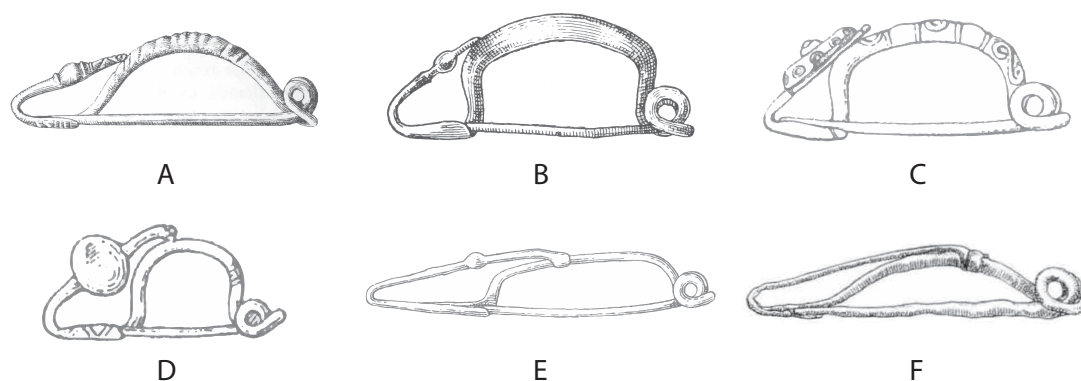


Fig. 2. Examples of typical La Tène brooches from Central Europe. A) Brooch with parabolic bow (LT A). B) Duchcov type (LT B1). C) Münsingen type (LT B1). D) Brooch with “ball” on foot. E) La Tène II type brooch. F) Mötschwill type. Sources of images: Čižmářová, 2005, Fig. 61:1, 2009, Fig. 40:1; 2013, Fig. 38:3; Gebhard, Fig. 24:372; Ludíkovský, 1964, Fig. 6:4; Meduna, 1965, p. 226, Fig. 4.

Chronology	Regions				Nomenclature (types and variants)		
	Jenišov Újezd (Bohemia)	51 funerary areas (Moravia)	Manching (Bavaria)	Total	Waldhauser et al. (1978)	Moravia	Gebhard (1991)
LT A	2	18	0	20	F 1100-1199	Spona se střežovitým/parabolickým lucíkem Spona s ptačí hlavičkou	-
LT B1	31	24	0	101	F1200-1399 (Duchcov)	Duchcovská spona	-
	6	40	0		F 1400-1499 (Münsingen)	Münsingenská spona	-
LT B2	22	28	0	50	F 2110-2510	Spona s (velkou/malou) kuličkou na patce	-
LT C1	21	17	5	43	F 3000-3800	Spona spojené konstrukce (/s kuličkou na patce)	G 13b, G13c
LT C2	0	6	28	34	-	Spona spojené konstrukce Mötschwilská spona	1a, 1b, 16, 17a, 17n, 18a, 18b, 20

Table 1. Brooches used in the study.

Chronology for the brooches found in tombs in Moravia and Jenišov Újezd (LT A – LT C1) was expressed as a discrete quantitative variable, based on relative chronological sequences. The detailed chronology of brooches from Moravia was determined based on their position in the chronological table established by Čižmář (1975). Three different seriations were used for the brooches from Jenišov Újezd: the human-based seriation produced by Waldhauser (1978), and two computational seriations produced at the same time (Hodson, 1978).

2.2. Morphometric treatment

2.2.1. Data acquisition

Procrustes Analysis requires landmarks to be identified on the objects to be analysed. In this study, illustrations in the literature depicting the lateral view of brooches were scanned for analysis. Of the three types of landmarks (Bookstein, 1991), the only ones that can be identified on brooches are Type 2 (extremities) and Type 3 (geometric constructs, e.g. angles, centres, or tangents). Fifteen landmarks were manually positioned on each brooch image (Fig. 3): the first one (L1) marks the point where the bow meets the head (spring), while the second (L2) is placed

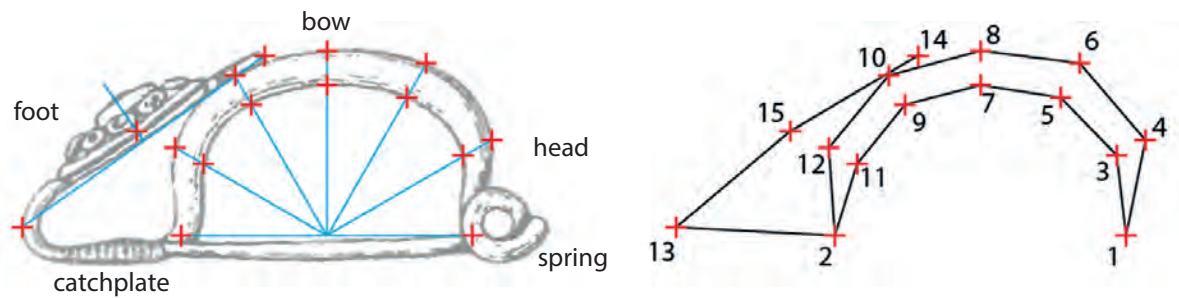


Fig. 3. Parts of brooches and landmark positions.

where the bow meets the foot (catchplate). Between these two points, ten other landmarks (L3 to L12) were placed on the bow, at 30° angles. The remaining three landmarks were placed on the foot, at each extremity (L13, L14) and in the centre (L15).

2.2.2. Procrustes Analysis

Morphometric data were processed by Generalised Procrustes Analysis (GPA; [Gower, 1975](#)), where the goal is to extract information about shape, independent of position, size, and orientation. The position effect is removed by translating centroids of all configurations (i.e. sets of landmarks) to the point of origin of the coordinate system. The size effect is removed by scaling all configurations to the same unit size ([Gower, 1971](#)), i.e. by dividing coordinates by the centroid size corresponding to the square root of the sum of squared distances from each landmark to the centroid of the configuration ([Bookstein, 1986](#); [Goodall, 1991](#); [Dryden and Mardia, 1998](#)). Finally, the orientation effect is removed by rotating configurations so that the summed squared distances between corresponding landmarks of all configurations are minimal ([Gower, 1975](#); [Rohlf, 1990](#); [Rohlf and Slice, 1990](#), [Slice, 2005](#)). In GPA, the dimensionality of the shape space is reduced by 4 degrees of freedom (2 for position, and 1 each for size and rotation).

2.3. Statistical Treatment

Shape space dimensionality can be reduced by applying Principal Component Analysis (PCA; [Jolliffe, 2002](#)) to the standardised Procrustes data. The advantage of PCA is that it provides a multidimensional space (or morphospace) where individual objects can easily be visualised. The centre of this space occupies the mean shape (Procrustes mean), calculated by averaging all superimposed coordinates of landmarks treated by GPA. Around this mean shape, all objects are distributed according to shape similarity. Based on Principal Component scores, the original shape of artefacts can be reconstructed.

Differences between the chrono-morpho-typological classes (i.e. LT A, LT B, etc.) and between geographic areas (Moravia and Jenišův Újezd) were tested by Linear Discriminant Analysis (LDA) and Multidimensional Analysis of Variance (MANOVA). The relationship between shape and chronology was examined by Multivariate Regression of the Procrustes coordinates ([Loy et al., 1998](#); [Monteiro, 1999](#); [Drake and Klingenberg, 2008](#)). The statistical significance of this relationship was calculated by permutation ([Good, 2010](#)). The regression procedure was used to trace and visualise trends related to shape changes, associated with time effects ([Klingenberg, 2013](#)).

The R statistical software ([R Core Team, 2017](#)) was used for all morphometric and statistical analyses, and visualisation, with 'MASS' ([Ripley et al., 2017](#)), 'rgl' ([Murdoch, 2017](#)) and 'geomorph' ([Adams and Otárola-Castillo, 2013](#)) packages, complemented by other functions ([Claude, 2008](#)).

3. Results

3.1. Investigation of chronology in the morphospace of all brooches

The chrono-morpho-typological classification was checked by projecting all brooches into a PCA-based morphospace (Fig. 4:A). The first three components accounted for 88.1% of the total variance. The first Principal Component (PC1) corresponds to foot elongation and bow flattening; the second (PC2) corresponds to foot reduction and bow elongation; the third (PC3) corresponds to catchplate elongation and bow reduction (Fig. 4:B).

All brooches within the morphospace seem to be well separated by chrono-typology (Fig. 4:A). In the first two PCA dimensions, the oldest brooches (LT A, LT B1) are in the upper left section, brooches from LT B2 are in the lower central section, and the most recent brooches (LT C2) are in the upper right section. Brooches from LT C1 express the greatest shape disparity and are less closely clustered. Overlapping brooches could represent transitional stages between consecutive archaeological phases.

This chrono-morpho-typological classification was statistically confirmed by both LDA and MANOVA ($p < 0.0001$), computed on shape variables. More than 90% of brooches are well classified by LDA, and 85% by LDA with cross-validation (Table 2).

		LDA Classification					LDA classification (cross-validation)				
		LT A	LT B1	LT B2	LT C1	LT C2	LT A	LT B1	LT B2	LT C1	LT C2
Archaeological classification	LT A	18	1		1		17	2		1	
	LT B1	3	94	3	1		6	90	3	2	
	LT B2		2	47	1			3	45	2	
	LT C1		3		38	2	1	3	1	34	4
	LT C2				7	27				8	26

Table 2. Concordance between the archaeological chrono-morpho-typology and LDA classifications.

3.2. Investigation of general changes in shape

The distribution of brooches grouped by their chronological attributes indicates the possible existence of a vector driven by time (Fig. 4). Its path was traced by a cubic spline (Hyman, 1983), obtained by the interpolation of points representing the centroids of the five chronological classes in the morphospace (Fig. 4:A). Based on this curve, a set of 100 shapes was reconstructed, representing the chrono-morpho-typological evolution of brooches (Fig. 5).

General changes in shape, potentially representing the inventiveness of ancient brooch manufacturers, can be characterised by landmark displacement and grid deformation. Such changes can easily be visualised as a series of frames (Fig. 5:B) or through animation, thus representing La Tène brooch shape evolution (see Supplementary materials S3). Graphics show that sharp parabolic-bow brooches with a small triangular foot (LT A) became gradually more oval to rectangular during LT B1, while the foot (mainly the catchplate) became longer. This bow rounding tendency, with additional thickening, continued in LT B2, until the bow became arched, while the size of the foot continued to increase. The transition to La Tène II type brooches (LT C1) was marked by the flattening of the whole shape of LT B2 brooches, until the foot finally became attached to the bow. The final transformation into LT C2 forms was marked by tilting and moving the centre of interest towards the head of the brooch, which became thicker and was more often decorated. This tendency can also be observed in brooches of La Tène D (e.g. Nauheim and spoon brooch types).

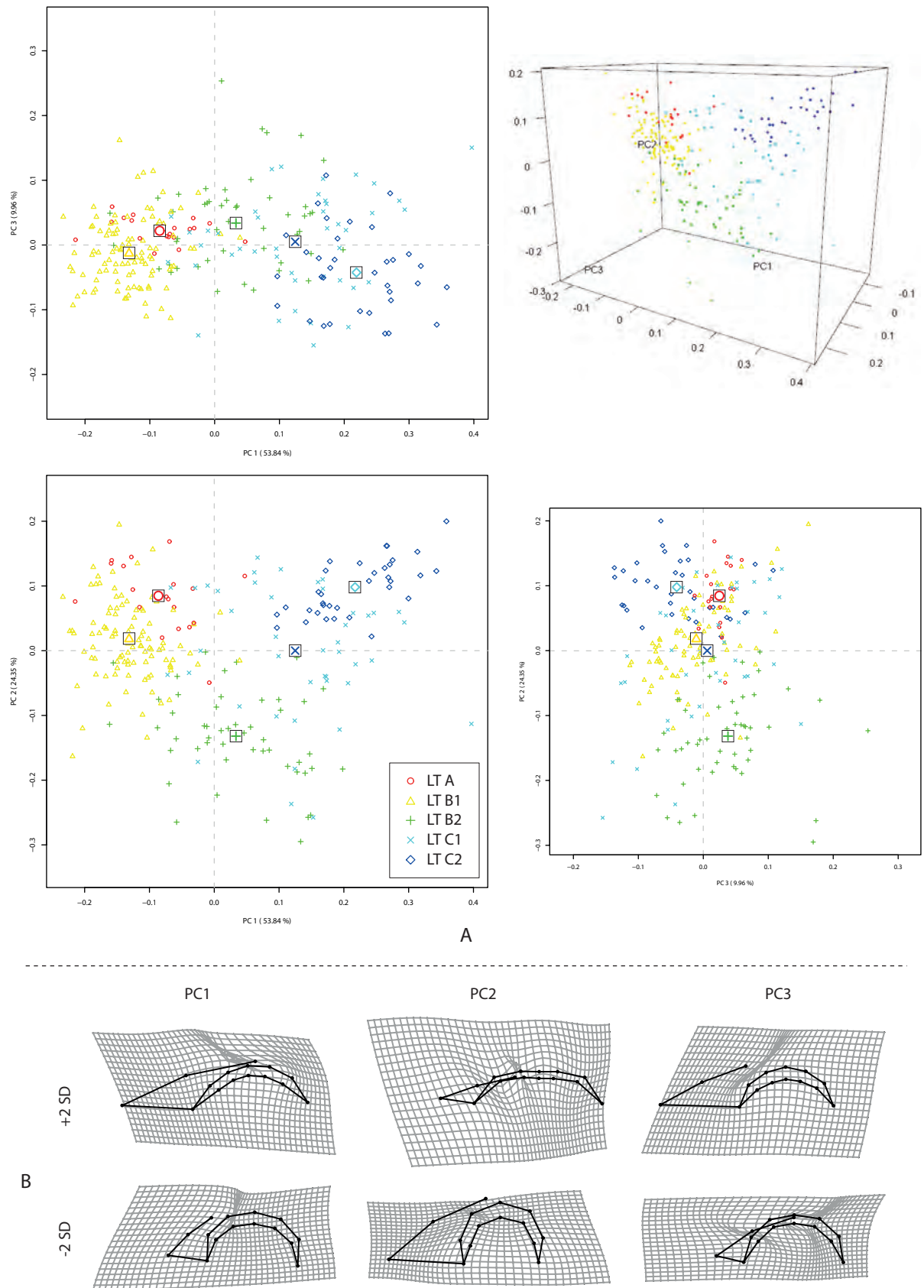


Fig. 4. Principal Component Analysis. A) Projection of 248 brooches into a PC3 vs PC2 vs PC1 morphospace. These three axes account for 89% of the total variance. Brooches are coloured according to typological classes, mean shapes are shown in squares. B) Visualisation of shape variation carried by first three Principal Components. Reconstructions are made with ± 0.2 standard deviation from the mean shape.

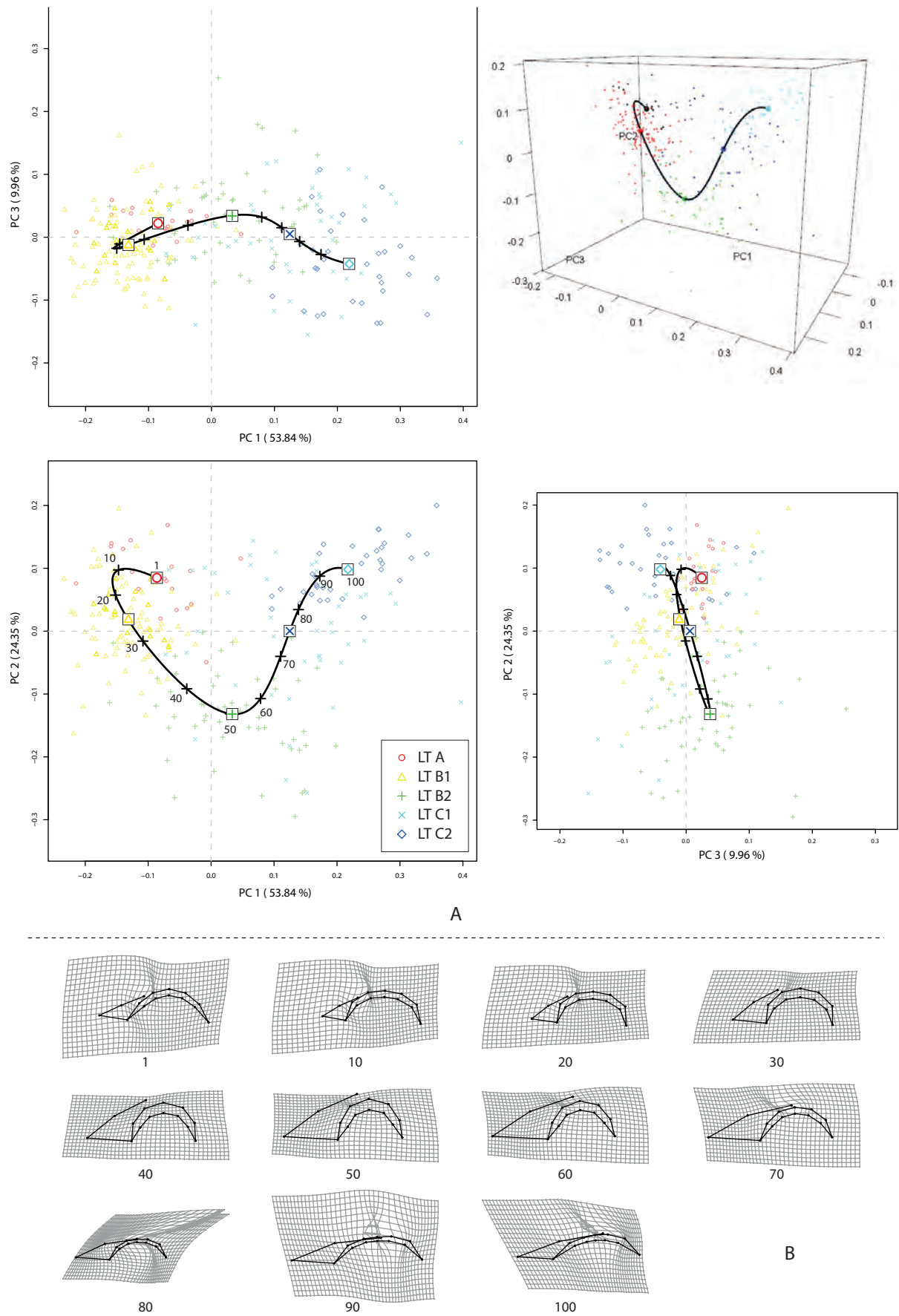


Fig. 5. Principal Component Analysis. A) Projection of 248 brooches into a PC3 vs PC2 vs PC1 morphospace. Black line represents continuous shape change vector, black crosses show reconstructed shapes. B) Reconstructed shapes (see [Supplementary materials S3](#) for animation)

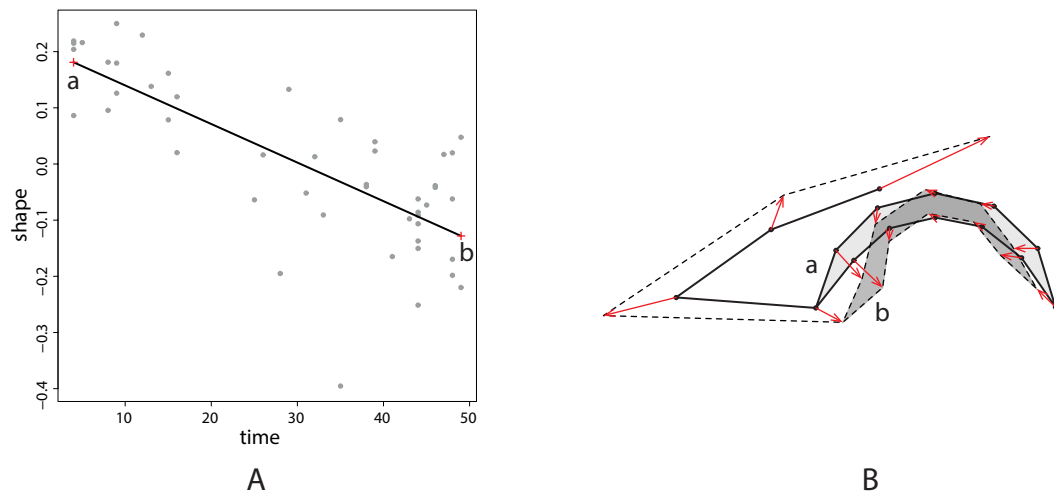


Fig. 6. Multivariate Regression of shape on time. A) Projection of the shape of 47 brooches (grey dots) on time. Time here corresponds to the position of the grave containing the brooch in the [Hodson \(1978\)](#) seriation table (Seriation 2). The black line represents the time factor. Red crosses show the reconstructed shape of the (a) “oldest”, and (b) “youngest” brooches. B) Superimposition of the “oldest” (a) and “youngest” (b) reconstructed brooches. Red lines with arrows correspond to shape changes over time (shape changes are shown twice as big as in reality). See [Supplementary materials S4](#) for an animated version of shape changes over time.

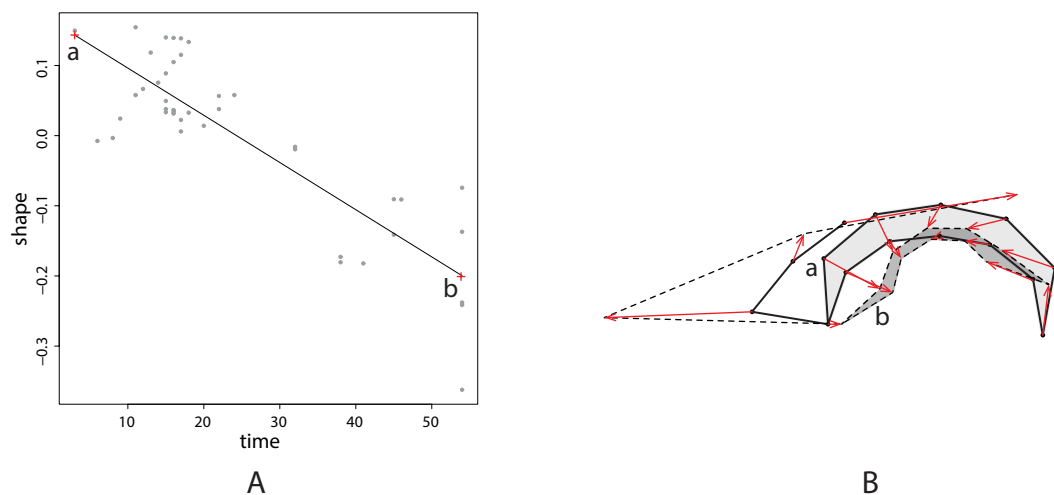


Fig. 7. Multivariate Regression of shape on time. A) Projection of the shape of 40 brooches (grey dots) on time. Time here corresponds to the position of the grave containing the brooch in the [Čižmář \(1975\)](#) seriation table. The black line represents the time factor. Red crosses show the reconstructed shape of the (a) “oldest”, and (b) “youngest” brooches. B) Superimposition of the “oldest” (a) and “youngest” (b) reconstructed brooches. Red lines with arrows correspond to shape changes over time (shape changes are shown twice as big as in reality). See [Supplementary materials S5](#) for an animated version of shape changes over time.

3.3. Investigation of the chronology of Jenišův Újezd and Moravia

The relationship between shape and time was further investigated on a subset of brooches used in chronological seriations for Moravian cemeteries and for the Jenišův Újezd cemetery. Time in the analyses was defined by the position of the grave containing the brooch in the seriations by Čižmář (Moravia), and by Waldhauser and by Hodson (Jenišův Újezd).

The graph of Multivariate Regression of shape over time shows that, in both regions, the time factor is related to foot lengthening and bow rounding (see [Fig. 6](#) for Moravia and [Fig. 7](#) for Jenišův Újezd).

Corpus	Classification		Df	SS	MS	Rsqr	F	Z	Pr (>F)
Jenišov Újezd	Waldhauser (1978)	Time	1	0.58137	0.58137	0.33085	22.744	10.901	0.002
		Residuals	46	1.17583	0.02556				
		Total	47	1.75720					
	Hodson (1978) Seriation 1	Time	1	0.49661	0.49661	0.26679	16.738	9.4248	0.002
		Residuals	46	1.36484	0.02967				
		Total	47	1.86145					
	Hodson (1978) Seriation 2	Time	1	0.58328	0.58328	0.31335	20.992	10.712	0.002
		Residuals	46	1.27817	0.02779				
		Total	47	1.86145					
Moravia	Čižmář (1975)	Time	1	0.44285	0.44285	0.42582	29.664	11.477	0.002
		Residuals	40	0.59715	0.01493				
		Total	41	1.03999					

Table 3. Results of Procrustes ANOVA on the relationship between shape and time for corpuses and classification schemes.

The Procrustes ANOVA (Klingenberg and McIntyre, 1998) showed that the relationship between time and shape is highly significant in all cases (Table 3). All four classifications, whether by specialists or by computer, are coherent and clearly demonstrate shape changes over time.

The relationship between shape and time in all classifications accounted for 44 to 58% of total shape variation. Residual shape variation can be attributed to other factors, such as the role of brooches in social stratification, regional costume differences, or differences in fabrication techniques.

3.4. Investigation of typological and regional differences in LT B1 brooches

The effect of typological and regional differences was tested on two contemporaneous and easily distinguished types of brooches (Duchcov and Münsingen) from two geographic regions (Moravia and Jenišův Újezd). The results of the LDA on this set of 101 brooches (Fig. 8) show that those from Moravia (green and blue symbols) differ from those from Jenišův Újezd (black and red symbols). The Duchcov type cannot be clearly distinguished from the Münsingen type. This observation was supported by the results of a Two-way non-parametric permutational MANOVA ($p < 0.203$). The geographic origin of brooches thus seems to have more impact than their type on differentiation. The analysis may have been biased by the fact that only 6 Münsingen brooches were found in Jenišův Újezd. However, additional tests computed solely on two types of brooches from Moravia confirmed that Duchcov and Münsingen types do not significantly differ in shape ($p < 0.399$).

Due to the small number of Münsingen brooches from Jenišův Újezd, detailed shape differences between brooches from the two regions were examined only for the Duchcov type. The Linear Discriminant model separated brooches from different zones, both statistically (95% and 73% of brooches were *a posteriori* well classified with LDA and LDA cross-validation) and visually (Fig. 9). The bows of brooches from Moravia are generally more robust and arched than those from Jenišův Újezd.

4. Discussion

In contrast with the complexity of established classifications, morphometric analysis based on several landmarks may seem simplistic. However, the advantage of morphometrics is that only one aspect of artefacts, i.e. shape, is investigated, allowing any other descriptors (e.g. decoration,

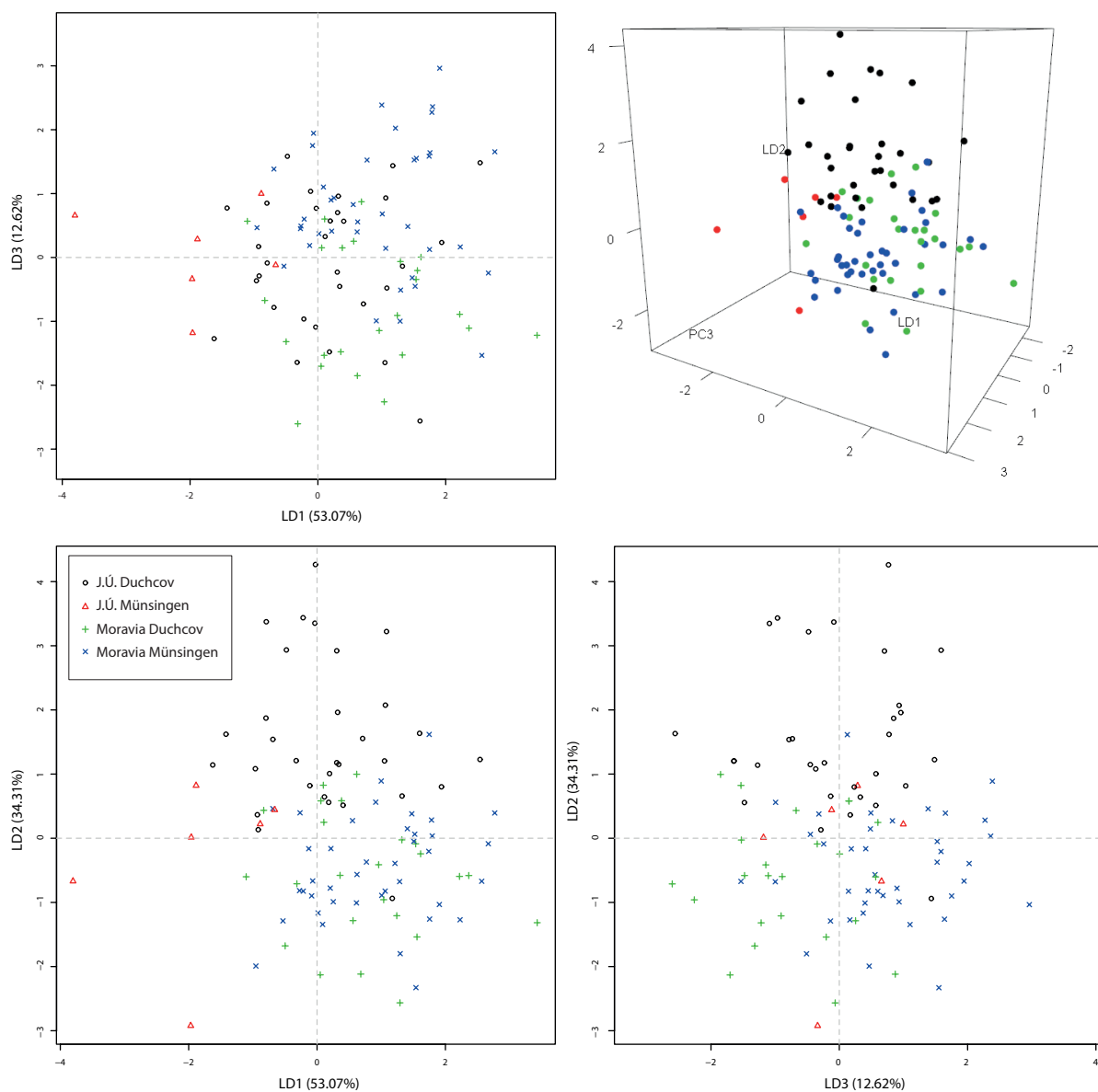


Fig. 8. Linear Discriminant Analysis. Projection of 101 brooches into the LD1 vs LD2 vs LD3 discriminant space.

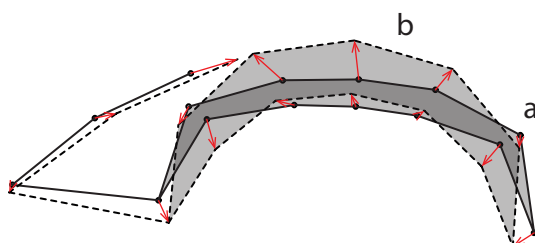


Fig. 9. Visualisation of differences between Duchcov brooches from (a) Jenišův Újezd and (b) cemeteries in Moravia.

chemical composition, *chaîne opératoire*, etc.) or factors of interest in archaeology (e.g. time, region, gender, social status, etc.) to be treated as independent variables. Morphometric shape variables are quantitative and continuous, allowing many inferential statistical techniques to be applied. From this point of view, morphometrics can bring archaeology closer to the natural and biological sciences, with all their methodological advantages. The effect of any factor on shape can easily be reconstructed and visualised, allowing complex concepts to be expressed in an easily understandable, clearly perceived form.

For the corpus in this study, morphometric analysis proved that brooches, classified in discrete chrono-morpho-typological classes, differ by their shape. Results confirmed that the two human-based and two computer-based funerary assemblage seriations all express the archaeological time factor satisfactorily. It can thus be concluded that intuitive morphological classifications, shared by several generations of archaeologists, can be as pertinent for dating brooches as computer-driven classifications.

This study demonstrates that archaeological time can be considered not only as a discrete quantitative variable (i.e. archaeological period or phase), but also as a continuous vector. This vector represents shape changes driven by brooches ordered by their presumed chronology. However, the curve passing through brooch mean shapes cannot be taken as a measure of physical time. It rather corresponds to the vector of artefact innovations by ancient brooch manufacturers, responding to the demands of La Tène society.

Shapes reconstructed on this vector visually highlight brooch parts that were objects of interest for modification over time. Interestingly, it seems that the arched brooches produced in LT B2 were no longer in favour in LT C1, and the design of the bow returned to shapes privileged in LT A. From this point of view, it can be expected that observing the shape evolution of artefacts over longer time sequences will reveal other periodic returns to a particular style or mode.

An ongoing problem in archaeology is the availability of sufficient data for statistical analysis. Shape differences between two types of brooches (Duchcov and Münsingen) from two regions (Moravia and Jenišův Újezd) could only be tested on a subset of 101 brooches from LT B1. Analysis revealed that geographic origin has more impact than type on brooch differentiation. The difficulty of type differentiation is coherent with the difficulty of classifying incomplete brooches (i.e. brooches without the characteristic knob for Duchcov, or the disc for Münsingen) observed in the literature. This result indicates that both types of brooches were constructed following the same basic procedures.

Although funerary practices and archaeological artefacts from Moravia should be culturally related to Bohemia in LT B1 ([Čížmář, 1993](#)), the results of morphometric analysis showed that Moravian Duchcov brooches differ from those found in Jenišův Újezd. It seems that the population of each region followed their own pattern of brooch production, possibly driven by factors related to regional dress and collective identity ([Adams, 2005](#)), but also because Jenišův Újezd is one of the furthest Bohemian sites from Moravia. Investigating whether the Duchcov brooches found in Moravia were produced under the influence of the Mid-Danube and Carpathian Basin, as proposed by [Ludikovský \(1964\)](#), or are related to neighbouring productions in Bohemia, could validate further morphometric inquiries into brooches over a wider geographic region.

Linear Discriminant models with relatively good classification power were obtained for each effect examined here. Based on these models, the chronology, typological class, and region of origin for any newly found brooch can be predicted with probabilistic control. If required, classification models can easily be improved by introducing new data, or by applying other statistical techniques suitable for the discrimination or classification of brooches.

5. Conclusion

The landmark-based GPA followed by Multivariate Regression techniques served to reconstruct, visualise, and interpret the general trends of shape variation and change in La Tène brooches from Central Europe.

Several multivariate analyses attested that both human- and computer-based chrono-morpho-typological classifications of brooches are coherent with shape change. Both methodological approaches thus seem to be reliable for brooch morphology investigation and classification. However, as specialist intuition is not easy to acquire or reproduce, and as the degree of certainty under which classes are created and artefacts are classified is difficult to control, the application of quantitative methods should be privileged as the most easily reproducible investigation technique.

The main contribution of geometric morphometrics to archaeology is that shape information about the artefact can be extracted, and treated as an independent variable on any dependent effects. Several tests applied here revealed no differences between the shape of Duchcov and Münsingen brooches, but attested to the regional differentiation of the Duchcov type. In the same way, any other descriptor or factor of interest in archaeological investigation can be tested on shape.

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Chapter 8

Conclusion

The aims of this doctoral project were to propose new methodological and technical solutions for the **acquisition**, **systematisation** and **interpretation** of archaeological artefacts, based on their **shape**. The acquisition process is discussed in the chapter dedicated to the DACORD program, while the systematisation and interpretation of archaeological artefacts based on shape are discussed in all the chapters.

Acquisition

The DACORD system was designed for data acquisition and its practical product is the illustration of archaeological pottery. The main advantage of this program is the time saved, which will also reduce costs, and allow more data to be processed. The workflows can be selected by the operator, and can be fully automated, thus improving production rates and reproducibility. From 3D model acquisition, through fragment orientation, to graphical illustration, the entire process takes no more than 10 minutes, whatever the complexity of the fragment. The system can handle all types of archaeological pottery, whether hand-made or wheel-turned, and can reproduce fragments from any part of the vessel (rim, body, and base), from any chronological period. The drawings produced by the DACORD system can serve to establish typologies, and can also be used for cataloguing and archiving data in digital and numerical formats, thus increasing the amount of shareable information available to the archaeological community. The great improvement that the DACORD system brings to archaeology is the increase in the availability, quality and quantity of valid information derived from the shape data recorded in the existing corpus of archaeological artefacts.

Although DACORD was originally designed to orient and draw archaeological pottery, additional tests (not presented here) have confirmed that the system can process an even wider range of archaeological artefacts, including lithics and Bronze Age axes.

Systematisation

Systematisation in this work was presented by proposing workflows to identify the best analogy for pottery fragments in referential databases, and by proposing the unsupervised classification of different types of archaeological artefacts, used here in two published case studies, on La Tène pottery vessels, and on Middle Bronze Age axes.

Three workflows were proposed for pottery fragment classification. Results confirmed that the best analogy for a given fragment can be identified in a referential database containing both existing vessels and potential vessel shapes, based on the overall shape variability of existing complete pottery vessels.

Results also showed that attribution and classification remain possible, even for small non-rim fragments. This ability to classify smaller fragments can considerably increase the quantity of data available for archaeological investigation. The importance for archaeology of small fragments, often discarded or at least only partially recorded, should now be reassessed, as the time factor has been considerably diminished. If the number of useful fragments available is substantially increased, the quantity of information will be multiplied, as even the tiny amount of information contained in an individual fragment will contribute to archaeological knowledge, when thousands more fragments have been processed. The challenge now is to reconstruct complete vessels from the new information thus acquired. By superimposing the alphashapes of the best candidates for a set of fragments, it is possible to identify the position, and therefore potentially the original shape of the complete vessel.

The question of which fragment is morphologically classifiable depends greatly on the shape variability of the complete vessels available in the literature. The focus of archaeological investigation should therefore seek to identify as many complete forms as possible, and to make this information available to the community.

The advantages and disadvantages of traditional and unsupervised shape-based classifications

were also discussed, based on two different corpuses of archaeological artefacts: pottery vessels from the Bibracte oppidum, and Middle Bronze Age axes from Western Europe. The established chrono-morpho-typology was also tested on set of La Tène brooches from Central Europe.

Results revealed that all traditional morpho-typologies examined in these case-studies are reliable, and coherent with classifications obtained by unsupervised clustering. Unsupervised geometric morphometric classifications can be used to classify archaeological artefacts as effectively as the intuitive morphological classifications that have been created, modified, and shared by several generations of archaeologists.

However, it was also shown that morphometric methods obtained results that could not be attained by classical approaches, such as visualising the mean shape, quantifying and visualising shape variations within a type, and above all quantifying the degree of certainty with which an artefact is attributed to a given class. The classifications proposed and used in these case-studies are reproducible, and can therefore allow objective inferences to be statistically tested.

The classifications proposed here are shape-based, but other descriptors can also be included in the classification process. However, as with all classification, the process is infinite and must be repeated as new information becomes available. Automating such a process presents obvious advantages, notably in the amount of time to be gained.

Archaeological interpretations

The general aims of this work were methodological rather than interpretative. However, several results obtained in the case-studies following the application of morphometric and statistical methods of analysis to archaeological data may serve in the interpretation of ancient social and cultural practices, and fabrication techniques.

In Chapter 2, the literature review indicates those areas where few studies have yet integrated geometric morphometric methods, and thus helps to identify potential research questions for the future.

Those reconstructed potential vessel shapes in the morphospace that were closest to existing complete vessels could well correspond to shapes technically within the scope of ancient manufacturers. Morphometrics may therefore contribute to the identification of some of the ideological and cognitive conceptions of ancient populations, based on visual representations.

Among pottery vessels in the Bibracte case-study, general type disparity may be related to functional and technical constraints, but could also be linked to the ostentatious nature of the vessels, as the most homogenous groups were types used for storing and serving liquids.

In the case-study of Middle Bronze Age axes, statistical analysis confirmed the existence of two separate traditions of production. Further investigation brought to light a more complex internal organisation, composed of six statistically different, shape-based groups. These six groups, although morphologically different, were geographically located in three larger zones (west, east and south). Exchanges were shown to exist between west and east, but almost no north-south exchanges could be attested. The existence of three separate zones suggests that the previous idea of a homogeneous Bronze Age “Atlantic world”, from the Atlantic coast to the North Sea, does not correspond to the finer structuration of geographical space, previously suggested, and confirmed in this case-study.

In the case study of La Tène brooches from Central Europe, the effect of time on the evolution of shape was clearly demonstrated by geometric morphometric methods. Archaeological time can be interpreted not only as a discrete variable, but also as a continuous vector. The brooch evolution demonstrated here may correspond to artefact innovation, probably in response to the demands of La Tène society. Shape changes seem to highlight a changing focus on brooch parts, which were perhaps enlarged to reflect changes in demand.

Implications and further development

The results obtained and presented here are scientific in nature, as statistical and probabilistic analyses are objective methods of assessment and testing.

The tools developed during this research project are practical applications for data acquisition and morphometric analysis, which should be of great interest for the archaeological community, allowing more data to be processed, more rapidly, and more economically.

The methods presented here are generalisable to other categories of artefacts, to study the level of production standardisation, and the evolution of artefact shape over space and time, thus providing more information about material and cultural exchanges.

The application of geometric morphometric methods along with other statistical methods used in the studies presented here may contribute to bringing archaeology ever closer to normative, standardised solutions for the processing of artefact shape data, and the production of typologies in the aim of overcoming current linguistic, spatiotemporal and memetic limitations.

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- Supplementary materials S1-4. Results of automatic workflow evaluation on real pottery fragments (well-oriented fragments).
- Supplementary materials S1-5. Results of automatic workflow evaluation on real pottery fragments (wrongly oriented).
- Supplementary materials S1-6. Results of semi-automatic workflow on real pottery fragments.
- Supplementary materials S2-1. Example of the result of pottery drawing.
- Supplementary materials S2-2. Example of the result of pottery drawing.
- Supplementary materials S2-3. Example of the result of pottery drawing.
- Supplementary materials S2-4. Example of the result of pottery drawing.
- Supplementary materials S2-5. Example of the result of pottery drawing.
- Supplementary materials S2-6. Example of the result of pottery drawing.
- Supplementary materials S2-7. Example of the result of pottery drawing.
- Supplementary materials S2-8. Example of the result of pottery drawing.
- Supplementary materials S2-9. Example of the result of pottery drawing.
- Supplementary materials S2-10. Example of the result of pottery drawing.
- Supplementary materials S2-11. Example of the result of pottery drawing.
- Supplementary materials S2-12. Example of the result of pottery drawing.
- Supplementary materials S2-13. Example of the result of pottery drawing.
- Supplementary materials S2-14. Example of the result of pottery drawing.
- Supplementary materials S2-15. Example of the result of pottery drawing.
- Supplementary materials S2-16. Example of the result of pottery drawing.
- Supplementary materials S2-17. Example of the result of pottery drawing.
- Supplementary materials S2-18. Example of the result of pottery drawing.
- Supplementary materials S2-19. Example of the result of pottery drawing.
- Supplementary materials S2-20. Example of the result of pottery drawing.
- Supplementary materials S2-21. Example of the result of pottery drawing.
- Supplementary materials S2-22. Example of the result of pottery drawing.
- Supplementary materials S2-23. Example of the result of pottery drawing.
- Supplementary materials S2-24. Example of the result of pottery drawing.
- Supplementary materials S2-25. Example of the result of pottery drawing.
- Supplementary materials S2-26. Example of the result of pottery drawing.
- Supplementary materials S2-27. Example of the result of pottery drawing.
- Supplementary materials S2-28. Example of the result of pottery drawing.
- Supplementary materials S2-29. Example of the result of pottery drawing.
- Supplementary materials S2-30. Example of the result of pottery drawing.
- Supplementary materials S3-1. Example of the result of pottery drawing.
- Supplementary materials S3-2. Example of the result of pottery drawing.
- Supplementary materials S3-3. Example of the result of pottery drawing.
- Supplementary materials S3-4. Example of the result of pottery drawing.
- Supplementary materials S3-5. Example of the result of pottery drawing.
- Supplementary materials S3-6. Example of the result of pottery drawing.
- Supplementary materials S3-7. Example of the result of pottery drawing.
- Supplementary materials S3-8. Example of the result of pottery drawing.

Supplementary materials S4-1. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-2. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-3. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-4. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-4. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-6. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-7. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-8. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-9. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-10. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-11. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-12. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-13. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-14. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-15. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-16. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-17. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-18. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-19. Example of the result of 3D pottery reconstruction.
 Supplementary materials S4-20. Example of the result of 3D pottery reconstruction.

Reconstructing the past from fragments (Chapter 4)

Supplementary materials S1. Pottery vessels used in the study.
 Supplementary materials S2. Evolution of Bayesian Information Criterion on group numbers for all unsupervised classification models.
 Supplementary materials S3. Histograms of probabilities for 44 target vessels.
 Supplementary materials S4. Visualisation of MBDA in the morphospace.
 Supplementary materials S5. Effect of number of Principal Components on reconstruction quality (click for animation).
 Supplementary materials S6. Effect of the number of Principal Components on reconstruction quality.
 Supplementary materials S7-F1-S1. Alphashape highlights distribution in the morphospace of the 1000 best candidates for fragment F1 of source vessel S1 (identified by a blue cross).
 Supplementary materials S7-F2-S1. Alphashape highlights distribution in the morphospace of the 1000 best candidates for fragment F2 of source vessel S1 (identified by a blue cross).
 Supplementary materials S7-F3-S1. Alphashape highlights distribution in the morphospace of the 1000 best candidates for fragment F3 of source vessel S1 (identified by a blue cross).
 Supplementary materials S7-F4-S1. Alphashape highlights distribution in the morphospace of the 1000 best candidates for fragment F4 of source vessel S1 (identified by a blue cross).
 Supplementary materials S7-F1-S2. Alphashape highlights distribution in the morphospace of the 1000 best candidates for fragment F1 of source vessel S2 (identified by a blue cross).
 Supplementary materials S7-F2-S2. Alphashape highlights distribution in the morphospace of the 1000 best candidates for fragment F2 of source vessel S2 (identified by a blue cross).
 Supplementary materials S7-F3-S2. Alphashape highlights distribution in the morphospace of the 1000 best candidates for fragment F3 of source vessel S2 (identified by a blue cross).
 Supplementary materials S7-F4-S2. Alphashape highlights distribution in the morphospace of the 1000 best candidates for fragment F4 of source vessel S2 (identified by a blue cross).
 Supplementary materials S7-F1-S3. Alphashape highlights distribution in the morphospace of the 1000 best candidates for fragment F1 of source vessel S3 (identified by a blue cross).
 Supplementary materials S7-F2-S3. Alphashape highlights distribution in the morphospace of the 1000 best candidates for fragment F2 of source vessel S3 (identified by a blue cross).
 Supplementary materials S7-F3-S3. Alphashape highlights distribution in the morphospace of the 1000 best candidates for fragment F3 of source vessel S3 (identified by a blue cross).
 Supplementary materials S7-F4-S3. Alphashape highlights distribution in the morphospace of the 1000

best candidates for fragment F4 of source vessel S3 (identified by a blue cross).

Supplementary materials S8-S1. Alphashapes highlight distribution in the morphospace of the 1000 best candidates for each of the four fragments of source vessel S1 (identified by a blue cross).

Supplementary materials S8-S2. Alphashapes highlight distribution in the morphospace of the 1000 best candidates for each of the four fragments of source vessel S2 (identified by a blue cross).

Supplementary materials S8-S3. Alphashapes highlight distribution in the morphospace of the 1000 best candidates for each of the four fragments of source vessel S3 (identified by a blue cross).

Supplementary materials S9-S1. A 3D alphashape highlights target vessels possessing the greatest number of hits (source vessel S1 broken into 10 fragments).

Supplementary materials S9-S2. A 3D alphashape highlights target vessels possessing the greatest number of hits (source vessel S2 broken into 10 fragments).

Supplementary materials S9-S3. A 3D alphashape highlights target vessels possessing the greatest number of hits (source vessel S3 broken into 10 fragments).

Supplementary materials S9-S4. A 3D alphashape highlights target vessels possessing the greatest number of hits (one randomly selected potential vessel, broken into 10 fragments, and then reconstructed in a 3D morphospace, using only the first 3 Principal Components).

Morphometrics of Second Iron Age ceramics strengths, weaknesses, and comparison with traditional typology (Chapter 5)

Supplementary materials S1. Literature containing images used in this study.

Supplementary materials S2. Traditional descriptive system for coding ceramics (modified from Vaginay and Guichard, 1988).

Supplementary materials S3. Interactive projection of items in a 3D morphospace (file is greater than 300 Mo so that opening and treatment can take some minutes).

Supplementary materials S4. Effect of the number of clusters on the amount of information gained using an unsupervised model-based clustering of the EFA-SS-e morphospace.

Supplementary materials S5. Visual representation of the unsupervised model-based clustering.

Unsupervised model-based clustering for typological classification of Middle Bronze Age flanged axes (Chapter 6)

Supplementary materials S1. References of flanged axes used in the corpus.

Supplementary materials S2. The relative cumulative power (in %).

Supplementary materials S3. Posterior probabilities of assignment of individual flanged axes and ΔBIC from unsupervised model-based classification, according to different numbers of clusters.

Brooches in motion: tracing shape evolution in La Tène brooches by geometric morphometrics (Chapter 7)

Supplementary materials S1. Catalogue of brooches used in the study.

Supplementary materials S2. Literature containing images used in this study.

Supplementary materials S3. Animation of general changes in brooch shape.

Supplementary materials S4. Multivariate Regression of shape on time (animation).

Supplementary materials S5. Multivariate Regression of shape on time (animation).

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